Hubbard-Holstein Mathematical Notes (Symbolic-Only Edition)

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March 20, 2026

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L=2 Noisy Surface Run Handoff

For the **local/offline L=2 HH noisy surface** (`phase3_v1` + `pareto_lean_l2` + `FakeNighthawk`), see MATH/l2_noisy_surface_run.md.

This is the terminal-agent handoff note for the three mitigation lanes:

- `none`
- `readout`
- `full`

It is intentionally the front-page pointer so the noisy-surface run recipe is visible before the rest of the manuscript.

[QPU] L=2 Winner Framing and Real QPU Settings (April 8, 2026)

> Front-page contract: distinguish the best raw HF-start ADAPT result from the separately validated real-QPU deployment anchor. Do not conflate either one with later prune/readapt JSON post-processing or with fixed-scaffold runtime submissions.

Two Front-Page Objects

- **Scientific raw HF-start winner.** This means direct ADAPT from the Hartree-Fock reference with `adapt_ref_json = null`, with no later JSON prune/recovery pass substituted in. Under the canonical `L=2, U=4.0, g=0.5` gate `|ΔE| \sim 10^{-4}`, the best substantiated raw HF-start line currently in hand is the direct `full_meta` artifact `hh_backend_adapt_fullhorse_powell_20260322T194155Z_fakenighthawk.json`, which reaches
$$|\Delta E| = 5.6178234645 \times 10^{-5}$$
at `81` transpiled two-qubit gates on `FakeMarrakesh`.
- **Validated real-QPU deployment anchor.** This is the separately validated April 5 `pareto_lean + phase3_v1 + SPSA` raw HF-start command that is currently trusted for rerunability on the live public CLI surface. Its scaffold burden is heavier, but it remains the validated deployment recipe:
$$|\Delta E| = 1.0822209459 \times 10^{-4},$$
with `218` transpiled two-qubit gates on the same `FakeMarrakesh` proxy.

Raw HF-Start Direct ADAPT Table (`|\Delta E|\sim 10^{-4}`)

Role	Artifact / surface	Pool	`\Delta E|`	Logical ops	Runtime params	Transpiled 2Q	Depth
Best raw HF-start ADAPT winner	`hh_backend_adapt_fullhorse_powell_20260322T194155Z_fakenighthawk`	`full_meta`	$5.6178234645 \times 10^{-5}$	27	27	81	151
Best reduced-pool direct line	`adapt_hh_L2_pareto_lean_phase3_powell_rerun_currentcode_20260321T171615Z`	`pareto_lean`	$5.6178234644 \times 10^{-5}$	14	25	87	208
Wide-shortlist direct comparison	`campaign_A_full_shortlist_phase3_v1`	`full_meta`	$5.6178234644 \times 10^{-5}$	14	30	116	310
Current-code reproduction of March 22 low-energy regime	`20260409_hh_full_hist_powell_distbarel_716_v1`	`full_meta` historicalized POWELL surface	$5.7917492583 \times 10^{-5}$	54	54	265	786
Best current native recovery branch on today's diagnostic surface (`children_off + repeats_off + hist`)	`20260409_hh_full_meta_children_off_repeats_off_hist_diag_bridge_diag_v1`	`full_meta` diagnostic bridge surface	$5.6178234861 \times 10^{-5}$	13	36	160	408

Role	Artifact / surface	Pool	|\Delta E|	Logical ops	Runtime params	Transpiled 2Q	Depth
Validated April 5 deployment anchor	pareto_lean + phase3_v1 + SPSA	pareto_lean	1.0822209459e-04	20	52	218	633

The final 218-2Q row is the same raw HF-start scaffold family as the confirmed April 5 authority command; it is heavier than the scientific raw winner, but it is the currently validated deployment anchor on the public CLI surface.

The post-processed 31/37/57-2Q prune/readapt artifacts are **not** the object a fresh real-QPU ADAPT run from HF would execute. They remain useful comparison context only.

Current native recovery note (April 9, 2026)

The new children_off + repeats_off + hist branch from 20260409_hh_l2_children_repeat_bridge_diag_v1 deserves front-page visibility because it is the best **current-code native recovery branch** on today's diagnostic/public surface:

$$|\Delta E| = 5.6178241861 \times 10^{-5}, \qquad F = 0.9999719078,$$

with 13 logical operators, 36 runtime parameters, and 160 transpiled two-qubit gates at depth 408 on FakeMarrakesh.

This is **not** the absolute historical raw HF-start winner; the March 22 81-2Q full_meta artifact remains lighter overall. What makes the April 9 branch important is different: it is the first current diagnostic line that re-enters the good native basin while simultaneously showing that disabling children and literal repeats helps native recovery.

It still does **not** recover the historical lean family exactly. Relative to the historical targets, it reaches 9/10 bridge overlap and 5/6 lean overlap, with the remaining extras concentrated in

- paop_full:paop_disp(site=1)
- paop_lf_full:paop_dbl_p(site=0->phonon=1)
- paop_lf_full:paop_dbl_p(site=1->phonon=0)
- paop_full:paop_hopdrag(0,1)

This diagnostic result matters because it isolates the current native failure mode more sharply than the older heavy broad-pool lines: the good basin is real, but some quadrature and inside-family extras are still acting as **path operators** rather than merely removable terminal clutter.

Current-code reproduction of the March 22 low-energy regime

The March 22 81-2Q full_meta line is still a real historical raw HF-start winner, but current code does **not** replay that exact burden from its historical settings block. What current code **does** reproduce is the same low-energy regime on a historicalized POWELL surface with a hard cap at the first in-run depth that re-enters the target band.

The saved current-code replay is

$$|\Delta E| = 5.7217492583 \times 10^{-5},$$

from 20260409_hh_l2_hist81_repro_powell_histbare_d16_v1, with 17 logical blocks, 54 runtime parameters, and 265 transpiled two-qubit gates at depth 786 on FakeMarrakesh.

So this run should be interpreted as a **current-code low-energy reproduction**, not as a faithful current reproduction of the historical 81-2Q scaffold burden.

Exact current-code replay CLI

```
python pipelines/hardcoded/adapt_pipeline.py \
  --L 2 --problem hh --t 1.0 --u 4.0 --dv 0.0 --omega0 1.0 --g-ep 0.5 --n-ph-max 1 \
  --boson-encoding binary --ordering blocked --boundary open --term-order sorted \
  --adapt-pool full_meta --adapt-continuation-mode phase3_v1 \
  --adapt-max-depth 16 --adapt-eps-grad 5e-7 --adapt-eps-energy 1e-9 \
  --adapt-inner-optimizer POWELL --adapt-state-backend compiled \
  --adapt-reopt-policy windowed --adapt-window-size 999999 --adapt-window-topk 999999 \
  --adapt-full-refit-every 8 --adapt-final-full-refit true \
  --adapt-beam-live-branches 1 --adapt-beam-children-per-parent 1 --adapt-beam-terminated-keep 1 \
  --phase1-lambda-F 1.0 --phase1-lambda-compile 0.05 --phase1-lambda-measure 0.02 --phase1-lambda-leak 0.0
  --phase1-score-z-alpha 0.0 --phase1-shortlist-size 256 --phase1-probe-max-positions 999999 \
  --phase1-plateau-patience 2 --phase1-trough-margin-ratio 1.0 \
  --phase1-prune-enabled --phase1-prune-mode live --phase1-prune-fraction 0.25 --phase1-prune-max-candidat
  --phase2-shortlist-fraction 1.0 --phase2-shortlist-size 128 \
  --phase2-lambda-H 1e-6 --phase2-rho 0.25 --phase2-gamma-N 1.0 \
  --phase2-enable-batching --phase2-batch-target-size 8 --phase2-batch-size-cap 16 --phase2-batch-near-deg
  --phase2-w-depth 0.0 --phase2-w-group 0.0 --phase2-w-shot 0.0 --phase2-w-optdim 0.0 --phase2-w-reuse 0.0
  --phase2-eta-L 0.0 --phase2-motif-bonus-weight 0.0 --phase2-duplicate-penalty-weight 0.0 \
  --phase2-compat-overlap-weight 0.0 --phase2-compat-comm-weight 0.0 --phase2-compat-curv-weight 0.0 --pha
  --phase2-frontier-ratio 1.0 --phase3-frontier-ratio 1.0 --phase2-remaining-evaluations-proxy-mode none \
  --phase3-tie-beam-max-branches 1 \
  --phase3-symmetry-mitigation-mode verify_only --phase3-enable-rescue \
  --phase3-lifetime-cost-mode phase3_v1 --phase3-runtime-split-mode shortlist_pauli_children_v1 \
  --phase3-backend-cost-mode transpile_single_v1 --phase3-backend-name FakeNighthawk \
  --phase3-backend-transpile-seed 7 --phase3-backend-optimization-level 1 \
  --adapt-seed 7 --adapt-finite-angle-fallback --adapt-finite-angle 0.1 --adapt-finite-angle-min-improveme
  --adapt-drop-floor 1e-11 --adapt-drop-patience 5 --adapt-drop-min-depth 12 --adapt-grad-floor -1.0 \
  --paop-r 1 --paop-normalization none \
  --initial-state-source adapt_vqe \
  --skip-pdf
```

Real QPU Deployment Anchor (April 5, 2026 Validated)

This is the recommended configuration for real QPU deployment. Validated via narrow 36-case local sweep confirming rerunnable deployment optimality in the `pareto_lean` + `phase3_v1` regime. This is the deployment anchor, not the absolute lowest raw HF-start two-qubit line.

Anchor Configuration

Use this exact command for real QPU L=2 ADAPT runs:

```
python -u -m pipelines.hardcoded.adapt_pipeline \
  --L 2 --problem hh --t 1.0 --u 4.0 --dv 0.0 \
  --omega0 1.0 --g-ep 0.5 --n-ph-max 1 \
  --boson-encoding binary --ordering blocked --boundary open --term-order sorted \
  --adapt-pool pareto_lean --adapt-continuation-mode phase3_v1 \
  --adapt-max-depth 160 --adapt-eps-grad 5e-7 --adapt-eps-energy 1e-9 --adapt-seed 5 \
  --adapt-inner-optimizer SPSA \
  --adapt-spsa-a 0.1 --adapt-spsa-c 0.02 --adapt-spsa-A 5.0 \
  --adapt-spsa-callback-every 5 --adapt-spsa-progress-every-s 30 \
  --adapt-maxiter 3200 \
  --adapt-state-backend compiled \
  --adapt-reopt-policy windowed --adapt-window-size 999999 --adapt-window-topk 999999 \
  --adapt-full-refit-every 8 --adapt-final-full-refit true \
  --adapt-beam-live-branches 2 --adapt-beam-children-per-parent 2 --adapt-beam-terminated-keep 2 \
  --phase1-prune-enabled --phase1-prune-fraction 0.25 --phase1-prune-max-candidates 6 --phase1-prune-max-r
```

```
  --phase1-probe-max-positions 999999 --phase1-trough-margin-ratio 1.0 --phase1-shortlist-size 64 \
  --phase2-shortlist-fraction 1.0 --phase2-shortlist-size 64 \
  --phase2-frontier-ratio 0.85 \
  --phase2-lambda-H 1e-06 --phase2-rho 0.25 --phase2-gamma-N 1.0 \
  --phase2-w-depth 0.1 --phase2-w-group 0.075 --phase2-w-shot 0.075 --phase2-w-optdim 0.05 --phase2-w-reus
  --phase2-eta-L 0.0 --phase2-motif-bonus-weight 0.05 --phase2-duplicate-penalty-weight 0.0 \
  --phase2-compat-overlap-weight 0.4 --phase2-compat-comm-weight 0.2 --phase2-compat-curv-weight 0.2 \
  --phase2-compat-sched-weight 0.2 --phase2-compat-measure-weight 0.2 \
  --phase2-remaining-evaluations-proxy-mode auto \
  --phase3-frontier-ratio 0.85 \
  --phase3-symmetry-mitigation-mode verify_only --phase3-enable-rescue \
  --phase3-backend-cost-mode proxy --phase3-runtime-split-mode off --phase3-lifetime-cost-mode off \
  --adapt-drop-floor 0.0005 --adapt-drop-patience 3 --adapt-drop-min-depth 12 \
  --initial-state-source adapt_vqe --skip-pdf --phase2-no-batching
```

Expected Performance

Metric	Value	Notes
Energy	0.15877612622031317	Exact ground state reference
$\|\Delta E\|$	**1.082e-4**	Excellent accuracy for NISQ
Logical scaffold size	20	Validated raw HF-start deployment scaffold
Runtime parameters	52	Efficient parameterization
Marrakesh proxy burden	`218` 2Q, depth `633`, size `939`	Same raw HF-start scaffold as the confirmed strong template
Wall-clock	~43s	Fast convergence

Validation Results (April 5 Sweep)

A 36-case narrow local sweep confirmed this configuration is optimal:

- [OK] **SPSA maxiter=3200**: Goldilocks zone (2400 undershoots, 4000 overshoots)
- [OK] **Frontier ratio 0.85**: All values (0.80–0.90) converge to same energy; 0.85 balances quality
- [OK] **Beam (2,2,2)**: Sufficient; wider beam (3,2,3) costs 50% more time, zero quality gain
- [OK] **Drop policy (0.0005, 3, 12)**: Optimal shallowness; tighter drop → shallower circuits for QPU

Critical Finding

Increasing SPSA `maxiter` beyond 3200 **degrades** performance (1.27–1.54e-4 vs 1.082e-4): - Suggests pool saturation: `pareto_lean` operators are exhausted at this budget - SPSA noise amplification: longer optimization in finite ansatz space doesn't help

Do not deviate from maxiter=3200 in this regime.

What NOT to Change

Based on sweep validation: - [X] Pool: Do not switch from `pareto_lean` - [X] Continuation: Do not move away from `phase3_v1` - [X] Runtime split: Keep `off` for L=2 - [X] SPSA budget: Do not increase `maxiter` beyond 3200 - [X] Frontier: Do not vary frontier ratio (insensitive in 0.80–0.90 range) - [X] Beam: Do not use (3,2,3); keep (2,2,2)

For Real QPU Deployment

1. Use the command above exactly as written
2. Expect $|\Delta E| \approx 1.08$e-4 (this is the best-known trustworthy result)
3. Ansatz depth = 20 is excellent for NISQ mitigation
4. No further local tuning will improve this configuration (sweep confirms)

Companion Driven Candidate

The leading **driven realtime-controller** candidate for the same L=2 HH scaffold family is now the heavier-target `secant_lead100` setting from `artifacts/agent_runs/20260405_controller_drive_heavier_secant_lead_scan_v1/json/s`

This should be tracked alongside the winning static/noiseless L=2 ADAPT anchor above when choosing real-QPU-facing settings, because it is currently the best controller law we have for recovering visible driven dynamics on the stronger `A=0.55`, `tbar=4.0`, `t_final=8.0` benchmark.

Current driven-controller headline metrics

Metric	Value	Notes
Controller law	`secant_lead100`	Signed-energy-lead-tapered secant + signed blend
Drive target	`A=0.55`, `omega=2.0`, `tbar=4.0`, `t0=0.0`	Heavier than the earlier fine-resolution regime
Mean `|ΔE|`	**1.633e-3**	Best whole-window value among current heavier-target candidates
Max `|ΔE|`	**6.501e-3**	Large improvement over prior signed-blend and raw secant anchors
Early-window MAE (`t 3.35`)	**1.176e-3**	Recovers the startup dip far better than `signed_mild_front_current`
Late-window MAE (`t 3.35`)	**1.964e-3**	Still slightly better than the previous late-time winner
Energy span ratio	`1.032`	Slight overshoot, but close to exact
Min fidelity	`0.9626`	Remains bounded with no append events

Leading QPU-facing pair

- **Static / ground-state anchor:** the validated `pareto_lean + phase3_v1` L=2 ADAPT command above.
- **Driven / dynamics anchor:** the heavier-target `secant_lead100` controller result above.

These are the two front-page candidates we should currently treat as the leading settings when optimizing toward a real QPU run.

Canonical naming note: `MATH/Math.md` is the markdown authoring source for manuscript sync, and `MATH/Math.tex` is the generated typesetting twin used to build `MATH/Math.pdf`. Regenerate the synchronized TeX/PDF pair with `python MATH/build_math_from_md.py` or `MATH/build_math.sh`. The archival repo-oriented manuscript is `MATH/archaic_repo_math.md`.

1. Parameter Manifest and Reader Contract

This duplicate is intentionally symbolic-first. It keeps the same broad manuscript shape as the existing mathematical notes,

1. primitives first,
2. composite operators second,
3. explicit substitutions third,
4. reduced master forms last,

but it suppresses almost all implementation labels, file names, mode strings, and executable-surface commentary. The goal is to make the document readable as mathematics rather than as a repository audit.

1.1 Required parameter manifest

A complete symbolic run or derivation should specify at least the following data.

- Model family: Hubbard or Hubbard-Holstein.
- Lattice size: L.
- Ordering map: interleaved or blocked, i.e. a choice of $p(i, \sigma)$.
- Boundary condition: open or periodic.
- Fermionic parameters: hopping t, onsite interaction U, and static site potentials v_i.
- Bosonic parameters: phonon frequency ω_0, electron-phonon coupling g, and cutoff $n_{\mathrm{ph,max}}$.
- Boson encoding choice: binary or unary.
- Variational family, when relevant: layerwise, physical-termwise, excitation-based, or adaptive.
- Control parameters, when relevant: optimizer family, propagation rule, drive waveform parameters, and benchmark grid.

1.2 Reader contract

This manuscript favors explicit substitution over compressed abstraction.

- If a primitive operator exists, it is written first.
- If a composite object is built from primitives, the primitive form is shown.
- If a later equation can be simplified by inserting an earlier primitive explicitly, that insertion is made.
- If multiple mathematical surfaces exist, such as different index orderings or boson encodings, the surface is named locally rather than hidden behind an unnamed default.

1.2.1 Quick notation glossary for adaptive and continuation sections

This subsection is a compact map for the symbols that recur in the adaptive-selection, split-aware, replay, and beam-style sections.

- g, m: generic candidate generators. The manuscript uses both letters in different sections; they play the same role.
- p: candidate insertion position in the ordered scaffold.
- $r = (m, p)$: a candidate-position record, i.e. a generator together with a chosen insertion position.
- $\mathcal{G}$: the full master generator pool.
- $\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \mathcal{G}^{(3)}$: progressively more restricted candidate universes, with

$$\mathcal{G}^{(3)} \subseteq \mathcal{G}^{(2)} \subseteq \mathcal{G}^{(1)} \subseteq \mathcal{G}.$$

- $\mathcal{P}_g$ or $\mathcal{P}_m$: admissible insertion positions for candidate g or m.
- $W(p)$: inherited refit window used if a candidate is inserted at position p.
- $W_{\mathrm{new}}(p)$: the newest-parameter portion of the refit window.
- $W_{|\theta|}(p)$: the older large-amplitude carry subset of the refit window.
- $\mathcal{P}^{(3)}_{\mathrm{avail}}$: the full available set of Phase-3 candidate-position records.
- $\mathcal{C}_{\mathrm{cheap}}$: the horizon-sized coarse shortlist obtained by cheap-score ranking over the full admissible Phase-3 candidate-position pool.
- $\mathcal{S}_2, \mathcal{S}_3$: the Phase-2 and Phase-3 shortlists that survive cheap screening before full reranking.
- $\mathcal{C}_{\mathrm{split}}(m)$: the family of split children generated from macro generator m.
- $\Gamma^{\mathrm{perm}}_{\mathrm{prune}}$: post-admission prune-permissibility gate deciding whether local scaffold simplification is attempted.
- $\mathcal{J}_{\mathrm{exp}}$: the post-admission set of locally expendable scaffold coordinates or generators considered by the prune pass.
- $\mathcal{O}_*$: the finally selected adaptive scaffold, i.e. the ordered generator sequence admitted by the adaptive stage.
- $\mathcal{M}_{\mathrm{warm}}, \mathcal{M}_{\mathrm{refine}}, \mathcal{M}_{\mathrm{replay}}$: archival appendix-only warm-start, optional refine, and replay manifolds.
- $\mathcal{B}$: a portable handoff/state bundle carrying manifest, amplitudes, energies, and continuation metadata.
- $\mathcal{C}$: continuation payload / provenance payload attached to a handoff bundle.
- Γ_{stage}: stage-admissibility gate.
- Γ_{sym}: symmetry-admissibility gate.
- $\mathfrak{b}$: a beam branch state. In §11 it indexes alternative ADAPT scaffolds; in §17 it indexes projected-dynamics branches over checkpoint segments.

- $\mathcal{H}_b$: admitted-record history carried by scaffold branch b.
- $\mathcal{A}_b$: branch-local admissible admission set retained for scaffold-beam expansion.
- $\mathcal{Q}(b)$: scaffold-beam proposal family, usually $\{\text{stop}\} \cup \mathcal{A}_b$.
- η_b: branch-local stage label for the scaffold beam, e.g. core/seed versus residual.
- $W_{\text{act}}(\mathfrak{b})$: the currently active position-probe window on branch b.
- $\mathfrak{F}_c$: live frontier at beam round c.
- $\mathfrak{T}_c$: terminal-branch pool at beam round c.
- $\mathfrak{W}_{\text{final}}$: final union of surviving frontier and terminal branches.
- J_b: cumulative branch objective tuple used to rank or prune branch b; its coordinates are section-specific.

1.2.2 Quick parameter glossary for adaptive and continuation sections

The symbols below are the main scalar controls that appear in the adaptive-selection, split-aware, and beam-style formulas.

- $\lambda_{2\text{q}}$: weight on two-qubit proxy burden.
- λ_d: weight on depth-style proxy burden.
- λ_θ: weight on marginal parameter-count burden.
- λ_F: cheap-stage normalization weight multiplying the tangent metric term.
- λ_H: regularization strength for inherited-window Hessian blocks.
- ρ: trust-region radius scale used in one-dimensional local improvement models.
- $N_{\text{cheap}}^{\max}$: absolute cap on the horizon-sized Phase-3 coarse shortlist.
- $N_{\max}$: maximum shortlist size.
- f_{short}: shortlist fraction used in Phase 3.
- w_R: weight on useful-horizon lifetime burden.
- w_D: weight on lifetime depth/compile burden.
- w_G: weight on newly introduced grouped-measurement burden.
- w_C: weight on newly introduced compile or circuit burden.
- w_c: weight on any extra cheap-path correction burden.
- γ_N: exponent applied to novelty factors in full rerank scores.
- ε: small positive stabilizer used in denominators and tie-safe quotients.
- ε_{nov}: novelty regularizer used when inverting or stabilizing tangent-overlap matrices.
- z_α: confidence multiplier used to convert a raw gradient estimate into a lower-confidence gradient.
- $\sigma(g,p)$ or σ_r: uncertainty estimate attached to the candidate-position gradient signal.
- τ_1: Phase-2 shortlist threshold.
- τ_{split}: split-replacement margin required before a split child or child-set beats its parent.
- τ_{drop}: minimum per-depth absolute-error improvement regarded as meaningful by the stage controller.
- M: patience window used when counting repeated small-drop depths.
- $D_{\text{left}}(t)$: remaining controller depth budget at selector step t.
- $S_t(k)$: survival factor for having a still-useful admission opportunity k selector steps ahead.
- $Q_t(k)$: conditional usefulness factor for the k-ahead admission opportunity.
- $\widehat{N}_{\text{rem}}(t)$: heuristic forecast of useful remaining admissions at selector step t.
- C_t: confidence attached to the current useful-runway forecast.
- $K_{\text{coarse}}(t)$: dynamic coarse-cap induced by the current useful-runway forecast.
- H_t: effective useful-depth horizon,
$$H_t = \min(D_{\text{left}}(t), \widehat{N}_{\text{rem}}(t)).$$
- $W_{\text{sat}}(t)$: saturation pressure induced by the current useful-runway forecast.
- π_t: selector mode tag, e.g. fast / mixed / high-fidelity.
- $\text{regime}_{\text{sat}}(t)$: hungry / watch / plateau saturation regime.
- η_{ret}: minimum retained-gain fraction required by the post-admission prune accept rule.
- β_{keep}: repo-facing retained-gain floor used in prune acceptance; in this manuscript it plays the same role as η_{ret}.
- τ_{miss}: projected-miss threshold used in manifold-growth logic.
- B_{child}: beam child cap per parent branch-expansion round.
- $B_{\text{target}}(t)$: active batch target size at selector step t.
- B_{live}: live-frontier beam width retained after pruning.
- B_{term}: cap on stored terminal branches.

- M_{probe}: cap on distinct probed insertion positions retained in order.
- $a_j(t)$: admission age of scaffold coordinate or generator j at selector step t.
- $\varrho_j(t)$: normalized admission-rank score of coordinate or generator j at selector step t.
- $\mathcal{P}_{\text{protect}}(t)$: hard-protected coordinate set excluded from the live prune pass.
- $c_j^{\text{cool}}(t)$: local prune cooldown on coordinate j.
- $S_\theta(j,t)$: optional small-angle prescreen score used only to save local prune work.
- τ_{stale}: minimum local expendability score required before a prune trial is even considered.
- Ξ_t^{trial}: exact rollback snapshot taken before a local prune/remove-refit trial.
- Δk_{cp}: checkpoint advance used in the projected-dynamics beam of §17, not in the scaffold beam of §11.
- Δk_{probe}: probe advance used in the projected-dynamics beam of §17, not in the scaffold beam of §11.
- C: final beam round index / number of retained beam-update rounds.

In §11, the index t denotes a selector/controller step rather than the real-time variable used later in the dynamics sections.

1.3 Non-negotiable representational conventions

1.3.1 Internal Pauli alphabet

In formulas below we use the standard alphabet

$$\{I, X, Y, Z\}.$$

If one prefers the lower-case alphabet $\{e, x, y, z\}$, the identification is simply

$$e \leftrightarrow I, \qquad x \leftrightarrow X, \qquad y \leftrightarrow Y, \qquad z \leftrightarrow Z.$$

1.3.2 Pauli-word and qubit ordering

Pauli words and computational-basis labels are written left-to-right as

$$q_{N_q-1} \cdots q_1 q_0,$$

with qubit q_0 the rightmost symbol and also the least-significant bit in basis-index arithmetic.

1.3.3 Canonical algebra sources

The canonical mathematical primitives are:

- Jordan-Wigner ladder operators $\hat{c}_p^\dagger$ and $\hat{c}_p$,
- fermion number operators $\hat{n}_p$,
- boson ladder operators $\hat{b}_i^\dagger$ and $\hat{b}_i$,
- Hermitian Pauli words and Pauli polynomials,
- ordered exponentials generated from those primitives.

1.3.4 Surface-specific defaults

This symbolic version does not pretend there is a single universal surface. The mathematics admits several equally valid choices:

- interleaved or blocked fermion ordering,
- open or periodic boundary conditions,
- binary or unary boson encoding,
- static or driven Hamiltonians,
- fixed-structure or adaptive variational manifolds.

Whenever a formula depends on the choice of surface, that dependence is shown locally.

1.4 Canonical mathematical anchors

The main objects carried through the manuscript are:

- the ordering map $p(i, \sigma)$,
- the ladder primitives $\hat{c}_p^\dagger$, $\hat{c}_p$, $\hat{b}_i^\dagger$, $\hat{b}_i$,
- the density operators $\hat{n}_{i\sigma}$, $\hat{n}_i$, and $\hat{D}_i$,
- the Hubbard and Hubbard-Holstein Hamiltonians,
- the reference state $|\psi_{\mathrm{ref}}\rangle$,
- the variational manifold $\mathcal{M}$ and its ansatz map $\theta \mapsto |\psi(\theta)\rangle$,
- the propagation rule $U(t)$,
- the continuation or handoff bundle $\mathcal{B}$.

2. Ordering, Indexing, and Register Layout

2.1 Site, spin, and mode indices

The site index is

$$i \in \{0, 1, \dots, L-1\},$$

and the spin label is stored as

$$\sigma \in \{\uparrow, \downarrow\} \equiv \{0, 1\}.$$

The fermion mode index is the Jordan-Wigner qubit index.

2.1.1 Interleaved ordering

The interleaved map is

$$p(i, \sigma) = 2i + \sigma,$$

so that

$$p(i, \uparrow) = 2i, \qquad p(i, \downarrow) = 2i + 1.$$

2.1.2 Blocked ordering

The blocked map is

$$p(i, \uparrow) = i, \qquad p(i, \downarrow) = L + i.$$

2.2 Pauli-word placement and basis-index extraction

If a Pauli letter acts on qubit q, then in a printed word of length N_q it sits at string position

$$\mathrm{pos}(q) = N_q - 1 - q.$$

If the computational-basis index is k, then the occupation bit on qubit q is

$$b_q(k) = \left\lfloor \frac{k}{2^q} \right\rfloor \bmod 2.$$

Thus the printed bitstring and integer basis index obey the same rightmost-q_0 convention.

2.3 Full HH register layout

The fermion register uses

$$N_{\mathrm{ferm}} = 2L$$

qubits.

If the local phonon cutoff is $n_{\mathrm{ph,max}}$, then the local phonon Hilbert dimension is

$$d = n_{\mathrm{ph,max}} + 1.$$

The number of phonon qubits per site is

$$q_{\mathrm{pb}} = \begin{cases} \max\{1, \lceil \log_2 d \rceil\}, & \text{binary}, \\ d, & \text{unary}. \end{cases}$$

Therefore the total HH qubit count is

$$N_q = 2L + Lq_{\mathrm{pb}}.$$

In qubit-index order, the register is naturally partitioned as

$$[\text{fermions} \mid \text{site-0 phonons} \mid \cdots \mid \text{site-(L-1) phonons}].$$

In printed order $q_{N_q-1} \cdots q_0$, the high-index phonon blocks appear on the left.

3. Fermionic Primitives and Direct Substitution

3.1 Jordan-Wigner ladder primitives

For mode p, the creation operator is

$$\hat{c}_p^\dagger = \frac{1}{2}(X_p - iY_p) \prod_{r=0}^{p-1} Z_r,$$

and the annihilation operator is

$$\hat{c}_p = \frac{1}{2}(X_p + iY_p) \prod_{r=0}^{p-1} Z_r.$$

In fully expanded printed-word form, the operator acts on the word ordered as $q_{N_q-1} \cdots q_0$ with the Jordan-Wigner Z-string extending through all modes below p.

3.2 Number primitive

The fermion number operator is

$$\hat{n}_p = \hat{c}_p^\dagger \hat{c}_p = \frac{I - Z_p}{2}.$$

3.3 Site densities and doublon operator

At site i, define

$$\hat{n}_i = \hat{n}_{i\uparrow} + \hat{n}_{i\downarrow},$$

and the doublon operator

$$\hat{D}_i = \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}.$$

By direct substitution,

$$\hat{D}_i = \frac{1}{4}(I - Z_{p(i,\uparrow)})(I - Z_{p(i,\downarrow)}).$$

3.4 Worked ordering substitutions

3.4.1 Interleaved

Under interleaved ordering,

$$\hat{n}_i = I - \frac{1}{2}(Z_{2i} + Z_{2i+1}),$$

and

$$\hat{D}_i = \frac{1}{4}(I - Z_{2i})(I - Z_{2i+1}).$$

For nearest-neighbor spin-up hopping in a one-dimensional chain,

$$\hat{c}_{i\uparrow}^\dagger \hat{c}_{i+1,\uparrow} + \hat{c}_{i+1,\uparrow}^\dagger \hat{c}_{i\uparrow} = \frac{1}{2}\Big(X_{2i} Z_{2i+1} X_{2i+2} + Y_{2i} Z_{2i+1} Y_{2i+2}\Big).$$

3.4.2 Blocked

Under blocked ordering,

$$\hat{n}_i = I - \frac{1}{2}(Z_i + Z_{L+i}),$$

and

$$\hat{D}_i = \frac{1}{4}(I - Z_i)(I - Z_{L+i}).$$

For spin-up nearest-neighbor hopping, the Jordan-Wigner string is empty because the two blocked spin-up modes are adjacent, so

$$\hat{c}^\dagger_{i\uparrow}\hat{c}_{i+1,\uparrow} + \hat{c}^\dagger_{i+1,\uparrow}\hat{c}_{i\uparrow} = \frac{1}{2}\Big(X_iX_{i+1} + Y_iY_{i+1}\Big).$$

The general blocked-ordering formula uses the same Jordan-Wigner interval rule as before, with an empty product for adjacent modes.

4. Boson Primitives and Encodings

4.1 Local boson Hilbert space

For each site i, the truncated boson Hilbert space is

$$\mathcal{H}_{b,i} = \mathrm{span}\{|n_i\rangle : 0 \le n_i \le n_{\mathrm{ph,max}}\}.$$

The ladder operators satisfy

$$\hat{b}^\dagger_i|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad \hat{b}_i|n\rangle = \sqrt{n}\,|n-1\rangle,$$

with

$$\hat{n}_{b,i} = \hat{b}^\dagger_i\hat{b}_i, \qquad \hat{x}_i = \hat{b}_i + \hat{b}^\dagger_i, \qquad \hat{P}_i = i(\hat{b}^\dagger_i - \hat{b}_i).$$

4.2 Binary encoding

In binary encoding, each local occupation number $n \in \{0, \dots, d-1\}$ is mapped to a binary word on $q_{\mathrm{pb}} = \lceil \log_2 d \rceil$ qubits:

$$|n\rangle \longmapsto |\,\mathrm{bin}(n)\rangle.$$

The truncated number operator is then

$$\hat{n}_{b,i} = \sum_{n=0}^{d-1} n\,|n\rangle\langle n|,$$

understood as an operator on the binary-encoded subspace.

4.3 Unary encoding

In unary encoding, each local occupation n is represented by a one-hot basis state on d qubits:

$$|n\rangle \longmapsto |0\cdots010\cdots0\rangle,$$

with the single 1 marking the occupied unary slot. The number operator remains

$$\hat{n}_{b,i} = \sum_{n=0}^{d-1} n\,|n\rangle\langle n|,$$

but now the basis states are embedded in a one-hot code space.

4.4 Phonon vacuum

4.4.1 Binary vacuum

The local binary vacuum is

$$|0\rangle_{b,i} \longmapsto |00\cdots0\rangle.$$

4.4.2 Unary vacuum

The local unary vacuum is the one-hot state carrying occupancy zero,

$$|0\rangle_{b,i} \longmapsto |10\cdots 0\rangle,$$

or whichever one-hot convention is chosen consistently for the lowest occupation.

5. Hubbard Hamiltonian by Explicit Substitution

Write the static Hubbard Hamiltonian as

$$\hat{H}_{\text{Hub}} = \hat{H}_t + \hat{H}_U + \hat{H}_v.$$

5.1 Kinetic term

The kinetic term is

$$\hat{H}_t = -t \sum_{\langle i,j\rangle,\sigma} \left(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\right).$$

For two fermion modes $p < q$, direct Jordan-Wigner substitution gives

$$\hat{c}^\dagger_p\hat{c}_q + \hat{c}^\dagger_q\hat{c}_p = \frac{1}{2}\left(X_pZ_{p+1}\cdots Z_{q-1}X_q + Y_pZ_{p+1}\cdots Z_{q-1}Y_q\right).$$

5.2 Onsite interaction

The onsite interaction term is

$$\hat{H}_U = U\sum_i \hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = U\sum_i \hat{D}_i.$$

Substituting the number primitive,

$$\hat{H}_U = \frac{U}{4}\sum_i (I - Z_{p(i,\uparrow)})(I - Z_{p(i,\downarrow)}).$$

5.3 Static potential term

The static potential term is

$$\hat{H}_v = \sum_{i,\sigma} v_i\hat{n}_{i\sigma}.$$

After substitution,

$$\hat{H}_v = \sum_i v_i\left(I - \frac{1}{2}(Z_{p(i,\uparrow)} + Z_{p(i,\downarrow)})\right).$$

5.4 Fully substituted Hubbard Hamiltonian

Combining the pieces,

$$\hat{H}_{\text{Hub}} = -t\sum_{\langle i,j\rangle,\sigma}\left(\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\right) + \frac{U}{4}\sum_i (I - Z_{p(i,\uparrow)})(I - Z_{p(i,\downarrow)}) + \sum_i v_i\left(I - \frac{1}{2}(Z_{p(i,\uparrow)} + Z_{p(i,\downarrow)})\right),$$

with the kinetic term expanded further into Pauli strings as needed.

6. Hubbard-Holstein Hamiltonian by Explicit Substitution

The driven Hubbard-Holstein Hamiltonian is

$$\hat{H}_{\mathrm{HH}}(t) = \hat{H}_t + \hat{H}_U + \hat{H}_v + \hat{H}_{\mathrm{ph}} + \hat{H}_g + \hat{H}_{\mathrm{drive}}(t).$$

The driven onsite-density contribution is written in fixed-label Pauli form as

$$\hat{H}_{\mathrm{drive}}(t) = \sum_{\lambda \in \mathcal{L}_{\mathrm{drive}}} c_\lambda(t) P_\lambda,$$

where the operator labels P_λ are static and only the coefficients $c_\lambda(t)$ vary in time. For the site-density drive used in the active HH dynamics stack,

$$\Delta c[Z_{p(i,\uparrow)}](t) = \Delta c[Z_{p(i,\downarrow)}](t) = -\frac{1}{2} v_i(t),$$

and, if one retains the scalar offset explicitly,

$$\Delta c[I](t) = \sum_i v_i(t).$$

6.1 Phonon energy

The phonon energy is

$$\hat{H}_{\mathrm{ph}} = \omega_0 \sum_i \left(\hat{n}_{b,i} + \frac{1}{2} I \right).$$

6.1.1 Unary explicit form

In unary form, $\hat{n}_{b,i}$ is diagonal on the one-hot occupation basis:

$$\hat{n}_{b,i} = \sum_{n=0}^{d-1} n \, |n\rangle_i \langle n|.$$

For the full qubit-level Pauli reduction in the spin-block JW + unary setting, including the explicit I/Z form of $\hat{H}_{\mathrm{ph}}$, see Section 6.2.3.4(a).

6.1.2 Binary explicit form

In binary form,

$$\hat{n}_{b,i} = \sum_{n=0}^{d-1} n \, |\,\mathrm{bin}(n)\rangle_i \langle \mathrm{bin}(n)|,$$

understood as an operator on the truncated binary code space.

6.2 Electron-phonon coupling

A natural density-shifted coupling is

$$\hat{H}_g = g \sum_i \hat{x}_i (\hat{n}_i - \bar{n} I),$$

where $\bar{n} = N_e/L$ is the reference density, often $\bar{n} = 1$ at half filling.

6.2.1 Unary explicit form

In unary encoding, $\hat{x}_i$ acts as nearest-neighbor hopping on the truncated local occupation chain:

$$\hat{x}_i = \sum_{n=0}^{d-2} \sqrt{n+1}\Big(|n+1\rangle_i\langle n| + |n\rangle_i\langle n+1|\Big).$$

Thus

$$\hat{H}_g = g\sum_i \hat{x}_i(\hat{n}_i - \bar{n}I)$$

with $\hat{x}_i$ understood in that unary basis. For the detailed spin-block JW + unary Pauli reduction of the Holstein sector, including the explicit ZXX/ZYY expansion, see Sections 6.2.3.2–6.2.3.4.

6.2.2 Binary explicit form

In binary encoding, the same operator is written on the binary code space:

$$\hat{x}_i = \sum_{n=0}^{d-2} \sqrt{n+1}\Big(|\operatorname{bin}(n+1)\rangle_i\langle \operatorname{bin}(n)| + |\operatorname{bin}(n)\rangle_i\langle \operatorname{bin}(n+1)|\Big),$$

with the same density factor $(\hat{n}_i - \bar{n}I)$.

6.2.3 Worked Holstein-only reduction after Jordan-Wigner fermions and unary phonons

Set

$$N_b = n_{\text{ph,max}}.$$

In this worked subsection, isolate the Holstein part of the Hubbard-Holstein Hamiltonian:

$$\hat{H}_{\text{Hol}} = \omega_0 \sum_{i=0}^{L-1} \hat{b}_i^\dagger \hat{b}_i \;+\; g\sum_{i=0}^{L-1}(\hat{b}_i^\dagger + \hat{b}_i)(\hat{n}_i - 1), \qquad \hat{n}_i = \hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}.$$

6.2.3.1 Qubit indexing used

For fermions, use spin-block Jordan-Wigner ordering:

$$q(i,\sigma) = i + \sigma L, \qquad \sigma = 0 \equiv\uparrow, \quad \sigma = 1 \equiv\downarrow .$$

Thus

$$q_{i\uparrow} = i, \qquad q_{i\downarrow} = i + L, \qquad i = 0, \dots, L-1.$$

For phonons, use a unary register with cutoff N_b. Per site i, allocate $N_b + 1$ qubits labelled by $n = 0, \dots, N_b$, and place all fermions first and phonons second:

$$q_{i,n}^{(b)} = 2L + i(N_b + 1) + n, \qquad n = 0, \dots, N_b.$$

Let X_q, Y_q, and Z_q denote the Pauli operators on qubit q.

6.2.3.2 Jordan-Wigner fermionic number operators

Jordan-Wigner gives

$$\hat{n}_{i\sigma} = \frac{I - Z_{q_{i\sigma}}}{2},$$

so

$$\hat{n}_i - 1 = \left(\frac{I - Z_{q_{i\uparrow}}}{2} + \frac{I - Z_{q_{i\downarrow}}}{2}\right) - 1 = -\frac{Z_{q_{i\uparrow}} + Z_{q_{i\downarrow}}}{2}.$$

6.2.3.3 Unary phonon ladder operators in Pauli form

Define the single-qubit combinations

$$(+)_q = \frac{X_q + iY_q}{2}, \qquad (-)_q = \frac{X_q - iY_q}{2}.$$

Then the unary truncated ladder operators are

$$\hat{b}_i^\dagger = \sum_{n=0}^{N_b-1} \sqrt{n+1}\,(+)_{q_{i,n}^{(b)}}\,(-)_{q_{i,n+1}^{(b)}},$$

$$\hat{b}_i = \sum_{n=1}^{N_b} \sqrt{n}\,(-)_{q_{i,n-1}^{(b)}}\,(+)_{q_{i,n}^{(b)}}.$$

A convenient real Pauli expansion is

$$\hat{b}_i^\dagger + \hat{b}_i = \frac{1}{2}\sum_{n=0}^{N_b-1} \sqrt{n+1}\Big(X_{q_{i,n}^{(b)}} X_{q_{i,n+1}^{(b)}} + Y_{q_{i,n}^{(b)}} Y_{q_{i,n+1}^{(b)}}\Big).$$

6.2.3.4 Explicit Holstein Hamiltonian in the qubit basis

(a) Phonon energy term On the unary one-hot code space, the phonon number operator is diagonal and may be written as

$$\hat{b}_i^\dagger \hat{b}_i \equiv \hat{n}_i^{(b)} = \sum_{n=0}^{N_b} n\,\frac{I - Z_{q_{i,n}^{(b)}}}{2} \qquad \text{(exact on the one-hot subspace).}$$

Therefore

$$\boxed{\hat{H}_{\text{ph}} = \omega_0 \sum_{i=0}^{L-1}\sum_{n=0}^{N_b} n\,\frac{I - Z_{q_{i,n}^{(b)}}}{2}}.$$

Equivalently, as constant plus Pauli-Z,

$$\hat{H}_{\text{ph}} = \underbrace{\omega_0 \frac{1}{2}\sum_i \sum_{n=0}^{N_b} n}_{\omega_0 L \frac{N_b(N_b+1)}{4}}\; I - \frac{\omega_0}{2}\sum_{i=0}^{L-1}\sum_{n=0}^{N_b} n\, Z_{q_{i,n}^{(b)}}.$$

The global constant shift may be dropped if desired.

(b) Electron-phonon coupling term Using

$$\hat{n}_i - 1 = -\frac{Z_{q_{i\uparrow}} + Z_{q_{i\downarrow}}}{2}$$

and the unary expansion of $\hat{b}_i^\dagger + \hat{b}_i$, one obtains

$$\boxed{\hat{H}_{e\text{-}ph} = -\frac{g}{4}\sum_{i=0}^{L-1}\sum_{n=0}^{N_b-1} \sqrt{n+1}(Z_{q_{i\uparrow}} + Z_{q_{i\downarrow}})\Big(X_{q_{i,n}^{(b)}} X_{q_{i,n+1}^{(b)}} + Y_{q_{i,n}^{(b)}} Y_{q_{i,n+1}^{(b)}}\Big)}.$$

The fully expanded Pauli-string sum is

$$\hat{H}_{e\text{-}ph} = -\frac{g}{4}\sum_{i=0}^{L-1}\sum_{n=0}^{N_b-1} \sqrt{n+1}\Big[Z_{q_{i\uparrow}} X_{q_{i,n}^{(b)}} X_{q_{i,n+1}^{(b)}} + Z_{q_{i\uparrow}} Y_{q_{i,n}^{(b)}} Y_{q_{i,n+1}^{(b)}} + Z_{q_{i\downarrow}} X_{q_{i,n}^{(b)}} X_{q_{i,n+1}^{(b)}} + Z_{q_{i\downarrow}} Y_{q_{i,n}^{(b)}} Y_{q_{i,n+1}^{(b)}}\Big].$$

Each summand is three-local.

Final Holstein Hamiltonian in JW + unary form Collecting the phonon-energy and electron-phonon pieces,

$$\boxed{\hat{H}_{\mathrm{Hol}}^{(\mathrm{JW+unary})} = \omega_0 \sum_{i=0}^{L-1} \sum_{n=0}^{N_b} n \frac{I - Z_{q_{i,n}^{(b)}}}{2} - \frac{g}{4} \sum_{i=0}^{L-1} \sum_{n=0}^{N_b-1} \sqrt{n+1} (Z_{q_{i\uparrow}} + Z_{q_{i\downarrow}}) \Big(X_{q_{i,n}^{(b)}} X_{q_{i,n+1}^{(b)}} + Y_{q_{i,n}^{(b)}} Y_{q_{i,n+1}^{(b)}} \Big)}.$$

6.3 Time-dependent density drive

A general density drive may be written as

$$\hat{H}_{\mathrm{drive}}(t) = \sum_i v_i(t) \hat{n}_i.$$

A useful waveform class is

$$v_i(t) = A s_i \sin(\omega t + \phi) \exp\left[-\frac{(t - t_0)^2}{2\bar{t}^2} \right],$$

where A is the amplitude, s_i is a static spatial pattern, ω is the drive frequency, ϕ is the phase, and $\bar{t}$ is the envelope width.

6.3.1 Generic density-drive waveform surface

After inserting $\hat{n}_i = I - \frac{1}{2}(Z_{p(i,\uparrow)} + Z_{p(i,\downarrow)})$, the drive separates into an identity offset and a sum of time-dependent Z-coefficients. Thus the time dependence resides in the coefficients rather than in the operator labels.

6.4 Two HH assembly surfaces

6.4.1 Exact Hamiltonian surface

The exact surface treats $\hat{H}_{\mathrm{HH}}(t)$ as an operator to be diagonalized, exponentiated, projected, or applied directly to statevectors.

6.4.2 Variational HH ansatz surface

The variational surface treats the same physical pieces as generator families from which one builds a structured manifold

$$|\psi(\theta)\rangle = U(\theta)|\psi_{\mathrm{ref}}\rangle.$$

6.5 Fully substituted HH master form

The full symbolic master form is therefore

$$\hat{H}_{\mathrm{HH}}(t) = \hat{H}_{\mathrm{Hub}} + \omega_0 \sum_i \left(\hat{n}_{b,i} + \frac{1}{2} I \right) + g \sum_i \hat{x}_i (\hat{n}_i - \bar{n} I) + \sum_i v_i(t) \hat{n}_i,$$

with $\hat{n}_i$, $\hat{n}_{b,i}$, and $\hat{x}_i$ expanded according to the chosen fermion ordering and boson encoding.

7. Reference States and Exact Sector ED

7.1 Fermionic Hartree-Fock determinant

Let $\mathcal{O}$ denote the occupied spin-orbital set in a Slater determinant reference. Then

$$|\Phi_{\mathrm{HF}}\rangle = \prod_{p \in \mathcal{O}} \hat{c}_p^\dagger |\mathrm{vac}_f\rangle.$$

7.2 Hubbard-Holstein reference state

The natural HH reference is the tensor product of the phonon vacuum and fermionic Hartree-Fock state:

$$|\psi_{\mathrm{ref}}^{\mathrm{HH}}\rangle = |\mathrm{vac}_{\mathrm{ph}}\rangle \otimes |\Phi_{\mathrm{HF}}\rangle.$$

7.3 Exact HH sector basis

Fixing the fermion particle numbers $(N_\uparrow, N_\downarrow)$, the exact sector basis is

$$\mathcal{H}_{\text{sector}} = \text{span}\{|f\rangle \otimes |\mathbf{n}_b\rangle : |f\rangle \in \mathcal{H}_{N_\uparrow, N_\downarrow},\ 0 \le n_{b,i} \le n_{\text{ph,max}}\}.$$

7.3.1 Binary index map

For binary boson encoding, one may order basis states by the pair

$$(f, \mathbf{n}_b),$$

where f is the fermionic bit pattern and $\mathbf{n}_b = (n_{b,0}, \dots, n_{b,L-1})$ is the vector of local phonon occupations, each encoded in binary.

7.3.2 Unary index map

For unary boson encoding, the same sector is indexed by

$$(f, \mathbf{e}_{n_{b,0}}, \dots, \mathbf{e}_{n_{b,L-1}}),$$

where $\mathbf{e}_n$ denotes the unary one-hot representation of occupation n.

7.4 Exact HH matrix elements

7.4.1 Diagonal elements

For basis state $|f\rangle \otimes |\mathbf{n}_b\rangle$, the diagonal contribution is

$$U \sum_i n_{i\uparrow}(f) n_{i\downarrow}(f) + \sum_i v_i\, n_i(f) + \omega_0 \sum_i \left(n_{b,i} + \frac{1}{2} \right).$$

7.4.2 Hopping matrix elements

If $|f'\rangle$ differs from $|f\rangle$ by a single allowed nearest-neighbor hop with the same phonon occupations, then

$$\langle f', \mathbf{n}_b | \hat{H}_t | f, \mathbf{n}_b \rangle = -t\,(-1)^{\chi(f;p,q)},$$

where $\chi(f; p, q)$ is the Jordan-Wigner parity count between the two fermion modes p and q.

7.4.3 Electron-phonon matrix elements

At fixed fermion configuration, the electron-phonon term changes the local phonon number by one. For site i,

$$\langle n_{b,i} + 1 | \hat{x}_i | n_{b,i} \rangle = \sqrt{n_{b,i} + 1}, \qquad \langle n_{b,i} - 1 | \hat{x}_i | n_{b,i} \rangle = \sqrt{n_{b,i}}.$$

Hence the off-diagonal electron-phonon matrix elements are proportional to

$$g\,(n_i(f) - \bar{n}) \sqrt{n_{b,i} + 1} \quad \text{or} \quad g\,(n_i(f) - \bar{n}) \sqrt{n_{b,i}},$$

according to whether the phonon occupation increases or decreases.

8. Statevector Action, Expectation, and Exponential Primitives

8.1 Basis-state primitive

A statevector on N_q qubits is written as

$$|\psi\rangle = \sum_{k=0}^{2^{N_q}-1} a_k |k\rangle = \sum_{b \in \{0,1\}^{N_q}} a_b |b\rangle.$$

8.2 Explicit Pauli-word action

Let P be a Pauli word. Then its action on a computational basis state can be written in the form

$$P|b\rangle = \omega_P(b)\,|b \oplus \chi_P\rangle,$$

where χ_P records which qubits are flipped by X or Y, and $\omega_P(b)$ is the phase determined by the Y and Z letters acting on the bitstring b.

8.3 Expectation values

If

$$\hat{H} = \sum_\alpha h_\alpha P_\alpha,$$

then the expectation value is

$$E(\psi) = \langle\psi|\hat{H}|\psi\rangle = \sum_\alpha h_\alpha \langle\psi|P_\alpha|\psi\rangle.$$

8.4 Pauli rotations

For any Hermitian Pauli word with $P^2 = I$,

$$e^{-i\theta P} = \cos\theta\, I - i\sin\theta\, P.$$

This identity is the primitive behind fast statevector application of Pauli exponentials.

8.5 Exponential of a Pauli polynomial

For a Pauli polynomial

$$A = \sum_\alpha a_\alpha P_\alpha,$$

one has

$$U_A(\theta) = e^{-i\theta A}.$$

If the P_α commute pairwise, then

$$U_A(\theta) = \prod_\alpha e^{-i\theta a_\alpha P_\alpha}.$$

If they do not commute, one may use either exact exponentiation on the working subspace or an ordered product formula.

8.6 Exact sector energy target

The natural exact comparison target in a fixed fermion sector is

$$E_{\text{exact,sector}} = \min_{\psi\in\mathcal{H}_{\text{sector}}} \langle\psi|\hat{H}|\psi\rangle.$$

This is distinct from the unrestricted full-Hilbert minimum when a sector constraint is imposed.

9. Fixed VQE Ansatz Families

This section records the main symbolic manifold families without tying them to any implementation names.

9.1 Hubbard layerwise ansatz

For the static Hubbard model, write

$$\hat{H}_{\text{Hub}} = \hat{H}_t + \hat{H}_U + \hat{H}_v.$$

A layerwise ansatz uses one parameter per physical group per layer:

$$U^{(\ell)}_{\text{Hub}} = \exp(-i\theta^{(\ell)}_t \hat{H}_t)\exp(-i\theta^{(\ell)}_U \hat{H}_U)\exp(-i\theta^{(\ell)}_v \hat{H}_v),$$

with the understanding that each grouped generator may itself be split into Pauli exponentials internally while sharing the same scalar parameter within the group.

9.2 Hubbard excitation-based surface

Relative to a Hartree-Fock occupied/virtual partition, single- and double-excitation generators take the form

$$G^{(\sigma)}_{i\to a} = i\left(\hat{c}^\dagger_{a\sigma}\hat{c}_{i\sigma} - \hat{c}^\dagger_{i\sigma}\hat{c}_{a\sigma}\right),$$

and

$$G^{(\sigma,\sigma')}_{ij\to ab} = i\left(\hat{c}^\dagger_{a\sigma}\hat{c}^\dagger_{b\sigma'}\hat{c}_{j\sigma'}\hat{c}_{i\sigma} - \text{h.c.}\right).$$

A layered excitation manifold is then

$$U_{\text{exc}}(\theta) = \prod_r e^{-i\theta_r G_r}.$$

9.3 HH layerwise ansatz

For the HH model, define grouped generators

$$G_t,\ G_U,\ G_v,\ G_{\text{ph}},\ G_g,\ G_{\text{drive}}.$$

One layer is

$$U^{(\ell)}_{\text{HH,lw}} = \exp(-i\theta_{t,\ell}G_t)\exp(-i\theta_{U,\ell}G_U)\exp(-i\theta_{v,\ell}G_v)\exp(-i\theta_{\text{ph},\ell}G_{\text{ph}})\exp(-i\theta_{g,\ell}G_g)\exp(-i\theta_{\text{drive},\ell}G_{\text{drive}}),$$

with inactive groups omitted.

9.4 HH physical-termwise ansatz

A physical-termwise layer sits between the grouped and fully split extremes. Let

$$\mathcal{A}_{\text{phys}} = \{A_m\}$$

be the collection of unsplit physical HH generators such as bond hopping terms, onsite interactions, local phonon energies, local couplings, and local drive terms. Then

$$U^{(\ell)}_{\text{HH,phys}} = \prod_{A_m\in\mathcal{A}_{\text{phys}}} e^{-i\phi_{m,\ell}A_m}.$$

Because the generators remain grouped at the physical-operator level, this fixed-family surface stays aligned with the same conserved-sector story as the HH layerwise surface. In this symbolic HH manuscript, the canonical fixed-family surface is intentionally restricted to the sector-preserving HH layerwise and HH physical-termwise families.

9.5 Reference-state and ansatz pairing principle

The reference state and variational family should be paired by three criteria:

1. the conserved sector they preserve,
2. the expressive scale needed by the target Hamiltonian,
3. the conditioning and optimization burden induced by the chosen parameterization.

10. Polaronic Operator Families

10.1 Primitive polaronic ingredients

10.1.1 Shifted density

If the total electron number is N_e, define the mean density

$$\bar{n} = \frac{N_e}{L},$$

and the shifted density

$$\tilde{n}_i = \hat{n}_i - \bar{n}I.$$

At half filling $N_e = L$, this reduces to

$$\tilde{n}_i = -\frac{1}{2}(Z_{p(i,\uparrow)} + Z_{p(i,\downarrow)}).$$

10.1.2 Phonon primitives P_i and x_i

Define

$$P_i = i(\hat{b}_i^\dagger - \hat{b}_i), \qquad x_i = \hat{b}_i + \hat{b}_i^\dagger.$$

10.1.3 Local squeeze primitive

A natural local squeeze generator is

$$S_i = i\left((\hat{b}_i^\dagger)^2 - \hat{b}_i^2\right).$$

10.1.4 Doublon primitive

The doublon projector is

$$D_i = \hat{n}_{i\uparrow}\hat{n}_{i\downarrow}.$$

10.1.5 Even hopping channel

A bond-even hopping operator is

$$T_{ij}^{(+)} = \sum_\sigma \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}\right).$$

10.1.6 Odd current channel

A bond-odd current operator is

$$J_{ij}^{(-)} = i\sum_\sigma \left(\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} - \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}\right).$$

10.2 Representative dressed channels

The exact family boundaries can vary, but the symbolic equivalents of the common channels can be written as follows.

10.2.1 Conditional displacement

$$G_i^{(\mathrm{cd})} = \tilde{n}_i P_i.$$

10.2.2 Doublon dressing

$$G_i^{(\mathrm{dd})} = D_i P_i.$$

10.2.3 Dressed hopping drag

$$G_{ij}^{(\mathrm{hd})} = T_{ij}^{(+)}(P_i - P_j).$$

10.2.4 Odd-channel drag

$$G_{ij}^{(\mathrm{od})} = J_{ij}^{(-)}(x_i - x_j).$$

10.2.5 Second-order even channel

$$G_{ij}^{(2)} = T_{ij}^{(+)}(x_i - x_j)^2.$$

10.2.6 Third-order odd current channel

$$G_{ij}^{(3)} = J_{ij}^{(-)}(x_i - x_j)^3.$$

10.2.7 Fourth-order even hopping channel

$$G_{ij}^{(4)} = T_{ij}^{(+)}(x_i - x_j)^4.$$

10.2.8 Bond-conditioned displacement and squeeze channels

With a bond-sensitive scalar observable B_{ij}, one may write

$$G_{ij}^{(\mathrm{bd})} = B_{ij}P_i, \qquad G_{ij}^{(\mathrm{bs})} = B_{ij}S_i.$$

A simple choice is $B_{ij} = \hat{n}_i - \hat{n}_j$.

10.2.9 Local squeeze channels

$$G_i^{(\mathrm{sq})} = S_i, \qquad \tilde{G}_i^{(\mathrm{sq})} = \tilde{n}_i S_i.$$

10.2.10 Extended cloud channels

A symbolic extended cloud family is

$$G_i^{(\mathrm{cloud})} = \tilde{n}_i \sum_{r\in\mathcal{N}(i)} \alpha_{ir} P_r,$$

with coefficients α_{ir} controlling cloud shape.

10.2.11 Doublon-translation and doublon-squeeze channels

$$G_{ij}^{(\mathrm{DT})} = D_i T_{ij}^{(+)}, \qquad G_i^{(\mathrm{DS})} = D_i S_i.$$

10.3 Pool-family map

A useful abstract partition is

$$\mathcal{G} = \mathcal{G}_{\mathrm{local}} \cup \mathcal{G}_{\mathrm{bond}} \cup \mathcal{G}_{\mathrm{squeeze}} \cup \mathcal{G}_{\mathrm{cloud}} \cup \mathcal{G}_{\mathrm{doublon}}.$$

These families can be opened gradually rather than all at once.

10.3.1 Compact displacement/squeeze macro families

Two especially compact subfamilies are

$$\mathcal{G}_{\mathrm{VLF}} = \{\tilde{n}_i P_i\}_i, \qquad \mathcal{G}_{\mathrm{SQ}} = \{S_i, \tilde{n}_i S_i\}_i.$$

They act as coarse polaronic and squeeze envelopes.

11. Adaptive Selection and Staged Continuation

11.1 Cumulative phase construction for adaptive selection

If a shortlisted macro generator m admits a split family $\mathcal{C}_{\mathrm{split}}(m) = \{m_1, \dots, m_{K_m}\}$, then each split child inherits the same position p and therefore defines a child record $r_j = (m_j, p)$. For simplicity we drop the j, and assume each m below is a unique generator.

Record-first view: the primary object is a candidate-position record $r := (m, p)$, with projections $\pi_m(m,p) := m$ and $\pi_p(m,p) := p$. For each phase $k \in \{1,2,3\}$, the one-way selector flow is

$$\mathcal{R}_k \longrightarrow \mathcal{S}_k \longrightarrow r_k^\star,$$

with

$$\mathcal{R}_k := \{(m,p) : m \in \mathcal{G}_k,\ p \in \mathcal{P}_m^k\}, \qquad \mathcal{S}_k := \{(m,p) \in \mathcal{R}_k : S_k(m,p) \geq \tau_k\} \quad \text{or} \quad \mathrm{Top}_{N_k}(S_k; \mathcal{R}_k), \qquad r_k^\star \in \mathcal{S}_k.$$

where S_k denotes the phase-k score used for shortlist construction. Generator image and position image: $\mathcal{G}^{(k)} := \pi_m(\mathcal{S}_k)$ and $\mathcal{P}_m^{(k)} := \{p : (m,p) \in \mathcal{S}_k\}$. Record inheritance: when later phases inherit the same candidate-position records, the clean handoff is $\mathcal{R}_{k+1} := \mathcal{S}_k$. Hence the cumulative record chain is $\mathcal{R}_3 \subseteq \mathcal{R}_2 \subseteq \mathcal{R}_1$, the cumulative

generator chain is $\mathcal{G}^{(3)} \subseteq \mathcal{G}^{(2)} \subseteq \mathcal{G}^{(1)} \subseteq \mathcal{G}$, the first chain follows from $\mathcal{R}_{k+1} := \mathcal{S}_k \subseteq \mathcal{R}_k$, and the retained position images inherit analogously via the corresponding record shortlists $\mathcal{S}_1 \to \mathcal{S}_2 \to \mathcal{S}_3$. Post-shortlist admission, batching, pruning, and beam continuation are then defined in terms of the final Phase-3 surface.

Seed broadening: in addition, we use a smooth broadening of the initial candidate record $\mathcal{R}$ as expected generator utility saturates, anti-comensurate to scaffold depth, where the initial seeds are low-order generators and late-stage seeds are higher-order excitation and coupling terms.

Chronology: thus the presentation chronologically accords with the algorithmic implementation: Phase 1 $\to$ Phase 2 $\to$ Phase 3, with each to-arrow filtering its preceding phase.

11.2 Phase 1

11.2.1 Core signal, append position, and one-coordinate local model

Core record, scaffold, and append slot: $r = (m, p)$, $\mathcal{O} = (a_1, \dots, a_n)$, $m \in \mathcal{G}$, and $p \in \mathcal{P}^1_m = \{0, \dots, n\}$. Here m is a candidate generator already admitted by the generator gate (so $m \in \mathcal{G}_1$), and p is the index of a slot in the pre-insertion, ordered scaffold $\mathcal{O}$. Window bias: note, the insertion window near the terminal scaffold point is biased to be small proportional to expected scaffold utility remaining.

Scaffold parameter space and energy map: WLOG consider $\mathcal{O}$, s.t. each a is assigned one scalar $\theta(\mod 2\pi)$, then define $\Theta_{\mathcal{O}} := \mathbb{R}^{|\mathcal{O}|}$, $E(\mathcal{O}, \theta) := \langle \psi(\mathcal{O}, \theta) | H | \psi(\mathcal{O}, \theta) \rangle$, and $\theta \in \Theta_{\mathcal{O}}$. Scaffold insertion and parameter insertion: define $\mathcal{O} \oplus_p m := (a_1, \dots, a_p, m, a_{p+1}, \dots, a_n)$ and $\theta \oplus_p \vartheta := (\theta_1, \dots, \theta_p, \vartheta, \theta_{p+1}, \dots, \theta_n) \in \Theta_{\mathcal{O} \oplus_p m}$.

Write $|\psi\rangle := U_{>p} U_{\leq p} |\psi_0\rangle$, $\widetilde{A}_r := U_{>p} A_m U^\dagger_{>p}$, $U_r(\alpha) := e^{-i\alpha \widetilde{A}_r}$, and $|\psi_r(\alpha)\rangle := U_r(\alpha)|\psi\rangle$. Let $Q_\psi := I - |\psi\rangle\langle\psi|$, $|d_r\rangle := \partial_\alpha |\psi_r(\alpha)\rangle|_{\alpha=0} = -i\widetilde{A}_r|\psi\rangle$, $\tau_r := Q_\psi |d_r\rangle$, $g_r := 2\Re\langle\psi|H|d_r\rangle$, $\sigma_r := \mathrm{SE}[\hat{g}(r)] \geq 0$, $g_{\mathrm{lcb}}(r) := \max\{|g_r| - z_\alpha \sigma_r, 0\}$, and $F_{\mathrm{raw}}(r) := \langle \tau_r, \tau_r \rangle + \varepsilon_F$. The live broad Phase-1 one-coordinate trust-region proxy is

$$\Delta E_{1,\mathrm{TR}}(r) := \max_{|\alpha| \leq \rho/\sqrt{F_{\mathrm{raw}}(r)}} \left[g_{\mathrm{lcb}}(r)|\alpha| - \frac{1}{2}\lambda_F F_{\mathrm{raw}}(r)\alpha^2 \right].$$

The strict one-coordinate insertion drop is retained as the exact auxiliary comparator

$$\delta E_1(m, p \mid \mathcal{O}, \theta) = E(\mathcal{O}, \theta) - \inf_{\vartheta \in \mathbb{R}} E(\mathcal{O} \oplus_p m, \theta \oplus_p \vartheta).$$

The infimum notation may be replaced by min or Min whenever the minimizer exists. Thus $\Delta E_{1,\mathrm{TR}}(r)$ is the live broad-pool second-order local insertion proxy, whereas $\delta E_1(m, p \mid \mathcal{O}, \theta)$ is the exact one-coordinate insertion drop. [1]

11.2.2 Gates, burdens, and feature ingredients

Base hardware ingredients: each candidate-position pair carries the proxy triple $C_{2\mathrm{q}}(m, p)$, $D(m, p)$, and $\Delta K(m, p)$, with fixed nonnegative weights $\lambda_{2\mathrm{q}}, \lambda_d, \lambda_\theta$.[2] Here $C_{2\mathrm{q}}$ is a two-qubit proxy cost, D is an expected depth-induced by m (at p) proxy, and ΔK is the marginal parameter count. Positivity assumptions: assume throughout that $\lambda_{2\mathrm{q}} \geq 0$, $C_{2\mathrm{q}}(m, p) \geq 0$, and $\lambda_\theta, \lambda_d, \Delta K(m, p), D(m, p) > 0$ on admissible records. [3]

11.2.3 Anticipated utility left and controller schedule

Let t denote the current selector step, let $D_{\mathrm{left}}(t)$ be the remaining controller depth budget, and let $\widehat{N}_{\mathrm{rem}}(t)$ be a heuristic forecast (expected count) of useful remaining admissions. For gain-process statistics, define $\Delta_{\mathrm{adm}}(s) :=$

[1] User interest: How does window parameterization width scale with cost?

[2] Admissibility gate: For each selector stage k, let $\Gamma_k : \mathcal{G}^{\mathrm{in}}_k \to \{0, 1\}$ be the generator-level admissibility gate and define $\mathcal{G}^{\mathrm{adm}}_k := \{m \in \mathcal{G}^{\mathrm{in}}_k : \Gamma_k(m) = 1\}$. Record domain: the working candidate-record domain is built only from this gated generator pool, $\mathcal{R}_k := \{(m, p) : m \in \mathcal{G}^{\mathrm{adm}}_k,\ p \in \mathcal{P}^{\mathrm{in},k}_m\}$. Selector action: all selectors below act on $\mathcal{R}_k$ itself, so admissibility is enforced by pre-score domain construction rather than post hoc score masking. Tie order: fix once and for all a total order $\prec$ on records (for example, lexicographic order on (m, p)). Tie-broken maximizer: for any score $S : \mathcal{R}_k \to \mathbb{R}$, define $\arg\max^{\prec}_{r \in \mathcal{R}_k} S(r) := \min_{\prec}\{r \in \mathcal{R}_k : S(r) = \max_{u \in \mathcal{R}_k} S(u)\}$, and similarly for $\arg\min^{\prec}$. Shortlist surface: define $\mathrm{Top}^{\prec}_K(S; \mathcal{R}_k)$ as the ordered list of the first K records obtained by sorting $\mathcal{R}_k$ in descending score, with ties broken by $\prec$. Unadorned arg max, arg min, and Top_K below follow this convention.

[3] Depth and parameter count are constants for our L=2 pareto lean runs.

$E(\theta^{(s-1)}) - E(\theta^{(s)})$, $\Delta^+_{\text{adm}}(s) := \operatorname{asinh}\Big(\frac{\Delta_{\text{adm}}(s)}{\tau_\Delta}\Big)$, and let $\mathcal{J}^+_t$ be the index set of past selector steps with valid admissions used for EWMA/MAD statistics.

Let D_{allow} denote the total allowed depth budget, let $\ell^{\text{plat}}_t := \max\{1, \lceil P\, D_{\text{left}}(t)/D_{\text{allow}}\rceil\}$ and $\ell^{\text{low}}_t := \max\{1, \lceil L\, D_{\text{left}}(t)/D_{\text{allow}}\rceil\}$, and define the relative-flattening and absolute weak-gain thresholds by $\tau^+_{\text{plat}}(t) := (\eta_0 + \eta_1(1 - D_{\text{left}}(t)/D_{\text{allow}}))m_t$ and $\tau^+_{\text{low}}(t) := \operatorname{asinh}(\tau_{\text{use},t}/\tau_\Delta)$. The corresponding depth-adaptive streak statistics are $\text{plateau_streak}(t) := \frac{P}{\ell^{\text{plat}}_t}\max\{q \in \{0,\dots,\ell^{\text{plat}}_t\} : \Delta^+_{\text{adm}}(t-j+1) \le \tau^+_{\text{plat}}(t)\ \forall 1 \le j \le q\}$ and $\text{low_streak}(t) := \frac{L}{\ell^{\text{low}}_t}\max\{q \in \{0,\dots,\ell^{\text{low}}_t\} : \Delta^+_{\text{adm}}(t-j+1) \le \tau^+_{\text{low}}(t)\ \forall 1 \le j \le q\}$.

For stagnation, gain, decay, and scale summaries, define $u_{\text{stag}}(t) := \text{clip}\Big(\max\Big(\frac{\text{plateau_streak}(t)}{P}, \frac{\text{low_streak}(t)}{L}\Big), 0, 1\Big)$, $m_t := \text{EWMA}_{s\in\mathcal{J}^+_t}(\Delta^+_{\text{adm}}(s))$, $\rho_t := \text{clip}\Big(\frac{\text{EWMA}_{\text{recent}}(\Delta_{\text{adm}})}{\text{EWMA}_{\text{older}}(\Delta_{\text{adm}})+\varepsilon}, \rho_{\min}, 1\Big)$, $\gamma_t := [-\log\rho_t]_+$, and $s_t := \max\Big(s_{\min}, \text{MAD}_{s\in\mathcal{J}^+_t}(\Delta^+_{\text{adm}}(s))\Big)$.

For the frontier statistic, let $s^{(1)}_{\text{raw},1}(t) \ge s^{(1)}_{\text{raw},2}(t) \ge 0$ denote the first two order statistics of the current broad Phase-1 score multiset $\{S_1(r;t) : r \in \mathcal{R}_1\}$ before any $\tau_1(t)$-thresholding or $N_1(t)$-truncation, with the conventions $s^{(1)}_{\text{raw},1}(t) = s^{(1)}_{\text{raw},2}(t) = 0$ if $\mathcal{R}_1 = \emptyset$ and $s^{(1)}_{\text{raw},2}(t) = 0$ if $|\mathcal{R}_1| = 1$. Define

$$u_{\text{front}}(t) := \frac{s^{(1)}_{\text{raw},2}(t) + \varepsilon}{s^{(1)}_{\text{raw},1}(t) + \varepsilon}.$$

Thus $u_{\text{front}}(t) \in [0,1]$, with larger values corresponding to a flatter, more crowded current frontier and smaller values corresponding to a clearly separated leading record.[4]

Then the survival hazard is $h_t(j) := \sigma\Big(\beta_0 + \beta_{\text{stag}}u_{\text{stag}}(t) + \beta_{\text{front}}u_{\text{front}}(t) + \eta_h(j-1)\Big)$.

$$S_t(k) := \prod_{j=1}^{k-1}(1 - h_t(j)), \qquad Q_t(k) := \Phi\left(\frac{m_t - \gamma_t(k-1) - \operatorname{asinh}\Big(\frac{\tau_{\text{use},t}}{\tau_\Delta}\Big)}{s_t}\right).$$

$$\widehat{N}_{\text{rem}}(t) := \sum_{k=1}^{D_{\text{left}}(t)} S_t(k)\, Q_t(k).$$

For pessimistic/optimistic runway envelopes, set $h^{\text{pess/opt}}_t(j) := \sigma(\operatorname{logit} h_t(j) \pm \Delta_h)$, $m^{\text{pess/opt}}_t := m_t \mp \Delta_m$, $s^{\text{pess/opt}}_t := s_t + \Delta_s$ (pess) or $s^{\text{pess/opt}}_t := \max(s_{\min}, s_t - \Delta_s)$ (opt), $S^{\text{pess/opt}}_t(k) := \prod_{j=1}^{k-1}(1 - h^{\text{pess/opt}}_t(j))$, and $Q^{\text{pess/opt}}_t(k) := \Phi\left(\frac{m^{\text{pess/opt}}_t - \gamma_t(k-1) - \operatorname{asinh}\Big(\frac{\tau_{\text{use},t}}{\tau_\Delta}\Big)}{s^{\text{pess/opt}}_t}\right)$. Then $\widehat{N}^{\text{low/high}}_{\text{rem}}(t) := \sum_{k=1}^{D_{\text{left}}(t)} S^{\text{pess/opt}}_t(k)\, Q^{\text{pess/opt}}_t(k)$, with $(\text{low}, \text{high}) \leftrightarrow (\text{pess}, \text{opt})$, and the confidence ratio is $C_t := \frac{\widehat{N}^{\text{low}}_{\text{rem}}(t)}{\widehat{N}^{\text{high}}_{\text{rem}}(t)+\varepsilon}$.

For the controller coordinate and directional transforms, define

$$R_t := \frac{\widehat{N}_{\text{rem}}(t)}{D_{\text{left}}(t) + \varepsilon} \in [0,1], \qquad e_t := R_t^a, \qquad g_t := (1 - R_t)^b, \qquad a, b > 0.$$

Here R_t is remaining useful runway as a fraction of the depth-budget permitted by hardware/run-time constraints, and g_t is the late-stage activation coordinate that scales controller variables depending on age, where age is a normalized quantity relative to/determined by expected utility of more generators.

Then $R_t \approx 1$ early and $R_t \approx 0$ late. Use e_t for early-dominant quantities and g_t for late-dominant quantities. In particular, define

$$\tau_{\text{pos}}(t) := \tau^{\min}_{\text{pos}} + (\tau^{\max}_{\text{pos}} - \tau^{\min}_{\text{pos}})e_t, \qquad \lambda_F(t) := \lambda^{\min}_F + (\lambda^{\max}_F - \lambda^{\min}_F)g_t, \qquad u_{\text{sat}}(t) := 1 - R_t.$$

[4] With the empty-pool convention $s^{(1)}_{\text{raw},1}(t) = s^{(1)}_{\text{raw},2}(t) = 0$, one gets $u_{\text{front}}(t) = 1$. This treats the absence of a usable current frontier as maximal compression rather than as a missing-data null signal. An explicit signal-to-noise controller statistic is a possible implementation here, but it may not be necessary if worsening signal-to-noise already manifests as frontier flattening, which the present expected-utility model scores directly through $u_{\text{front}}(t)$. Here $h_t(j)$ is the conditional probability that usefulness terminates at future offset j, given survival through offset $j-1$; accordingly, $S_t(k)$ is the survival probability to the start of offset k.

The quantity $\widehat{N}_{\mathrm{rem}}(t)$ is controller telemetry rather than a physical observable: it summarizes recent selector progress and remaining useful runway, but it is not itself a hard admissibility certificate.[5]

For inherited refit windows, use the maturity-shrinking retention law

$$\rho_W(t) := \rho_W^{\min} + (\rho_W^{\max} - \rho_W^{\min})e_t, \qquad 0 \le \rho_W^{\min} \le \rho_W^{\max} \le 1,$$

so that the retained old-coordinate fraction is largest early and contracts as useful runway is exhausted.

For measurement effort, use the Phase-1 baseline schedule

$$N_{\mathrm{shot},1}(t) := \Big\lceil N_{\mathrm{shot}}^{\min} + (N_{\mathrm{shot}}^{\max} - N_{\mathrm{shot}}^{\min})\Big(\alpha_{\mathrm{front}}\, u_{\mathrm{front}}(t) + \alpha_{\mathrm{util}}\,(1 - R_t)\Big)\Big\rceil,$$

with $N_{\mathrm{shot}}^{\min} \le N_{\mathrm{shot}}^{\max}$, $\alpha_{\mathrm{front}}, \alpha_{\mathrm{util}} \ge 0$, and $\alpha_{\mathrm{front}} + \alpha_{\mathrm{util}} = 1$. Thus Phase 1 increases shot count when the frontier flattens and when useful runway is being exhausted.

For $k \in \{2,3\}$, once the phase-k uncertainty summary $u_{\sigma,k}(t)$ is available, use

$$N_{\mathrm{shot},k}(t) := \Big\lceil N_{\mathrm{shot}}^{\min} + (N_{\mathrm{shot}}^{\max} - N_{\mathrm{shot}}^{\min})\Big(\alpha_{\mathrm{front}}\, u_{\mathrm{front}}(t) + \alpha_{\mathrm{util}}\,(1 - R_t) + \alpha_\sigma\, u_{\sigma,k}(t)\Big)\Big\rceil, \qquad k \in \{2,3\},$$

with $\alpha_{\mathrm{front}}, \alpha_{\mathrm{util}}, \alpha_\sigma \ge 0$ and $\alpha_{\mathrm{front}} + \alpha_{\mathrm{util}} + \alpha_\sigma = 1$.[6] Thus Phase 2 and Phase 3 inherit the same controller baseline and then increase shot count further when geometric uncertainty rises.

For phase thresholds, use the live monotone law

$$\tau_k(t) := \tau_k^{\min} + (\tau_k^{\max} - \tau_k^{\min})e_t, \qquad \tau_k^{\min} \le \tau_k^{\max}, \qquad k \in \{1,2,3\},$$

so that shortlist admission is stricter early and more permissive later.

For phase shortlist caps, use the live bounded law

$$N_k(t) := \lceil N_k^{\min} + (\, N_k^{\max} - N_k^{\min}\,)g_t \rceil, \qquad N_k^{\min} \le N_k^{\max}, \qquad k \in \{1,2,3\}.$$

Thus $N_k(t)$ and $\tau_k(t)$ are both live shortlist-control quantities: $N_k(t)$ ensures budget constraints are met, whereas $\tau_k(t)$ helps ensure logical admissibility constraints proceed with time.

11.2.4 Phase-local cost and structural burden definitions

For any current phase-input record family $\mathcal{R}_k(t)$, with $\mathcal{R}_1(t) := \mathcal{R}_1$ and $k \in \{1,2,3\}$, and any scalar record feature $x : \mathcal{R}_k(t) \to \mathbb{R}_{\ge 0}$, define the empirical quantiles $q_\alpha^x(t) := \mathrm{Quantile}_\alpha(\{x(u) : u \in \mathcal{R}_k(t)\})$ and $m_x(t) := q_{0.5}^x(t)$, together with the robust scales $s_x^{\mathrm{MAD}}(t) := 1.4826\ \mathrm{median}_{u \in \mathcal{R}_k(t)}\big|x(u) - m_x(t)\big|$, $s_x^{\mathrm{pct}}(t) := \frac{q_{0.9}^x(t) - q_{0.1}^x(t)}{2.563}$, and $s_x(t) := \max\big(\varepsilon_x,\ s_x^{\mathrm{MAD}}(t),\ s_x^{\mathrm{pct}}(t)\big)$. Thus $s_x(t)$ is a percentile/MAD hybrid reference scale, with percentile fallback when the MAD degenerates on sparse discrete features.

Define the one-sided robust normalized excess by $z_x(r;t) := \mathrm{asinh}\Big(\frac{[x(r) - m_x(t)]_+}{s_x(t)}\Big)$, with $[y]_+ := \max(y, 0)$. When a direct ratio form is preferred, use the reference level $x_{\mathrm{ref}}(t) := m_x(t) + s_x(t)$.

For a candidate-position record $r = (m, p)$, let $\mathcal{L}_r := \mathrm{supp}(m) = \{j : m_j \ne I\}$, $\mathrm{wt}(r) := |\mathcal{L}_r|$, and $\#_{X,Y}(r) := \big|\{j \in \mathcal{L}_r : m_j \in \{X, Y\}\}\big|$.

Define feature-specific robust excesses by $z_{\mathrm{wt}}(r;t) := z_x(r;t)$ with $x = \mathrm{wt}$ and $z_{XY}(r;t) := z_x(r;t)$ with $x = \#_{X,Y}$. The structural burden term is

$$D_{\mathrm{struct}}(r;t) := \beta_{\mathrm{wt}}\, z_{\mathrm{wt}}(r;t) + \beta_{XY}\, z_{XY}(r;t),$$

with nonnegative coefficients $\beta_{\mathrm{wt}}, \beta_{XY} \ge 0$.[7]

[5] Interpretation: the model has three layers — (i) gain process (m_t, s_t), (ii) horizon-decay process (γ_t), and (iii) survival process $(h_t(j))$; together these drive the runway forecast $\widehat{N}_{\mathrm{rem}}(t)$.

[6] A one-weight simplification is the special case $\alpha_{\mathrm{front}} = \alpha_{\mathrm{util}} = \frac{1-\alpha_\sigma}{2}$.

[7] Provenance note: generator novelty $G_{\mathrm{new}}(r)$, family novelty $\mathrm{groups}_{\mathrm{new}}(r)$ with single-valued $\mathfrak{g} : \mathcal{G} \to \mathcal{C}$, lifetime/staleness $\mathbf{1}_{\mathrm{lifetime}}(r;t)$ with $t_{\mathrm{first}}(m)$ meaning first time ever seen in the run, the auxiliary raw burden $D_{\mathrm{raw}}(r;t) := D_{\mathrm{struct}}(r;t) + d_{\mathrm{pos}}(r)$ with $d_{\mathrm{pos}}(r) := |p - p_{\mathrm{app}}|$, and the normalized descendants $\bar{D}_{\mathrm{life}}^{\mathrm{struct}}, \bar{G}_{\mathrm{new}}, \bar{C}_{\mathrm{new}}, \bar{c}, \bar{P}_{\mathrm{opt}}$ are omitted from the live hardware cost in the present draft and retained only as controller-side provenance/policy bookkeeping.

For candidate-specific measurement burden, let $O_r := i[H, \widetilde{A}_r] = \sum_{b\in\mathcal{B}_r} O_{r,b}$, with $O_{r,b} := \sum_{\ell\in B_{r,b}} c_{r,b\ell} P_{r,b\ell}$, where $\mathcal{B}_r$ is the qubit-wise commuting partition of the candidate's gradient observable and $\mathcal{B}_r^{\text{new}} \subseteq \mathcal{B}_r$ is the subset not already covered by measurement reuse. Here $\widetilde{A}_r$ is the dressed candidate generator at record $r = (m, p)$, $\mathcal{B}_r$ is the partition of $O_r = i[H, \widetilde{A}_r]$ into qubit-wise commuting Pauli-string measurement groups, and $\mathcal{B}_r^{\text{new}}$ is the subset that still requires new measurement effort. When exact state-dependent variances are unavailable, use the commutator-based proxy

$$\hat{v}_{r,b} := \sum_{\ell\in B_{r,b}} |c_{r,b\ell}|^2, \qquad \widehat{\widehat{C}}_{\text{shot}}(r) := \frac{1}{\text{shot}_{\text{ref}}} \cdot \frac{\left(\sum_{b\in\mathcal{B}_r^{\text{new}}} \sqrt{\hat{v}_{r,b}}\right)^2}{\sigma_\star^2}.$$

Here $\sigma_\star$ is the target standard error for the candidate gradient estimate. The positive constant shot_{ref} is a baseline measurement budget against which candidate-specific shot cost is compared.

For the shared phase-k cost, define $z_{2\text{q}}(r;t) := z_x(r;t)$ with $x = C_{2\text{q}}$, $z_d(r;t) := z_x(r;t)$ with $x = D$, $z_\theta(r;t) := z_x(r;t)$ with $x = \Delta K$, $z_{\text{shot}}(r;t) := z_x(r;t)$ with $x = \widehat{\widehat{C}}_{\text{shot}}$, and

$$K_k(r;t) := \lambda_{2\text{q}}\, z_{2\text{q}}(r;t) + \lambda_d\, z_d(r;t) + \lambda_\theta\, z_\theta(r;t) + \lambda_{\text{shot}}\, z_{\text{shot}}(r;t) + D_{\text{struct}}(r;t).$$

Explicitly,

$$\begin{aligned} K_k(r;t) = \lambda_{2\text{q}}\, z_{2\text{q}}(r;t) + \lambda_d\, z_d(r;t) + \lambda_\theta\, z_\theta(r;t) + \lambda_{\text{shot}}\, z_{\text{shot}}(r;t) \\ + \beta_{\text{wt}}\, z_{\text{wt}}(r;t) + \beta_{XY}\, z_{XY}(r;t), \qquad k \in \{1,2,3\}. \end{aligned}$$

11.2.5 Final Phase I Scoring For Phase 1, the set $\mathcal{R}_1 := \{(m,p) : m \in \mathcal{G}_1,\ p \in \mathcal{P}_m^1\}$ is scored by

$$S_1(r;t) := \frac{\Delta E_{1,\text{TR}}(r)}{1 + K_1(r;t)}, \qquad r \in \mathcal{R}_1.$$

Here $\Delta E_{1,\text{TR}}(r)$ is the Phase-1 one-coordinate trust-region proxy defined in §11.2.1, and the phase-1 cost is evaluated on the current Phase-1 input family. This Phase-1 score is intentionally broad as a screening surface: it privileges cheap local second-order insertion estimates while the controller still has long useful runway. The corresponding broad Phase-1 probing/evaluation budget is set by the baseline controller shot law $N_{\text{shot},1}(t)$.

In the present draft, the live Phase-1 numerator is the one-coordinate trust-region proxy $\Delta E_{1,\text{TR}}(r)$, so that the broad screen remains in the same gradient-plus-second-order family later refined in Phases 2 and 3. The strict one-coordinate insertion drop $\delta E_1(r \mid \mathcal{O}, \theta)$ is retained only as an auxiliary exact comparator for selective frontier confirmation, ablation studies, or future confirm-stage refinement. A broader early-stage local-refit numerator remains only a non-live hook. To make such a refinement live, one would first have to specify (i) the admissible refit-window family, (ii) the corresponding constrained local-refit drop, and (iii) a controller activation law independent of the current raw Phase-1 score multiset, so that the live Phase-1 law is not self-referential.

11.2.6 Position probing, trough detection, and effective selector

The Phase-1 scoring mechanism produces the retained record shortlist for further refined scoring. The retained model set is $\mathcal{G}^{(1)} := \pi_m(\mathcal{S}_1(t))$, the retained position fibers are $\mathcal{P}_m^{(1)} := \{p : (m,p) \in \mathcal{S}_1(t)\}\ \forall m \in \mathcal{G}^{(1)}$, the Phase-2 record pool is $\mathcal{R}_2 := \mathcal{S}_1(t)$, and the acceptance set is $\mathcal{A}_1(t) := \{r = (m,p) \in \mathcal{R}_1 : S_1(r;t) \geq \tau_1(t)\} = \{r_{(1)}, \dots, r_{(M)}\}$, with $M := |\mathcal{A}_1(t)|$. The ordered Phase-1 scores are $s_k^{(1)}(t) := S_1(r_{(k)};t)$, with $s_1^{(1)}(t) \geq s_2^{(1)}(t) \geq \cdots \geq s_M^{(1)}(t)$, and the candidate score distinction is $u_{\text{front},k}^{(1)}(t) := \frac{s_{k+1}^{(1)}(t)+\varepsilon}{s_k^{(1)}(t)+\varepsilon}$, with $k = 1, \dots, M-1$ and $0 < \eta_1 < 1$.

$$k_1^\star(t) := \begin{cases} 0, & M = 0, \\ \min\left\{k \in \{1, \dots, \min(M-1, N_1(t)-1)\} : u_{\text{front},k}^{(1)}(t) \leq \eta_1\right\}, & \text{if this set is nonempty}, \\ \min\{M, N_1(t)\}, & \text{otherwise}, \end{cases}$$

$$\mathcal{D}_1(t) := \{r_{(j)}(t) : 1 \leq j \leq k_1^\star(t)\}, \qquad \mathcal{S}_1(t) := \mathcal{A}_1(t) \cap \mathcal{D}_1(t) = \mathcal{D}_1(t).$$

Here η_1 is a controller parameter chosen either from policy intuition or from empirical calibration.

11.3 Phase II

Phase II is the raw geometric selector stage. It acts on the Phase-1 survivors, restores the coarse geometric signal that distinguishes directional usefulness beyond insertion-drop alone, and produces the filtered shortlist $\mathcal{S}_2$.

11.3.1 Raw geometry, restored novelty, and auxiliary overlap window

The Phase-2 selector works over candidate-position records inherited from Phase 1. In the record-inheritance mode used here, the inherited and admissible domain is $\mathcal{R}_2 := \mathcal{S}_1(t)$. For $r = (m, p)$, let $W(r; t) \subseteq \{1, \dots, n+1\}$ denote the inherited overlap window attached to r. In the present three-phase architecture this window enters the Phase-2 selector authoritatively only through overlap against the current horizontal tangent family; the same inherited window is then carried forward to Phase 3, where reduced-window elimination and final reranking are performed.[8]

Reusing the record-local insertion objects already introduced in Phase 1, let $|d_r^{(2)}\rangle := -\widetilde{A}_r^2|\psi\rangle$ and $\chi_r := \mathrm{clip}\Big(1 - \frac{z_\alpha\,\sigma_r}{|g_r|+\varepsilon_g}, 0, 1\Big)$.[9] For phase-dependent shot allocation in later phases, define

$$u_{\sigma,k}(t) := \mathrm{clip}\left(\mathrm{median}_{r\in\mathcal{R}_k(t)}\,\frac{z_\alpha\,\sigma_r}{|g_r|+\varepsilon_g},\ 0,\ 1\right), \qquad k \in \{2, 3\},$$

where in Phase 2 the record-local gradient and uncertainty estimates are evaluated using $N_{\mathrm{shot},2}(t)$, and in Phase 3 the pair (g_r, σ_r) is inherited from the Phase-II raw geometry carried by the surviving records.

The raw geometric quantities are

$$\begin{aligned}
F_{\mathrm{raw}}(r) &:= \langle \tau_r, \tau_r\rangle + \varepsilon_F, \qquad \varepsilon_F > 0,\\
\mathcal{N}_2(r;t) &:= 1 - \max_{j\in W(r;t)} \frac{\big|\langle d_r|Q_\psi|\partial_{\theta_j}\psi\rangle\big|}{\sqrt{F_{\mathrm{raw}}(r)}}\,\frac{1}{\sqrt{\langle\partial_{\theta_j}\psi|Q_\psi|\partial_{\theta_j}\psi\rangle + \varepsilon_F}},\\
h_r &:= 2\Re\Big(\langle d_r|H|d_r\rangle + \langle\psi|H|d_r^{(2)}\rangle\Big),\\
\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r) &:= \max_{|\alpha|\le\rho/\sqrt{F_{\mathrm{raw}}(r)}} \left[g_{\mathrm{lcb}}(r)|\alpha| - \frac{1}{2}\max(h_r, 0)\alpha^2\right].
\end{aligned}$$

Here $F_{\mathrm{raw}}(r)$ is the raw tangent norm, $\mathcal{N}_2(r;t)$ is the restored Phase-2 novelty score, h_r is the raw local curvature, and $\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r)$ is the raw trust-region gain before reduced-window elimination.

The authoritative Phase-II score is

$$S_2(r;t) := \frac{\Delta E_{\mathrm{TR}}^{\mathrm{raw}}(r)\,\chi_r\,\mathcal{N}_2(r;t)}{1 + K_2(r;t)}, \qquad r \in \mathcal{R}_2.$$

Thus Phase II remains the raw geometric selector stage: it restores directional discrimination beyond insertion-drop alone, damps that geometry when the candidate direction is poorly resolved against its own gradient noise, and filters the Phase-1 survivors before any reduced-window refit quantity is made authoritative.

For continuity with the older local-refit notation, retain the auxiliary windowed drop

$$\mathcal{C}_W(r;\theta) := \{\tilde\theta \in \Theta_{\mathcal{O}\oplus_p m} : \tilde\theta_j = (\theta \oplus_p 0)_j\ \forall j \notin W(r;t)\}$$

and

$$\delta E_2(r \mid \mathcal{O}, \theta) := E(\mathcal{O}, \theta) - \inf_{\tilde\theta\in\mathcal{C}_W(r;\theta)} E(\mathcal{O}\oplus_p m, \tilde\theta).$$

This δE_2 quantity is retained only as an auxiliary comparator for the inherited window. It may be used for notation continuity, diagnostics, or ablation studies, but it does not define a separate selector phase, it does not enter $S_2(r;t)$, and it does not modify the Phase-II shortlist rule. Any authoritative reduced-window rerank remains deferred to Phase 3.

[8] Here p indexes the cut after U_p. Thus $U_{>p} := U_K \cdots U_{p+1}$ and $U_{\le p} := U_p \cdots U_1$, and insertion at cut p means $|\psi_r(\alpha)\rangle = U_{>p} e^{-i\alpha A_m} U_{\le p}|\psi_0\rangle = U_K \cdots U_{p+1}\, e^{-i\alpha A_m}\, U_p \cdots U_1|\psi_0\rangle$. So the new generator is inserted between U_p and U_{p+1}.

[9] Let $\hat g(r)$ denote the empirical gradient estimator for record $r = (m, p)$ obtained from N_r independent shot-level repetitions. If $X_1^{(r)}, \dots, X_{N_r}^{(r)}$ are the repetition-level gradient samples, then $\hat g(r) := \frac{1}{N_r}\sum_{i=1}^{N_r} X_i^{(r)}$ and $\sigma_r := \mathrm{SE}[\hat g(r)] = \sqrt{\mathrm{Var}(\hat g(r))}$. Under independent repetitions, $\mathrm{Var}(\hat g(r)) = \frac{\mathrm{Var}(X_1^{(r)})}{N_r}$, so in practice one estimates $\sigma_r \approx \frac{s_r}{\sqrt{N_r}}$, where s_r is the sample standard deviation of the repetition-level gradients. Thus σ_r is the standard error of the mean gradient estimate, not the sample standard deviation itself.

11.3.2 Shortlist rule

The Phase-II scoring mechanism produces the retained geometric shortlist for the later reduced-window stage. The acceptance set is $\mathcal{A}_2(t) := \{r = (m,p) \in \mathcal{R}_2 : S_2(r;t) \geq \tau_2(t)\} = \{r_{(1)}, \dots, r_{(M)}\}$, with $M := |\mathcal{A}_2(t)|$. Here S_2 is the raw geometric gain times restored novelty score. Let $s_k^{(2)}(t) := S_2(r_{(k)};t)$, such that $s_1^{(2)}(t) \geq s_2^{(2)}(t) \geq \cdots \geq s_M^{(2)}(t)$, with superscript denoting the phase scoring regime. Rate the candidate's score distinction by $u_{\mathrm{front},k}^{(2)}(t) := \frac{s_{k+1}^{(2)}(t)+\varepsilon}{s_k^{(2)}(t)+\varepsilon}$, with $k = 1, \dots, M-1$ and $0 < \eta_2 < 1$. Then

$$k_2^\star(t) := \begin{cases} 0, & M = 0, \\ \min\left\{k \in \{1, \dots, \min(M-1, N_2(t)-1)\} : u_{\mathrm{front},k}^{(2)}(t) \leq \eta_2\right\}, & \text{if this set is nonempty}, \\ \min\{M, N_2(t)\}, & \text{otherwise}, \end{cases}$$

$$\mathcal{D}_2(t) := \{r_{(j)} : 1 \leq j \leq k_2^\star(t)\}, \qquad \mathcal{S}_2(t) := \mathcal{A}_2(t) \cap \mathcal{D}_2(t) = \mathcal{D}_2(t).$$

Of course, $\mathcal{G}^{(2)} := \pi_m(\mathcal{S}_2(t))$, and $\mathcal{P}_m^{(2)} := \{p : (m,p) \in \mathcal{S}_2(t)\}$.

11.4 Phase 3 reduced-window rerank and post-shortlist admission

This section refines the surviving Phase-2 records with the more expensive reduced-window geometry, then uses that final rerank surface for shortlist formation, batched admission, pruning, and continuation.

11.4.1 Reduced-window geometry, curvature, and trust-region gain

Phase 3 starts from the Phase-II raw quantities τ_r, g_r, h_r, $g_{\mathrm{lcb}}(r)$, $F_{\mathrm{raw}}(r)$, and $\mathcal{N}_2(r;t)$, then refines them using the inherited refit window. The coarse novelty score $\mathcal{N}_2(r;t)$ is not recomputed here; instead the candidate direction is reduced against the inherited window before the expensive rerank surface is formed.

11.4.2 Refit window, reduced geometry, and Phase-3 record family

In the record-inheritance mode used here, the Phase-3 input set is $\mathcal{R}_3(t) := \mathcal{S}_2(t)$. The inherited refit window attached to position p at selector step t is denoted $W(p;t)$, with decomposition $W(p;t) = W_{\mathrm{new}}^{(p)} \cup W_{\mathrm{age}}^{(p)}$, where $W_{\mathrm{new}}^{(p)}$ keeps the newest coordinates and $W_{\mathrm{age}}^{(p)}$ preserves the maturity-scaled set of oldest retained old coordinates defined later by the branch-local admission order. For each shortlisted record $r = (m,p)$, let $W_r := W(r;t)$ be the inherited window that would actually be refit if m were admitted at position p, and let $\{\tau_j\}_{j \in W_r}$ be the current horizontal tangents in that inherited window. Phase 3 evaluates its inherited reduced-window geometry under the later-phase controller shot law $N_{\mathrm{shot},3}(t)$.

Using the Phase-II raw tangent data, define $(Q_r)_{jk} = \Re\langle \tau_j, \tau_k \rangle$, $(q_r)_j = \Re\langle \tau_j, \tau_r \rangle$, $E_0 := E(\theta; \mathcal{O})$, $(b_r)_j := \partial_\alpha \partial_{\theta_j} E(\alpha, \delta\theta_{W_r}; r)\big|_{\alpha=0,\, \delta\theta_{W_r}=0}$ for $j \in W_r$, and $(H_{W_r})_{jk} := \partial_{\theta_j} \partial_{\theta_k} E(\alpha, \delta\theta_{W_r}; r)\big|_{\alpha=0,\, \delta\theta_{W_r}=0}$ for $j,k \in W_r$. The local reduced-window energy model is

$$E(\alpha, \delta\theta_{W_r}; r) \approx E_0 + g_r \alpha + \tfrac{1}{2} h_r \alpha^2 + \alpha b_r^\top \delta\theta_{W_r} + \tfrac{1}{2} \delta\theta_{W_r}^\top H_{W_r} \delta\theta_{W_r}.$$

For reduction and reduced novelty, $\mathbf{R}_r = \frac{1}{2}(H_{W_r} + H_{W_r}^\top) + \lambda_H I$, $\tilde{h}_r = h_r - b_r^\top \mathbf{R}_r^{-1} b_r$, $F_r^{\mathrm{red}} = F_{\mathrm{raw}}(r) - 2 q_r^\top \mathbf{R}_r^{-1} b_r + b_r^\top \mathbf{R}_r^{-1} Q_r \mathbf{R}_r^{-1} b_r$, $F_{r,\mathrm{safe}}^{\mathrm{red}} = \max(F_r^{\mathrm{red}}, \varepsilon_{\mathrm{red}})$ with $\varepsilon_{\mathrm{red}} > 0$, $q_r^{\mathrm{red}} = q_r - Q_r \mathbf{R}_r^{-1} b_r$, and $z_r := \frac{(q_r^{\mathrm{red}})^\top (Q_r + \varepsilon_{\mathrm{nov}} I)^{-1} q_r^{\mathrm{red}}}{F_{r,\mathrm{safe}}^{\mathrm{red}}}$, $\mathcal{N}_3(r;t) := \mathrm{clip}_{[0,1]}(1 - z_r)$. The reduced trust-region gain is

$$\Delta E_{\mathrm{TR}}(r) := \max_{|\alpha| \leq \rho/\sqrt{F_{r,\mathrm{safe}}^{\mathrm{red}}}} \left[g_{\mathrm{lcb}}(r)|\alpha| - \tfrac{1}{2} \max(\tilde{h}_r, 0) \alpha^2 \right].$$

11.4.3 Phase-3 rerank score and effective selector The authoritative Phase-3 rerank score is

$$S_3(r;t) := \frac{\Delta E_{\mathrm{TR}}(r)\, \chi_r\, \mathcal{N}_3(r;t)}{1 + K_3(r;t)} + \beta_{\mathrm{motif}} S_{\mathrm{motif}}(r) - \beta_{\mathrm{dup}} D_{\mathrm{dup}}(r), \qquad r \in \mathcal{R}_3(t).$$

Here S_{motif} measures motif compatibility or transferable local pattern alignment, and D_{dup} penalizes near-duplicate continuation directions. These are controller-side feature functionals normalized to comparable scale and layered

on top of the explicit reduced geometric product $\Delta E_{\mathrm{TR}}(r)\,\chi_r\,\mathcal{N}_3(r;t)$; they are not part of the phase-invariant cost denominator.

The retained Phase-3 shortlist is obtained from $\mathcal{A}_3(t) := \{r = (m,p) \in \mathcal{R}_3(t) : S_3(r;t) \geq \tau_3(t)\} = \{r_{(1)}, \dots, r_{(M)}\}$, with $M := |\mathcal{A}_3(t)|$. Let $s_k^{(3)}(t) := S_3(r_{(k)};t)$, such that $s_1^{(3)}(t) \geq s_2^{(3)}(t) \geq \cdots \geq s_M^{(3)}(t)$. Rate the candidate's score distinction by $u_{\mathrm{front},k}^{(3)}(t) := \frac{s_{k+1}^{(3)}(t)+\varepsilon}{s_k^{(3)}(t)+\varepsilon}$, with $k = 1, \dots, M-1$ and $0 < \eta_3 < 1$. Then

$$k_3^\star(t) := \begin{cases} 0, & M = 0, \\ \min\Big\{k \in \{1, \dots, \min(M-1, N_3(t)-1)\} : u_{\mathrm{front},k}^{(3)}(t) \leq \eta_3\Big\}, & \text{if this set is nonempty}, \\ \min\{M, N_3(t)\}, & \text{otherwise}, \end{cases}$$

$$\mathcal{D}_3(t) := \{r_{(j)} : 1 \leq j \leq k_3^\star(t)\}, \qquad \mathcal{S}_3(t) := \mathcal{A}_3(t) \cap \mathcal{D}_3(t) = \mathcal{D}_3(t).$$

Of course, $\mathcal{G}^{(3)} := \pi_m(\mathcal{S}_3(t))$, and $\mathcal{P}_m^{(3)} := \{p : (m,p) \in \mathcal{S}_3(t)\}$.

11.4.4 Reduced-plane geometric batch selection after shortlisting

After the Phase-3 shortlisting has produced the record family $\mathcal{S}_3(t) \subseteq \mathcal{R}_3(t)$, batching is resolved by reduced-state-space geometry on a common inherited tangent plane, rather than by heuristic pairwise compatibility penalties. Support-overlap, commutation, and related proxy features may still be used as optional prescreens, but they are not part of the principled batching law below. Assume throughout that $\mathcal{S}_3(t) \neq \emptyset$; otherwise no Phase-3 batch is formed.

Near-degenerate shell and structural feasibility: let $s_t(r) := S_3(r;t)$ and $r_{\max}(t) \in \arg\max_{r \in \mathcal{S}_3(t)} s_t(r)$. Fix a near-degenerate shell parameter $\eta_{\mathrm{nd}} \in (0,1]$ and a batch cap $B_{\max}(t) \in \mathbb{N}$. Define the near-degenerate record shell $\mathcal{N}_3(t) := \big\{r \in \mathcal{S}_3(t) : s_t(r) \geq \eta_{\mathrm{nd}}\, s_t(r_{\max}(t))\big\}$ and the corresponding near-degenerate batch family $\mathfrak{B}_{\mathrm{nd}}(t) := \{B \subseteq \mathcal{N}_3(t) : r_{\max}(t) \in B, \quad 1 \leq |B| \leq B_{\max}(t)\}$. Define a hard batch-structural gate $\Gamma_{\mathrm{batch}}^{\mathrm{struct}}(B;t) \in \{0,1\}$ to mean that the records in B are jointly insertable on the current branch/state, satisfy all hard cross-record constraints, and determine a well-defined batched scaffold update $\mathcal{O}_t \oplus B$ in the fixed canonical insertion order. The structurally feasible batch family is $\mathfrak{B}_{\mathrm{struct}}(t) := \{B \subseteq \mathcal{S}_3(t) : \Gamma_{\mathrm{batch}}^{\mathrm{struct}}(B;t) = 1\}$. A controller-side transient beam fallback may also be defined on the same rerank shell.[10]

Common-window reduced geometry: write the current state as $|\psi_t\rangle := |\psi(\mathcal{O}_t, \theta_t)\rangle$, $\quad Q_{\psi_t} := I - |\psi_t\rangle\langle\psi_t|$, where Q_{ψ_t} is the horizontal projector. For any batch $B \in \mathfrak{B}_{\mathrm{struct}}(t)$, define the common inherited refit window $W_B(t) := \bigcup_{r \in B} W_r(t)$ and the corresponding inherited horizontal tangent matrix $T_B(t) := [\,t_j(t)\,]_{j \in W_B(t)}$, $\quad t_j(t) := Q_{\psi_t}\,\partial_{\theta_j}|\psi_t\rangle$. Let $E_t(\alpha, \delta\theta_{W_B}; B)$ denote the local energy obtained by inserting the records in B with batch amplitudes $\alpha = (\alpha_r)_{r \in B}$ and varying the inherited coordinates on $W_B(t)$ by $\delta\theta_{W_B}$. For each $r \in B$, let $|\psi_{t,r}(\alpha_r)\rangle$ denote the state obtained by inserting only record r at amplitude α_r into the current scaffold while keeping inherited coordinates fixed. Its zero-initialized horizontal candidate tangent is $d_r(t) := Q_{\psi_t}\,\partial_{\alpha_r}|\psi_{t,r}(\alpha_r)\rangle\Big|_{\alpha_r=0}$. Let $\mathsf{H}_{W_B}(t)$ denote the inherited-window Hessian block of $E_t(\alpha, \delta\theta_{W_B}; B)$ with respect to the coordinates on $W_B(t)$ at $(\alpha, \delta\theta_{W_B}) = (0,0)$, and set $M_B(t) := \frac{1}{2}\Big(\mathsf{H}_{W_B}(t) + \mathsf{H}_{W_B}(t)^\top\Big) + \lambda_H I$, $\quad \lambda_H > 0$. For each $r \in B$, define the mixed candidate-window curvature vector by $\big(b_r^{(B)}(t)\big)_j := \partial_{\alpha_r}\partial_{\theta_j} E_t(\alpha, \delta\theta_{W_B}; B)\Big|_{(\alpha,\delta\theta_{W_B})=(0,0)}$, $\quad j \in W_B(t)$. The reduced candidate tangent after linear relaxation of the common inherited window is then $u_r^{(B)}(t) := d_r(t) - T_B(t)\,M_B(t)^{-1} b_r^{(B)}(t)$, hence the effective batch plane is $\Pi_B(t) := \mathrm{span}\{u_r^{(B)}(t) : r \in B\} \subseteq T_{\psi_t}\mathbb{P}(\mathcal{H})$. Its reduced Gram matrix is $\big(G_B(t)\big)_{rs} := \Re\langle u_r^{(B)}(t), u_s^{(B)}(t)\rangle$, $\quad r,s \in B$, and in particular $\big(G_B(t)\big)_{rr} = F_r^{\mathrm{red},(B)}(t)$ is the reduced norm of record r in the common batch reduction. Next define the symmetrized raw candidate-candidate curvature on the same common reduction by $h_{rs}^{(B)}(t) := \frac{1}{2}\,(\partial_{\alpha_r}\partial_{\alpha_s} + \partial_{\alpha_s}\partial_{\alpha_r}) E_t(\alpha, \delta\theta_{W_B}; B)\Big|_{(\alpha,\delta\theta_{W_B})=(0,0)}$, $\quad r,s \in B$, and then the reduced curvature matrix $\big(\widetilde{H}_B(t)\big)_{rs} := h_{rs}^{(B)}(t) - \big(b_r^{(B)}(t)\big)^\top M_B(t)^{-1} b_s^{(B)}(t)$. For $r = s$, the entry $\big(\widetilde{H}_B(t)\big)_{rr}$

[10] One convenient rule is to set $\mathcal{T}_{\mathrm{tie}}(t) := \{\, r \in \mathcal{S}_3(t) : s_t(r) \geq \max(\eta_{\mathrm{tie}}\, s_t(r_{\max}(t)),\, s_t(r_{\max}(t)) - \delta_{\mathrm{tie}})\,\}$. If $|\mathcal{T}_{\mathrm{tie}}(t)| > 1$ and the controller elects not to resolve the shell by an immediate one-scaffold batch, it may retain a highest-scoring subset $\mathcal{A}_{\mathrm{tie}}(t) \subseteq \mathcal{T}_{\mathrm{tie}}(t)$ with $1 \leq |\mathcal{A}_{\mathrm{tie}}(t)| \leq B_{\mathrm{tie}}(t)$ and pass one retained alternative per record into the scaffold-beam machinery of §11.5.2.

is the usual single-record reduced curvature in the common batch window; for $r \neq s$, the entry $\left(\widetilde{H}_B(t)\right)_{rs}$ is the mixed second-order coupling after inherited-window relaxation. The diagnostic coherences are $\mu_{\tan}(B;t) := \max_{r\neq s\in B} \frac{|(G_B(t))_{rs}|}{\sqrt{(G_B(t))_{rr}(G_B(t))_{ss}}+\varepsilon}$ and $\mu_{\mathrm{curv}}(B;t) := \max_{r\neq s\in B} \frac{|(\widetilde{H}_B(t))_{rs}|}{\sqrt{\left((\widetilde{H}_B(t))_{rr}\right)^{(+)}\left((\widetilde{H}_B(t))_{ss}\right)^{(+)}}+\varepsilon}$, $\quad x^{(+)} := \max\{x,0\}$; these may be used as optional early pruning diagnostics, but the actual batch objective is the joint reduced trust-region gain below.

Joint reduced gain and additivity: let $g_{B,\mathrm{lcb}}(t) := \left(g_{\mathrm{lcb}}(r;t)\right)_{r\in B}$, and let $\left(\widetilde{H}_B(t)\right)^{(+)}$ denote the positive-semidefinite part of $\widetilde{H}_B(t)$. The batch trust-region objective below is therefore a conservative convexified local gain model on the reduced batch plane:

$$\Delta E_B^{\mathrm{TR}}(t) := \max_{\alpha\in\mathbb{R}^{|B|},\ \alpha^\top G_B(t)\alpha\le\rho_B(t)^2} \left[g_{B,\mathrm{lcb}}(t)^\top\alpha - \frac{1}{2}\alpha^\top\left(\widetilde{H}_B(t)\right)^{(+)}\alpha\right].$$

Here $\rho_B(t) > 0$ is the Phase-3 batch trust-region radius. If record identity already includes the favorable orientation, one may additionally impose $\alpha \geq 0$ componentwise. For each $r \in B$, define the contextual single-record gain in the same common reduction by $\Delta E_{r|B}^{\mathrm{TR}}(t) := \max_{\beta\in\mathbb{R},\ \beta^2(G_B(t))_{rr}\le\rho_B(t)^2} \left[g_{\mathrm{lcb}}(r;t)\,\beta - \frac{1}{2}\left((\widetilde{H}_B(t))_{rr}\right)^{(+)}\beta^2\right]$. Then the additivity defect is $\delta_{\mathrm{add}}(B;t) := \left[1 - \frac{\Delta E_B^{\mathrm{TR}}(t)}{\sum_{r\in B}\Delta E_{r|B}^{\mathrm{TR}}(t)+\varepsilon}\right]_+$. Small $\delta_{\mathrm{add}}(B;t)$ means that the joint reduced model is approximately additive on the common batch plane.

Batch choice and greedy implementation: the geometric admissible batch family is $\mathfrak{B}_{\mathrm{geom}}(t) := \Big\{B \in \mathfrak{B}_{\mathrm{nd}}(t) \cap \mathfrak{B}_{\mathrm{struct}}(t) : \lambda_{\min}(G$ When a more aggressive pruning rule is desired, one may additionally require $\mu_{\tan}(B;t) \leq \tau_{\tan}, \quad \mu_{\mathrm{curv}}(B;t) \leq \tau_{\mathrm{curv}}$. The principled post-shortlist batch choice is

$$B_t^\star \in \arg\max_{B\in\mathfrak{B}_{\mathrm{geom}}(t)} \Delta E_B^{\mathrm{TR}}(t).$$

If $\mathfrak{B}_{\mathrm{geom}}(t) = \emptyset$, no Phase-3 batch is admitted from the current shortlist. A greedy implementation avoids exhaustive search: initialize $B_t^{(1)} := \{r_{\max}(t)\}$. Given a current batch $B_t^{(k)}$, define the admissible add-on family $\mathcal{A}_k(t) := \Big\{r \in \mathcal{N}_3(t) \;\; B_t^{(k)} : B_t^{(k)} \cup \{r\} \in \mathfrak{B}_{\mathrm{nd}}(t) \cap \mathfrak{B}_{\mathrm{struct}}(t)\Big\}$. For each $r \in \mathcal{A}_k(t)$, define the marginal joint gain $\Delta_{\mathrm{marg}}(r \mid B_t^{(k)};t) := \Delta E^{\mathrm{TR}}_{B_t^{(k)}\cup\{r\}}(t) - \Delta E^{\mathrm{TR}}_{B_t^{(k)}}(t)$. Choose $r_k^{\mathrm{best}}(t) \in \arg\max_{r\in\mathcal{A}_k(t)} \Delta_{\mathrm{marg}}(r \mid B_t^{(k)};t)$ subject to $\lambda_{\min}(G_{B_t^{(k)}\cup\{r\}}(t)) \geq \tau_{\mathrm{rank}} \frac{\operatorname{tr} G_{B_t^{(k)}\cup\{r\}}(t)}{|B_t^{(k)}|+1}, \quad \delta_{\mathrm{add}}(B_t^{(k)} \cup \{r\};t) \leq \tau_{\mathrm{add}}$, and, when enabled, $\mu_{\tan}(B_t^{(k)} \cup \{r\};t) \leq \tau_{\tan}, \quad \mu_{\mathrm{curv}}(B_t^{(k)} \cup \{r\};t) \leq \tau_{\mathrm{curv}}$. If no such r exists, or if the best feasible marginal gain is nonpositive, stop and set $B_t^\star := B_t^{(k)}$. Otherwise append the best add-on, $B_t^{(k+1)} := B_t^{(k)} \cup \{r_k^{\mathrm{best}}(t)\}$, and continue until no further feasible positive-gain append exists or until $|B_t^{(k)}| = B_{\max}(t)$. The deployed Phase-3 action is the final batch $B_t^\star$. If $|B_t^\star| = 1$, this reduces to the usual single-record admission. If $|B_t^\star| > 1$, the controller inserts the batch on the current branch in the canonical batch order and performs the common-window refit on $W_{B_t^\star}(t)$.

11.5 Post m_p Application Entails Windowed-Refitting

In ADAPT-VQE the variational problem is $\min_\theta E_L(\theta)$ with $E_L(\theta) = \langle\phi_0|(\prod_{\ell=L}^1 e^{i\theta_\ell G_\ell})H(\prod_{\ell=1}^L e^{-i\theta_\ell G_\ell})|\phi_0\rangle$, and the measurement bottleneck arises because one iteration must estimate not only E_L but also pool-selection scores $s_\mu(\theta) = |\langle\phi_0|(\prod_{\ell=L}^1 e^{i\theta_\ell G_\ell}), i[H, A_\mu], (\prod_{\ell=1}^L e^{-i\theta_\ell G_\ell})|\phi_0\rangle|$ for $\mu \in \mathcal{P}$, reoptimization gradients $\partial_{\theta_j} E_L(\theta)$, and, for geometry-aware updates, metric entries $M_{jk}(\theta) = \Re\langle\partial_{\theta_j}[(\prod_{\ell=1}^L e^{-i\theta_\ell G_\ell})|\phi_0\rangle]|\partial_{\theta_k}[(\prod_{\ell=1}^L e^{-i\theta_\ell G_\ell})|\phi_0\rangle]\rangle$, so at precisions $\epsilon_s, \epsilon_g, \epsilon_M$ the shot cost obeys $N_{\mathrm{iter}} \gtrsim \sum_{\mu\in\mathcal{P}} \mathrm{Var}(\hat{s}_\mu)/\epsilon_s^2 + \sum_{j=1}^L \mathrm{Var}(\widehat{\partial_{\theta_j} E_L})/\epsilon_g^2 + \sum_{j,k=1}^L \mathrm{Var}(\hat{M}_{jk})/\epsilon_M^2$; the proposed strategy is to choose disjoint sets $F, A \subset 1, \dots, L$ with $F \cup A = 1, \dots, L$, impose $\dot{\theta}_j = 0$ for $j \in F$, and optimize only $j \in A$, which reduces the reparameterization sector to $\sum_{j\in A} \mathrm{Var}(\widehat{\partial_{\theta_j} E_L})/\epsilon_g^2 + \sum_{j,k\in A} \mathrm{Var}(\hat{M}_{jk})/\epsilon_M^2$ and therefore removes the growth coming from repeatedly updating old coordinates, but it leaves the pool-screening term $\sum_{\mu\in\mathcal{P}} \mathrm{Var}(\hat{s}_\mu)/\epsilon_s^2$ unchanged and it does not shorten the physical circuit, since every shot still executes the full depth-L scaffold with entangling-gate count $N_{2q}(L)$ and runtime $t_{\mathrm{circ}}(L)$, so hardware fidelity still degrades roughly as $e^{-t_{\mathrm{circ}}(L)/T_1} e^{-t_{\mathrm{circ}}(L)/T_2} (1-\varepsilon_{2q})^{N_{2q}(L)}$; consequently the freezing strategy cures only the reoptimization component of the ADAPT bottleneck, and the limiting cost after that is either the unchanged operator-pool screening overhead

or, if screening is separately controlled, the conventional VQE bottleneck of resolving E_L on a deep noisy ansatz, with shots scaling as $N_E \gtrsim \mathrm{Var}(H_{\mathrm{meas}} \mid \prod_{\ell=1}^{L} e^{-i\theta_\ell G_\ell}, \phi_0, \mathrm{noise})/(\Delta E)^2$ to distinguish an energy change ΔE.

Let the length-(m) scaffold be $U^{(m)}(\theta) = \prod_{j=1}^{m} e^{-i\theta_j G_j}$, where (G_j) is the (j)-th appended generator and (θ_j) its coefficient. The reparameterization motif is to optimize only an active subset $(\mathcal{A}_m \subseteq 1, \dots, m)$, chosen by an activity functional $a_j^{(m)} := \lambda_{\mathrm{rec}}\, r_j^{(m)} + \lambda_{\mathrm{mob}}\, v_j^{(m)} - \lambda_{\mathrm{age}}\, \phi(m-j) - \lambda_{\mathrm{stag}}\, s_j^{(m)}$, and then set $(\mathcal{A}_m = \{j : a_j^{(m)} > \tau_m\})$. Here $(r_j^{(m)})$ is a recency weight, large for recently appended generators; $(v_j^{(m)} := \sum_{k=j}^{m-1} |\Delta\theta_j^{(k)}|)$ is a lifetime mobility measure, large for parameters that have historically moved substantially; $(\phi(m-j))$ is an age penalty, biasing against early-added generators; and $(s_j^{(m)})$ is a stagnation indicator, large when a parameter has remained nearly inert over recent refits. Thus the inclusion bias is toward recent and historically labile generators, while the exclusion bias is toward old and typically stagnant generators.

This is the same scaffold language used above: $U^{(m)}(\theta)$ is the scaffold $\mathcal{O}$, the active set $\mathcal{A}_m$ plays the role of the inherited refit window $W(r;t)$, the frozen set is $\{1, \dots, n+1\} \;\; W(r;t)$, and the activity functional $a_j^{(m)}$ is the same maturity-facing bookkeeping later summarized by the eligible set $\mathcal{M}_t$.

Concomitantly, as scaffold maturity increases, the controller shot laws $N_{\mathrm{shot},1}(t)$ and $N_{\mathrm{shot},k}(t)$ should also increase so as to preserve the decision-level signal-to-noise ratio on the active block. Formally, if δ_t denotes a typical useful signal scale among the unfrozen directions and σ_t the corresponding estimator spread, then one chooses the phase-appropriate shot law so that $\mathrm{SNR}_t \sim \frac{\delta_t \sqrt{N_{\mathrm{shot},1}(t)}}{\sigma_t} \geq \kappa$ in Phase 1 and $\mathrm{SNR}_{t,k} \sim \frac{\delta_{t,k}\sqrt{N_{\mathrm{shot},k}(t)}}{\sigma_{t,k}} \geq \kappa$ for $k \in \{2,3\}$ in later phases. Hence the control law is: as scaffold maturity grows, shrink the reoptimized subspace by freezing archaic or stagnant directions while increasing shots on the surviving active directions to maintain resolvable updates.

11.5.1 Branch-local mature-scaffold pruning: permissibility, frozen-ablation ranking, and exact rollback

All quantities in this subsection are branch-local. On a single active branch we suppress the branch label. Let $(\mathcal{O}_t^-, \theta_t^-, E_t^-)$ be the committed branch state immediately before the accepted Phase-3 payload $\mathcal{A}_t^\star$ is applied, and let $(\mathcal{O}_t^+, \theta_t^+, E_t^+)$ be the committed post-admission state after the ordinary local refit on that same branch has completed. The admitted-step gain is $\Delta_{\mathrm{adm}}(t) := E_t^- - E_t^+$.

Each committed coordinate of $\mathcal{O}_t^+$ carries the metadata $(m_j, \iota_j, r_j, \kappa_j, c_j)$: generator label m_j, branch-local admission step ι_j, admission record r_j, stored selector burden $\kappa_j := K_3(r_j; \iota_j)$, and prune-cooldown counter $c_j(t) \in \{0, 1, \dots, c_{\mathrm{cool}}\}$, transported with that coordinate under later scaffold edits. Here $t_{\mathrm{first}}(m_j)$ denotes the first time the generator m_j was ever seen in the run.

Use the maturity-increasing prune-pressure law $\lambda_{\mathrm{prune}}(t) := \lambda_{\mathrm{prune}}^{\min} + (\lambda_{\mathrm{prune}}^{\max} - \lambda_{\mathrm{prune}}^{\min}) g_t$, with $0 \leq \lambda_{\mathrm{prune}}^{\min} \leq \lambda_{\mathrm{prune}}^{\max}$. Let the baseline gain-resolution scale be $\Delta_{\mathrm{scr}}(t) := \max(\tau_{\mathrm{use},t}, \mathrm{EWMA}_{\mathrm{recent}}(\Delta_{\mathrm{adm}}), \Delta_{\mathrm{num}})$, use the noise-aware screening scale $\Delta_{\mathrm{scr}}^{\mathrm{noise}}(t) := (1 + u_{\sigma,3}(t))\Delta_{\mathrm{scr}}(t)$, and define the admitted-gain signal-to-noise ratio by $\mathrm{SNR}_{\mathrm{adm}}(t) := \Delta_{\mathrm{adm}}(t) \big/ (\Delta_{\mathrm{scr}}^{\mathrm{noise}}(t) + \varepsilon)$. A prune pass may open only after an accepted admission and its ordinary local refit have already committed. Let $\mathcal{T}_{\mathrm{chk}} \subseteq \mathbb{N}$ be the scheduled checkpoint set, define $C_t^{\mathrm{perm}} := (u_{\mathrm{sat}}(t) \geq \tau_{\mathrm{med}}) \wedge \Big(\mathrm{SNR}_{\mathrm{adm}}(t) \leq \kappa_{\mathrm{pr}} \;\vee\; t \in \mathcal{T}_{\mathrm{chk}}\Big)$, and $\Gamma_{\mathrm{prune}}^{\mathrm{perm}}(t) := \mathbf{1}[\text{accepted admission at } t]\mathbf{1}[\text{post-admission local refit completed}]\mathbf{1}[|\mathcal{O}_t^+| \geq 2]\mathbf{1}[C_t^{\mathrm{perm}}]$. Thus pruning is a maturity-controlled maintenance action that opens only when the scaffold is already in the mature regime and the admitted gain is no longer cleanly resolved above the noise-aware screening floor, or when a maturity-gated checkpoint explicitly requests a simplification test.

Newly admitted coordinates are protected for ℓ_{protect} admitted steps via $\mathcal{P}_{\mathrm{protect}}(t) := \{\, j : \; t - \iota_j < \ell_{\mathrm{protect}} \,\}$. Let $s_j^{\mathrm{stag}}(t)$ denote the branch-local t-indexed lift of the earlier coordinate stagnation indicator $s_j^{(m)}$, with larger values meaning less recent parameter motion and therefore greater local stagnation. The mature prune-eligible set is

$$\mathcal{M}_t := \Big\{ j : \; t - t_{\mathrm{first}}(m_j) \geq A_{\mathrm{stale}}, \; s_j^{\mathrm{stag}}(t) \geq s_{\mathrm{stale}}, \; j \notin \mathcal{P}_{\mathrm{protect}}(t), \; c_j(t) = 0 \Big\}.$$

Hence prune eligibility is branch-local and conservative: old by first-seen age, stale by recent inactivity, not recently admitted, and not under temporary cooldown.

Small angle is used only as a cheap nomination device. If $\mathcal{M}_t \neq \emptyset$, set $\theta_{\text{small}}(t) := \max(\theta_{\text{abs}}, c_\theta \operatorname{median}_{j\in\mathcal{M}_t} |\operatorname{wrap}(\theta_j(t))|)$ and $\mathcal{J}_\angle(t) := \{ j \in \mathcal{M}_t : |\operatorname{wrap}(\theta_j(t))| \leq \theta_{\text{small}}(t) \}$; let $\mathcal{P}_{\text{probe}}(t) \subseteq \mathcal{J}_\angle(t)$ be the set of the $\min\{K_p, |\mathcal{J}_\angle(t)|\}$ smallest values of $|\operatorname{wrap}(\theta_j(t))|$ over $j \in \mathcal{J}_\angle(t)$. If $\mathcal{M}_t = \emptyset$, set $\mathcal{P}_{\text{probe}}(t) = \emptyset$. Thus old/stale and small-angle coordinates are nominated cheaply, but neither condition is itself an admissible proof of redundancy.

For each $j \in \mathcal{P}_{\text{probe}}(t)$, let $\mathcal{O}^+_{t,-j}$ denote the scaffold obtained from $\mathcal{O}^+_t$ by deleting coordinate j, and let $(\theta^+_t)_{-j}$ be θ^+_t with the same entry removed. The frozen-ablation energy and immediate regression are $E^{\text{frz}}_{-j} := E(\mathcal{O}^+_{t,-j}, (\theta^+_t)_{-j})$ and $L^{\text{frz}}_j(t) := E^{\text{frz}}_{-j} - E^+_t$. Use the cheap prune score

$$P^{\text{pr}}_j(t) := \frac{[L^{\text{frz}}_j(t)]_+}{1+\kappa_j}, \qquad j \in \mathcal{P}_{\text{probe}}(t),$$

and, when $\mathcal{P}_{\text{probe}}(t) \neq \emptyset$, choose

$$j^\star_t := \arg\min_{j\in\mathcal{P}_{\text{probe}}(t)} P^{\text{pr}}_j(t).$$

Thus the cheap ranking reuses the same K-style burden logic already used for admission: low retained frozen-ablation utility relative to stored selector burden means good prune-test priority, so smaller $P^{\text{pr}}_j(t)$ is better.

Before the authoritative trial, store the exact rollback snapshot $\Xi^{\text{pr}}_t := (\mathcal{O}^+_t, \theta^+_t, E^+_t, \{c_i(t)\}_i, \text{current local optimizer state})$. Only one generator is tested per prune pass. If $\Gamma^{\text{perm}}_{\text{prune}}(t) = 0$ or $\mathcal{P}_{\text{probe}}(t) = \emptyset$, no prune trial is launched and the committed branch state remains $(\mathcal{O}^+_t, \theta^+_t, E^+_t)$.

For the authoritative trial on $j^\star_t$, let $n^+_t := |\mathcal{O}^+_t|$, $\omega^{(-j)}_{\text{eff}} := \max\{1, \min(\omega_{\text{pr}}, n^+_t - 1)\}$, $\ell_j := \max\left\{1, \min\left(j - \left\lfloor \frac{\omega^{(-j)}_{\text{eff}}-1}{2} \right\rfloor,\ n^+_t - \omega^{(-j)}_{\text{eff}}\right)\right\}$ $W_{\text{loc}}(j) := \{\ell_j, \dots, \ell_j + \omega^{(-j)}_{\text{eff}} - 1\}$, $C^{(-j)}_{\text{old}} := \{1, \dots, n^+_t - 1\}$ $W_{\text{loc}}(j)$, and $n^{\text{pr}}_{\text{old}}(t,j) := \left\lceil \rho_W(t)\, |C^{(-j)}_{\text{old}}| \right\rceil$. Let $W^{(-j)}_{\text{age}}(t)$ be the set of the $n^{\text{pr}}_{\text{old}}(t,j)$ oldest surviving coordinates in $C^{(-j)}_{\text{old}}$ under the stored admission-order values ι_i. The prune-refit window is

$$W_{\text{pr}}(j;t) := W_{\text{loc}}(j) \cup W^{(-j)}_{\text{age}}(t).$$

Hence the prune refit window contracts automatically as maturity increases, because $\rho_W(t)$ decreases as useful runway is exhausted.

The authoritative post-ablation refit is

$$\theta^{(-j)}_t := \arg\min_{\tilde\theta\in\Theta_{\mathcal{O}^+_{t,-j}}} \left\{ E(\mathcal{O}^+_{t,-j}, \tilde\theta) : \tilde\theta_i = ((\theta^+_t)_{-j})_i \ \forall i \notin W_{\text{pr}}(j;t) \right\},$$

with corresponding post-refit energy $E^{(-j)}_t := E(\mathcal{O}^+_{t,-j}, \theta^{(-j)}_t)$. This ablation-plus-local-refit result is the authoritative prune test. No extra geometric score is required at this stage.

Define the retained admitted-step gain after removing j by $\Delta^{(-j)}_{\text{keep}}(t) := E^-_t - E^{(-j)}_t$. Accept the prune of $j^\star_t$ iff $E^{(-j^\star_t)}_t \leq E^+_t + \Delta_{\text{safe}}$ and $\Delta^{(-j^\star_t)}_{\text{keep}}(t) \geq \eta_{\text{ret}}\Delta_{\text{adm}}(t)$. Thus the final accept rule is recoverability-based: the reduced scaffold may be kept only when the ablation-plus-refit energy regression is numerically small and the pruned scaffold still retains a controlled fraction of the gain produced by the last admitted payload. The burden term κ_j is used only for cheap ranking; the final accept rule remains energy-based and conservative.

If the prune is accepted, commit $(\mathcal{O}^+_{t,-j^\star_t}, \theta^{(-j^\star_t)}_t, E^{(-j^\star_t)}_t)$ as the new branch state and delete the metadata carried by coordinate $j^\star_t$. If the prune is rejected, restore the branch state exactly to Ξ^{pr}_t, set $c_{j^\star_t}(t) \leftarrow c_{\text{cool}}$, and launch no second prune trial at the same selector step. On later admitted steps the cooldown counter decrements by one until it reaches zero. Hence prune failure never invalidates the already committed admission, and rollback is exact.

11.5.2 Beam-adapt over scaffold alternatives: branch state, pruning, and effective selector

In this section beam-adapt is **purely structural**: each branch carries an alternative ADAPT scaffold together with its locally refit amplitudes. There is no time checkpoint, probe rollout, or projected-dynamics integral in this section; those belong to the separate time-dynamics beam surface developed later.

The cleanest way to read a live scaffold branch is

$$\begin{aligned}\mathfrak{b} = (&\text{identity/history; current scaffold and amplitudes;}\\ &\text{current energy/depth; cumulative selector totals/controller telemetry/status}).\end{aligned}$$

More explicitly, write

$$\mathfrak{b} = (\mathrm{id}_b, \pi_b, \mathcal{H}_b, \mathcal{O}_b, \theta_b, E_b, d_b, S_b^{\mathrm{cum}}, K_b^{\mathrm{cum}}, \mathcal{T}_b^{\mathrm{ctrl}}, \mathrm{status}_b, \mathrm{term}_b),$$

with $\mathcal{H}_b = (r_{b,1}, \dots, r_{b,d_b})$ and $r_{b,j} = (m_{b,j}^{\mathrm{adm}}, p_{b,j}^{\mathrm{adm}})$. Here id_b and π_b are the branch and parent identifiers, $\mathcal{H}_b$ is the ordered history of admitted records, $\mathcal{O}_b$ is the scaffold currently carried by branch b, θ_b is the refit parameter vector on that scaffold, $E_b = E(\theta_b; \mathcal{O}_b)$ is the current variational energy, d_b is the branch ADAPT depth, $S_b^{\mathrm{cum}} = \sum_{j=1}^{d_b} s_{b,j}$ is the cumulative admitted selector value, $K_b^{\mathrm{cum}} = \sum_{j=1}^{d_b} K_{b,j}^{\mathrm{sel}}$ is the cumulative admitted selector burden, $\mathcal{T}_b^{\mathrm{ctrl}}$ stores branch-local controller telemetry, and $(\mathrm{status}_b, \mathrm{term}_b)$ record live/terminal status and termination label. The increment $K_{b,j}^{\mathrm{sel}}$ is the selector-cost contribution attached to the j-th admission on branch b; for a newly materialized child this contribution is the branch-local Phase-3 cost $K_3(r; \mathfrak{b})$. Thus K_b^{cum} is cumulative selector-side cost bookkeeping rather than a raw hardware count.

If beam search is entered from an incumbent scaffold $(\mathcal{O}^{(0)}, \theta^{(0)})$, the root branch is $\mathfrak{b}_{\mathrm{root}} = (\mathrm{id}_0, \emptyset, \mathcal{H}^{(0)}, \mathcal{O}^{(0)}, \theta^{(0)}, E(\theta^{(0)}; \mathcal{O}^{(0)}), d_0, S$ with $d_0 = |\mathcal{H}^{(0)}|$, $S_{\mathrm{root}}^{\mathrm{cum}} = \sum_{j=1}^{d_0} s_j^{(0)}$, and $K_{\mathrm{root}}^{\mathrm{cum}} = \sum_{j=1}^{d_0} K_j^{\mathrm{sel},(0)}$, where $K_j^{\mathrm{sel},(0)}$ denotes the selector-cost contribution attached to the j-th historical admission on the incumbent scaffold. If the beam is started from a fresh scaffold, then $\mathcal{H}^{(0)} = \emptyset$, $d_0 = 0$, and both cumulative sums vanish.

For clock and budget, set $t_b := d_b$ and $D_{\mathrm{left}}(\mathfrak{b}) := d_{\max} - d_b$. Now apply the branch lift rule: for any live-state quantity X from Phase 3 whose definition is written in terms of $(\mathcal{H}, \mathcal{O}, \theta, t, \mathcal{T}^{\mathrm{ctrl}})$, define $X(\mathfrak{b}) := X\big|_{(\mathcal{H},\mathcal{O},\theta,t,\mathcal{T}^{\mathrm{ctrl}}) \mapsto (\mathcal{H}_b,\mathcal{O}_b,\theta_b,t_b,\mathcal{T}_b^{\mathrm{ctrl}})}$; for any record-local quantity $X(r; t)$, define $X(r; \mathfrak{b}) := X(r; t)\big|_{(\mathcal{H},\mathcal{O},\theta,t,\mathcal{T}^{\mathrm{ctrl}}) \mapsto (\mathcal{H}_b,\mathcal{O}_b,\theta_b,t_b,\mathcal{T}_b^{\mathrm{ctrl}})}$. In particular, $R(\mathfrak{b})$, $u_{\mathrm{sat}}(\mathfrak{b})$, $\widehat{N}_{\mathrm{rem}}(\mathfrak{b})$, $K_3(r; \mathfrak{b})$, and $g_{\mathrm{lcb}}(r; \mathfrak{b})$ are branch-lifted evaluations of their previously defined live-state counterparts.

The Phase-3 chain is branch-local by construction: all of its dependencies factor through the state tuple $(\mathcal{O}_b, \theta_b, t_b, \mathcal{T}_b^{\mathrm{ctrl}})$. Hence each live branch induces its own raw candidate-position surface $\mathcal{R}_b^{\mathrm{raw}}$, its own branch-local shortlist $\mathcal{S}_3(\mathfrak{b})$, and its own retained admission set $\mathcal{A}_b$.

For stage and enabled family, let $\eta_b \in \{\mathrm{core/seed}, \mathrm{residual}\}$ and

$$\mathcal{G}_{\eta_b}^{(3)}(\mathfrak{b}) = \begin{cases} \mathcal{G}_{\mathrm{core/seed}}^{(3)}(\mathfrak{b}), & \eta_b = \mathrm{core/seed}, \\ \mathcal{G}_{\mathrm{residual}}^{(3)}(\mathfrak{b}), & \eta_b = \mathrm{residual}. \end{cases}$$

Let $\mathcal{P}_{\mathrm{residual}}(\mathfrak{b}) \subseteq \mathcal{G}^{(3)}$ denote the residual-only generator family on branch b. Then the controller-stage gate and branch-local available Phase-3 generator family are

$$\Gamma_{\mathrm{stage},b}(m) = \begin{cases} 1, & \eta_b = \mathrm{residual}, \\ 1, & \eta_b = \mathrm{core/seed} \text{ and } m \notin \mathcal{P}_{\mathrm{residual}}(\mathfrak{b}), \\ 0, & \eta_b = \mathrm{core/seed} \text{ and } m \in \mathcal{P}_{\mathrm{residual}}(\mathfrak{b}), \end{cases} \qquad \mathcal{G}_{\mathrm{avail}}^{(3)}(\mathfrak{b}) = \{m \in \mathcal{G}^{(3)} : \Gamma_{\mathrm{stage},b}(m) = 1\}.$$

For scaffold length, append slot, and active probe window, let $n_b = |\mathcal{O}_b|$, $p_{\mathrm{app}}(\mathfrak{b}) = n_b$, and $W_{\mathrm{act}}(\mathfrak{b}) \subseteq \{0, \dots, n_b - 1\}$. The ordered probe list is $\mathcal{L}_{\mathrm{probe}}(\mathfrak{b}) := (p_{\mathrm{app}}(\mathfrak{b}), 0, W_{\mathrm{act}}(\mathfrak{b}))$, and the distinct retained probe positions are $\mathcal{P}_{\mathrm{probe}}(\mathfrak{b}) = \mathrm{Head}_{M_{\mathrm{probe}}}\Big(\mathrm{Dedup}_{\mathrm{ord}}(\mathcal{L}_{\mathrm{probe}}(\mathfrak{b}))\Big) \subseteq \{0, \dots, n_b\}$. For symmetry and admissible positions, let

$$\Gamma_{\mathrm{sym},b}(m, p) = \begin{cases} 0, & \text{the symmetry audit for } (m, p; \mathcal{O}_b) \text{ returns a hard violation}, \\ 1, & \text{otherwise}, \end{cases}$$

so the branch-local raw candidate-position surface is

$$\mathcal{R}_b^{\mathrm{raw}} := \{\, r = (m, p) : m \in \mathcal{G}_{\mathrm{avail}}^{(3)}(\mathfrak{b}),\ p \in \mathcal{P}_{\mathrm{probe}}(\mathfrak{b}),\ \Gamma_{\mathrm{sym},b}(m, p) = 1 \,\}.$$

The resulting local shortlist and retained admission set are

$$\mathcal{S}_3(\mathfrak{b}) := \mathrm{Top}_{N_3(t_b)}(S_3(\cdot; \mathfrak{b}); \mathcal{R}_b^{\mathrm{raw}}), \qquad \mathcal{A}_b := \mathrm{Top}_{K_b}(S_3(\cdot; \mathfrak{b}); \{r \in \mathcal{S}_3(\mathfrak{b}) : S_3(r; \mathfrak{b}) > 0\}).$$

When a branch-local rerank shell is judged tie-degenerate, the retained admission set may be replaced by a transient tie-shell selector.[11]

[11] For example, if $r_{b,\max} \in \arg\max_{r \in \mathcal{S}_3(\mathfrak{b})} S_3(r; \mathfrak{b})$, one may define $\mathcal{T}_{\mathrm{tie}}(\mathfrak{b}) := \{r \in \mathcal{S}_3(\mathfrak{b}) : S_3(r; \mathfrak{b}) \geq \max(\eta_{\mathrm{tie}} S_3(r_{b,\max}; \mathfrak{b}),\ S_3(r_{b,\max}; \mathfrak{b}) - \delta_{\mathrm{tie}})\}$ and replace $\mathcal{A}_b$ by a retained subset $\mathcal{A}_b^{\mathrm{tie}} \subseteq \mathcal{T}_{\mathrm{tie}}(\mathfrak{b})$ with $1 \leq |\mathcal{A}_b^{\mathrm{tie}}| \leq B_{\mathrm{tie}}(t_b)$, chosen by descending $S_3(\cdot; \mathfrak{b})$.

% [ARCHAIC/REVIEW: split-aware promotion machinery.] % Origin: §11.1 split family definition. % Status: removed from the main beam surface and retained only for provenance. % If reinstated, it should act on children of a single generator family at fixed position p, % not on unrelated branch children from different global generators.

(ii) Materialize either stop or one new admission. The proposal family is $\mathcal{Q}(b) = \{\text{stop}\} \cup \mathcal{A}_b$. The symbol stop means `terminate this scaffold as-is,'' while each $r=(m,p)\in\mathcal{A}_b$ means`insert one new generator into this branch and refit.'' If $\mathcal{O}_b = (m_{b,1}, \dots, m_{b,n_b})$ and $0 \le p \le n_b$, then the scaffold insertion map is $\mathrm{Insert}(\mathcal{O}_b, m, p) = (m_{b,1}, \dots, m_{b,p}, m, m_{b,p+1}, \dots, m_{b,n_b})$, the insertion index transport is $I_p(j) = \begin{cases} j, & 1 \le j \le p, \\ j+1, & p < j \le n_b, \end{cases}$, and the zero-lifted parameter vector is $\theta_b^{\uparrow(r)} = (\theta_{b,1}, \dots, \theta_{b,p}, 0, \theta_{b,p+1}, \dots, \theta_{b,n_b})$.

For contiguous inserted-window cap ω_{new}, let $\omega_{\text{eff}}^{(p)} := \max\{1, \min(\omega_{\text{new}}, n_b+1)\}$ and $\ell_p := \max\Big\{1, \min\Big(p+1-\Big\lfloor \frac{\omega_{\text{eff}}^{(p)}-1}{2} \Big\rfloor,\ n_b+2-$

Then $W_{\text{new}}^{(p)} = \{\ell_p,\ \ell_p+1,\ \dots,\ \ell_p+\omega_{\text{eff}}^{(p)}-1\}$, so $|W_{\text{new}}^{(p)}| = \omega_{\text{eff}}^{(p)}$ and $p+1 \in W_{\text{new}}^{(p)}$. Let the retained old-coordinate fraction shrink with maturity according to $\rho_W(t) := \rho_W^{\min} + (\rho_W^{\max} - \rho_W^{\min})e_t$, with $0 \le \rho_W^{\min} \le \rho_W^{\max} \le 1$, and define the old-coordinate complement by $C_{\text{old}}^{(p)} := \{1, \dots, n_b+1\} \quad W_{\text{new}}^{(p)}$, so $|C_{\text{old}}^{(p)}| = n_b + 1 - \omega_{\text{eff}}^{(p)}$. Then the retained old-coordinate count is $n_{\text{old}}(t,p) := \lceil \rho_W(t)\,|C_{\text{old}}^{(p)}| \rceil = \lceil \rho_W(t)\,(n_b+1-\omega_{\text{eff}}^{(p)}) \rceil$. For each current scaffold coordinate $j \in \{1, \dots, n_b\}$, let $a_{b,j}^{\text{adm}}$ denote the admission step at which that coordinate entered the current scaffold on branch b, with smaller values meaning older coordinates, and let the insertion-lifted admission-order vector be $a_b^{\uparrow(r)} := (a_{b,1}^{\text{adm}}, \dots, a_{b,p}^{\text{adm}}, +\infty, a_{b,p+1}^{\text{adm}}, \dots, a_{b,n_b}^{\text{adm}})$, so the newly inserted slot is never selected as an old coordinate. The retained old-coordinate window is then $W_{\text{age}}^{(p)} := \{\, j \in C_{\text{old}}^{(p)} : \#\{\, i \in C_{\text{old}}^{(p)} : (a_b^{\uparrow(r)})_i < (a_b^{\uparrow(r)})_j \,\} < n_{\text{old}}(t,p) \,\}$, namely the set of the $n_{\text{old}}(t,p)$ oldest old coordinates outside $W_{\text{new}}^{(p)}$, and hence the inherited refit window on the inserted scaffold is $W_r = W(r;t) := W_{\text{new}}^{(p)} \cup W_{\text{age}}^{(p)}$.

Then, for $r = (m,p)$ and $\mathcal{O}_b \oplus_p m := \mathrm{Insert}(\mathcal{O}_b, m, p)$, the branch-local refit map is $\mathrm{Refit}_{W_r}(\theta_b; r) := \arg\min_{\tilde\theta \in \Theta_{\mathcal{O}_b \oplus_p m}} \Big\{ E(\mathcal{O}_b \oplus_p m, \tilde\theta) : \ \tilde\theta_j = (\theta_b^{\uparrow(r)})_j \ \forall j \notin W_r \Big\}$.

The transfer map is $\mathcal{T}(\mathfrak{b}, \text{stop}) = \mathfrak{b}'$ with $\mathcal{H}_{b'} = \mathcal{H}_b$, $\mathcal{O}_{b'} = \mathcal{O}_b$, $\theta_{b'} = \theta_b$, $E_{b'} = E_b$, $d_{b'} = d_b$, $S_{b'}^{\text{cum}} = S_b^{\text{cum}}$, $K_{b'}^{\text{cum}} = K_b^{\text{cum}}$, $\text{status}_{b'} = \text{terminal}$, and $\text{term}_{b'} = \begin{cases} \text{empty}, & \mathcal{A}_b = \emptyset, \\ \text{stop}, & \mathcal{A}_b \neq \emptyset. \end{cases}$ For $r = (m,p)$, $\mathcal{T}(\mathfrak{b}, r) = \mathfrak{b}'$ with $\mathcal{O}_{b'} = \mathrm{Insert}(\mathcal{O}_b, m, p)$, $\theta_{b'} = \mathrm{Refit}_{W_r}(\theta_b; r)$, $\mathcal{H}_{b'} = (\mathcal{H}_b, r)$, $E_{b'} = E(\theta_{b'}; \mathcal{O}_{b'})$, $d_{b'} = d_b + 1$, $S_{b'}^{\text{cum}} = S_b^{\text{cum}} + S_3(r; \mathfrak{b})$, $K_{b'}^{\text{cum}} = K_b^{\text{cum}} + K_3(r; \mathfrak{b})$, $\text{status}_{b'} = \begin{cases} \text{terminal}, & d_{b'} \ge d_{\max} \text{ or } \mathcal{A}_{b'} = \emptyset, \\ \text{frontier}, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} \neq \emptyset, \end{cases}$, and $\text{term}_{b'} = \begin{cases} \text{depth_cap}, & d_{b'} \ge d_{\max}, \\ \text{empty}, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} = \emptyset, \\ \emptyset, & d_{b'} < d_{\max} \text{ and } \mathcal{A}_{b'} \neq \emptyset. \end{cases}$ where $\mathcal{A}_{b'}$ is recomputed from step (i) on the child branch itself. So a non-stop child is exactly one more ADAPT admission applied to the parent scaffold, followed by the inherited-window refit already defined above, and its cumulative K-coordinate is updated by the admitted branch-local Phase-3 selector burden $K_3(r; \mathfrak{b})$. If $\mathcal{A}_b = \emptyset$, then $\mathcal{Q}(b) = \{\text{stop}\}$ and the branch contributes only a terminal child.

(iii) Classify, deduplicate, and prune the scaffold children. The frontier and terminal descendants of branch b are $\mathfrak{C}_b^{\text{front}} = \{\mathfrak{b}' : \mathfrak{b}' = \mathcal{T}(\mathfrak{b}, r),\ r \in \mathcal{A}_b,\ \text{status}_{b'} = \text{frontier}\}$ and $\mathfrak{C}_b^{\text{term}} = \{\mathfrak{b}' : \mathfrak{b}' = \mathcal{T}(\mathfrak{b}, q),\ q \in \mathcal{Q}(b),\ \text{status}_{b'} = \text{terminal}\}$. Two branches are treated as duplicates when they represent the same scaffold and essentially the same refit point, for example under the fingerprint $\mathrm{rnd}_{10}(x) := 10^{-10}\,\mathrm{round}(10^{10}x)$, $\mathrm{round}_{10}(\theta_b) := (\mathrm{rnd}_{10}(\theta_{b,1}), \dots, \mathrm{rnd}_{10}(\theta_{b,n_b}))$, and $\mathrm{fp}(\mathfrak{b}) = (d_b, \mathrm{labels}(\mathcal{O}_b), \mathrm{round}_{10}(\theta_b))$. Among near-equivalent branches, keep the one with best lexicographic prune key $\pi_{\text{prune}}(\mathfrak{b}) = (E_b, -S_b^{\text{cum}}, K_b^{\text{cum}}, |\mathcal{O}_b|, \mathrm{labels}(\mathcal{O}_b), \mathrm{round}_{10}(\theta_b), \mathrm{id}_b)$.

The beam operators are therefore $\mathrm{Dedup}(\mathcal{X}) = \{\text{best branch per fingerprint under } \pi_{\text{prune}}\}$ and $\mathrm{Prune}(\mathcal{X}; B) = \mathrm{Top}_B^{\min}(\pi_{\text{prune}}; \mathrm{Dedup}(\mathcal{X}))$. Hence the live and terminal pools update by $\mathfrak{F}_{c+1} = \mathrm{Prune}\Big(\bigcup_{b \in \mathfrak{F}_c} \mathfrak{C}_b^{\text{front}}; B_{\text{live}}\Big)$ and $\mathfrak{T}_{c+1} = \mathrm{Prune}\Big(\mathfrak{T}_c \cup \bigcup_{b \in \mathfrak{F}_c} \mathfrak{C}_b^{\text{term}}; B_{\text{term}}\Big)$.

(iv) Choose the final scaffold branch. After C scaffold-beam rounds, the surviving candidate set is $\mathfrak{W}_{\text{final}} = \mathfrak{F}_C \cup \mathfrak{T}_C$, and the effective beam selector is $\mathfrak{b}_* = \arg\min_{\mathfrak{b} \in \mathfrak{W}_{\text{final}}} \pi_{\text{prune}}(\mathfrak{b})$.

A compact scaffold-beam algorithm block is therefore

$$
\begin{aligned}
&\text{A: } \mathfrak{F}_0 = \{\mathfrak{b}_{\text{root}}\}, \qquad \mathfrak{T}_0 = \emptyset,\\
&\text{B: } \mathcal{R}_b^{\text{raw}} \to \mathcal{S}_3(\mathfrak{b}) \to \mathcal{A}_b \qquad (\mathfrak{b} \in \mathfrak{F}_c),\\
&\text{C: } \mathcal{Q}(b) = \{\text{stop}\} \cup \mathcal{A}_b, \qquad \mathfrak{C}_b = \{\mathcal{T}(\mathfrak{b}, q) : q \in \mathcal{Q}(b)\},\\
&\text{D: } \mathfrak{C}_b^{\text{front}} = \{\mathfrak{b}' \in \mathfrak{C}_b : \text{status}_{b'} = \text{frontier}\}, \qquad \mathfrak{C}_b^{\text{term}} = \{\mathfrak{b}' \in \mathfrak{C}_b : \text{status}_{b'} = \text{terminal}\},\\
&\text{E: } \mathfrak{F}_{c+1} = \text{Prune}\left(\bigcup_{b \in \mathfrak{F}_c} \mathfrak{C}_b^{\text{front}}; B_{\text{live}}\right), \qquad \mathfrak{T}_{c+1} = \text{Prune}\left(\mathfrak{T}_c \cup \bigcup_{b \in \mathfrak{F}_c} \mathfrak{C}_b^{\text{term}}; B_{\text{term}}\right),\\
&\text{F: } \mathfrak{W}_{\text{final}} = \mathfrak{F}_C \cup \mathfrak{T}_C, \qquad \mathfrak{b}_* = \arg\min_{\mathfrak{b} \in \mathfrak{W}_{\text{final}}} \pi_{\text{prune}}(\mathfrak{b}).
\end{aligned}
$$

11.6 Active HH pool surface and selected scaffold Identify the nested active HH pool with the retained generator images by $\Omega_{\text{HH}}^{(k)} := \mathcal{G}^{(k)}$ for $k \in \{1, 2, 3\}$. The active HH adaptive surface is therefore carried by $\Omega_{\text{HH}}^{(3)} \subseteq \Omega_{\text{HH}}^{(2)} \subseteq \Omega_{\text{HH}}^{(1)}$, with $\mathcal{G}_{\text{adapt}}^{(k)} = \{\tau_m\}_{m \in \Omega_{\text{HH}}^{(k)}}$.

If beam continuation is used, set $\mathcal{O}_* := \mathcal{O}_{\mathfrak{b}_*}$ and $\theta_*^{\text{adapt}} := \theta_{\mathfrak{b}_*}$; otherwise $(\mathcal{O}_*, \theta_*^{\text{adapt}})$ denotes the final admitted scaffold and its refit amplitudes on the main branch. The resulting state is $|\psi_*\rangle = U(\theta_*^{\text{adapt}}; \mathcal{O}_*)|\phi_0\rangle$, so the active HH scaffold manifold selected by ADAPT may be written as $\mathcal{M}_{\text{scaf}}(\mathcal{O}_*) = \{\, U(\vartheta; \mathcal{O}_*)|\phi_0\rangle : \vartheta \in \mathbb{R}^{|\mathcal{O}_*|} \,\}$.

12. SPSA and Optimizer Semantics

12.1 SPSA schedules

The standard SPSA schedules are

$$
c_k = \frac{c}{(k+1)^\gamma}, \qquad a_k = \frac{a}{(A+k+1)^\alpha}.
$$

12.2 Two-point stochastic gradient

At iteration k, with Rademacher direction Δ_k, evaluate

$$
y_+ = f(x_k + c_k \Delta_k), \qquad y_- = f(x_k - c_k \Delta_k),
$$

and form the gradient estimate

$$
\hat{g}_k = \frac{y_+ - y_-}{2c_k}\, \Delta_k.
$$

12.3 Update and projection

The update is

$$
x_{k+1} = x_k - a_k \hat{g}_k.
$$

If projection or clipping is imposed, it is applied to both the perturbed points and the updated iterate.

12.4 Repeat aggregation

If the same point is sampled multiple times, the repeated evaluations may be aggregated by

$$
\text{mean} \qquad \text{or} \qquad \text{median}\,.
$$

12.5 Return policy

Two standard return policies are:

- return an averaged late-trajectory iterate,
- return the best sampled point observed during the run.

12.6 Surface note

Stochastic inner optimizers are especially natural for large, noisy, or nondifferentiable effective surfaces, while deterministic optimizers are often effective on smaller and smoother fixed-structure manifolds.

17A. Projective McLachlan Real-Time Dynamics and Exact-Forecast Checkpoint-Adaptive Snake Control

The canonical real-time law is piecewise smooth in physical time and piecewise discrete in scaffold structure. On each open checkpoint interval the scaffold is frozen and the state follows the projective McLachlan projection of the sampled Schrödinger drift. At the checkpoint itself one chooses exactly one structural action: keep the scaffold, append one new ordered block, or delete one old block. In what follows, each object is defined once, the main quantities are expanded linearly down to the operator/expectation-value primitives, and the condensed shorthand is given only after the fully expanded form has already appeared.

The driven $L = 2$ controller printed at the front of this manuscript is one calibrated instance of the law below, but the formulas are written for a general scaffold length, a general driven Hamiltonian, a general finite forecast horizon, and a general left/midpoint/right drive-sampling rule.

17A.1 Fixed-scaffold projective McLachlan geometry

Here $n_k := |\mathcal{O}_k|$ is the current scaffold length, $d := \dim \mathcal{H}$ is the Hilbert-space dimension, $a \in \{1, \dots, d\}$ is the Hilbert-space row index, and $i, j \in \{1, \dots, n_k\}$ are scaffold-coordinate indices. The quantities introduced in this subsection are the scaffold state, its projected tangents, the projected Schrödinger drift, the tangent metric, the force vector, the regularized normal matrix, the quadratic McLachlan fit, and the projected velocity on the frozen scaffold.

$$
\begin{aligned}
|\psi_k\rangle &:= |\psi(\theta_k; \mathcal{O}_k)\rangle = U(\theta_k; \mathcal{O}_k)|\psi_{\mathrm{ref}}\rangle, \qquad U(\theta_k; \mathcal{O}_k) := \prod_{j=n_k}^{1} e^{-i\theta_{k,j} A_{k,j}}, \\
U_{k,>j} &:= \prod_{\ell=n_k}^{j+1} e^{-i\theta_{k,\ell} A_{k,\ell}}, \qquad U_{k,<j} := \prod_{\ell=j-1}^{1} e^{-i\theta_{k,\ell} A_{k,\ell}}, \qquad \widetilde{A}_{k,j} := U_{k,>j} A_{k,j} U_{k,>j}^{\dagger}, \\
\tau_{k,j} &:= \partial_{\theta_{k,j}} |\psi_k\rangle = -iU_{k,>j} A_{k,j} U_{k,<j} |\psi_{\mathrm{ref}}\rangle = -i\widetilde{A}_{k,j}|\psi_k\rangle, \\
\bar{\tau}_{k,j} &:= \tau_{k,j} - \langle \psi_k, \tau_{k,j}\rangle |\psi_k\rangle = Q_{\psi_k} \tau_{k,j} = -i(\widetilde{A}_{k,j} - \langle \widetilde{A}_{k,j}\rangle_{\psi_k})|\psi_k\rangle, \qquad Q_{\psi_k} := I - |\psi_k\rangle\langle\psi_k|, \\
\bar{T}_k &:= \left[(\bar{T}_k)_{aj}\right]_{a=1,\dots,d}^{j=1,\dots,n_k} = \left[(\bar{\tau}_{k,j})_a\right]_{a=1,\dots,d}^{j=1,\dots,n_k} \\
&= \begin{bmatrix} -i((\widetilde{A}_{k,1} - \langle \widetilde{A}_{k,1}\rangle_{\psi_k})\psi_k)_1 & \cdots & -i((\widetilde{A}_{k,n_k} - \langle \widetilde{A}_{k,n_k}\rangle_{\psi_k})\psi_k)_1 \\ \vdots & \ddots & \vdots \\ -i((\widetilde{A}_{k,1} - \langle \widetilde{A}_{k,1}\rangle_{\psi_k})\psi_k)_d & \cdots & -i((\widetilde{A}_{k,n_k} - \langle \widetilde{A}_{k,n_k}\rangle_{\psi_k})\psi_k)_d \end{bmatrix}, \\
\bar{b}_k &:= Q_{\psi_k}(-iH_k^{\sharp}|\psi_k\rangle) = -i(H_k^{\sharp} - E_{\psi_k,k}^{\sharp})|\psi_k\rangle, \qquad E_{\psi_k,k}^{\sharp} := \langle\psi_k|H_k^{\sharp}|\psi_k\rangle, \\
&= \begin{bmatrix} (-i(H_k^{\sharp} - E_{\psi_k,k}^{\sharp})\psi_k)_1 \\ \vdots \\ (-i(H_k^{\sharp} - E_{\psi_k,k}^{\sharp})\psi_k)_d \end{bmatrix}.
\end{aligned}
$$

The metric, force, regularized system, quadratic fit, and projected velocity are then defined in one chain:

$$
\begin{aligned}
G_k &:= \left[(G_k)_{ij}\right]_{i,j=1}^{n_k} = \Re(\bar{T}_k^\dagger \bar{T}_k) = \left[\Re\left(\sum_{a=1}^{d} (\bar{T}_k)_{ai}^* (\bar{T}_k)_{aj}\right)\right]_{i,j=1}^{n_k} \\
&= \left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}\widetilde{A}_{k,j}|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}\langle\widetilde{A}_{k,j}\rangle_{\psi_k}\Big)\right]_{i,j=1}^{n_k}, \\
f_k &:= \left[(f_k)_i\right]_{i=1}^{n_k} = \Re(\bar{T}_k^\dagger \bar{b}_k) = \left[\Re\left(\sum_{a=1}^{d} (\bar{T}_k)_{ai}^* (\bar{b}_k)_a\right)\right]_{i=1}^{n_k} \\
&= \left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}H_k^\sharp|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}E_{\psi_k,k}^\sharp\Big)\right]_{i=1}^{n_k}, \\
K_k &:= \left[(K_k)_{ij}\right]_{i,j=1}^{n_k} = G_k + \Lambda_k = \left[(G_k)_{ij} + (\Lambda_k)_{ij}\right]_{i,j=1}^{n_k} \\
&= \left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}\widetilde{A}_{k,j}|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}\langle\widetilde{A}_{k,j}\rangle_{\psi_k}\Big) + (\Lambda_k)_{ij}\right]_{i,j=1}^{n_k}, \\
\mathfrak{m}_k(v) &:= \|\bar{T}_k v - \bar{b}_k\|_2^2 + v^\top \Lambda_k v \\
&= \left\|\sum_{j=1}^{n_k} v_j[-i(\widetilde{A}_{k,j} - \langle\widetilde{A}_{k,j}\rangle_{\psi_k})|\psi_k\rangle] + i(H_k^\sharp - E_{\psi_k,k}^\sharp)|\psi_k\rangle\right\|_2^2 \\
&= \sum_{i,j=1}^{n_k} v_i (K_k)_{ij} v_j - 2\sum_{i=1}^{n_k} v_i (f_k)_i + \langle\psi_k|(H_k^\sharp - E_{\psi_k,k}^\sharp)^2|\psi_k\rangle, \\
\mathcal{Q}_k(v) &:= 2v^\top f_k - v^\top G_k v \\
&= 2\Re\left\langle \sum_{j=1}^{n_k} v_j[-i(\widetilde{A}_{k,j} - \langle\widetilde{A}_{k,j}\rangle_{\psi_k})|\psi_k\rangle], -i(H_k^\sharp - E_{\psi_k,k}^\sharp)|\psi_k\rangle \right\rangle \\
&\quad - \left\|\sum_{j=1}^{n_k} v_j[-i(\widetilde{A}_{k,j} - \langle\widetilde{A}_{k,j}\rangle_{\psi_k})|\psi_k\rangle]\right\|_2^2, \\
\dot{\theta}_k &:= K_k^+ f_k \\
&= \left(\left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}\widetilde{A}_{k,j}|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}\langle\widetilde{A}_{k,j}\rangle_{\psi_k}\Big) + (\Lambda_k)_{ij}\right]_{i,j=1}^{n_k}\right)^+ \left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}H_k^\sharp|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}E_{\psi_k,k}^\sharp\Big)\right]_{i=1}^{n_k}, \\
\mathrm{Var}_{\psi_k}(H_k^\sharp) &:= \|\bar{b}_k\|_2^2 = \langle\psi_k|(H_k^\sharp - E_{\psi_k,k}^\sharp)^2|\psi_k\rangle.
\end{aligned}
$$

Equivalently, the regularized fixed-scaffold McLachlan solve may be written in condensed form as $K_k\dot{\theta}_k = f_k$, while the intrinsic unregularized projective optimum is $\dot{\theta}_k^{\mathrm{proj}} := G_k^+ f_k$ with $\mathcal{Q}_k(\dot{\theta}_k^{\mathrm{proj}}) = f_k^\top G_k^+ f_k$.

17A.2 Piecewise-constant driven Hamiltonian on each checkpoint interval

Here H_{stat} is the static Pauli polynomial, $H_{\mathrm{drv}}(t)$ is the driven Pauli polynomial, $I_k := [\tau_k, \tau_{k+1}]$ is the kth checkpoint interval, $\Delta\tau_k := \tau_{k+1} - \tau_k$, $M_k \geq 1$ is the exact-reference micro-step count, and $\sigma_k \in \{0, \frac{1}{2}, 1\}$ selects left, midpoint, or right time sampling. The driven Hamiltonian is kept simple and inline:

$H(t) := H_{\mathrm{stat}} + H_{\mathrm{drv}}(t) = \sum_{\alpha\in\mathcal{J}_{\mathrm{stat}}} h_\alpha P_\alpha + \sum_{\beta\in\mathcal{J}_{\mathrm{drv}}} c_\beta(t) P_\beta$, $t_k^\sharp := t_0 + \tau_k + \sigma_k\Delta\tau_k$, and $H_k^\sharp := H(t_k^\sharp) = \sum_{\alpha\in\mathcal{J}_{\mathrm{stat}}} h_\alpha P_\alpha + \sum_{\beta\in\mathcal{J}_{\mathrm{drv}}} c_\beta(t_k^\sharp) P_\beta$.

With this sampled Hamiltonian, the force entries of §17A.1 become $(f_k)_i = \sum_{\alpha\in\mathcal{J}_{\mathrm{stat}}} h_\alpha \Re\langle\bar{\tau}_{k,i}, -iP_\alpha\psi_k\rangle + \sum_{\beta\in\mathcal{J}_{\mathrm{drv}}} c_\beta(t_k^\sharp)\Re\langle\bar{\tau}_{k,i}, -iP_\beta\psi_k\rangle$, while G_k is unchanged because it depends only on tangent overlaps. For the refined exact reference path one sets $\delta\tau_k := \Delta\tau_k/M_k$, $t_{k,\mu}^\sharp := t_0 + \tau_k + (\mu + \sigma_k)\delta\tau_k$, $H_{k,\mu}^\sharp := H(t_{k,\mu}^\sharp)$, and $|\psi_{k,\mu+1}^{\mathrm{ex}}\rangle := \exp(-i\delta\tau_k H_{k,\mu}^\sharp)|\psi_{k,\mu}^{\mathrm{ex}}\rangle$. Thus midpoint sampling is simply the special case $\sigma_k = \frac{1}{2}$.

17A.3 Residual geometry and local inadequacy of the scaffold

This subsection measures how much of the sampled horizontal Schrödinger drift is not captured by the current scaffold tangent plane. The main quantities are the regularized residual $\bar{r}_k$, the realized step defect $\epsilon_{\mathrm{step},k}$, the intrinsic projective defect $\epsilon_{\mathrm{proj},k}$, and the normalized miss ratio $\rho_{\mathrm{miss},k}$.

$$
\begin{aligned}
\bar{r}_k &:= \bar{T}_k \dot{\theta}_k - \bar{b}_k \\
&= \sum_{j=1}^{n_k} (\dot{\theta}_k)_j [-i(\widetilde{A}_{k,j} - \langle \widetilde{A}_{k,j} \rangle_{\psi_k})|\psi_k\rangle] + i(H_k^\sharp - E^\sharp_{\psi_k,k})|\psi_k\rangle, \\
\epsilon^2_{\mathrm{step},k} &:= \|\bar{r}_k\|_2^2 \\
&= \left\| \sum_{j=1}^{n_k} (\dot{\theta}_k)_j [-i(\widetilde{A}_{k,j} - \langle \widetilde{A}_{k,j} \rangle_{\psi_k})|\psi_k\rangle] + i(H_k^\sharp - E^\sharp_{\psi_k,k})|\psi_k\rangle \right\|_2^2, \\
\epsilon^2_{\mathrm{proj},k} &:= \min_{v \in \mathbb{R}^{n_k}} \|\bar{T}_k v - \bar{b}_k\|_2^2 \\
&= \mathrm{Var}_{\psi_k}(H_k^\sharp) - \sum_{i,j=1}^{n_k} (f_k)_i (G_k^+)_{ij} (f_k)_j \\
&= \langle \psi_k | (H_k^\sharp - E^\sharp_{\psi_k,k})^2 | \psi_k \rangle - \sum_{i,j=1}^{n_k} \Re\Big(\langle \psi_k | \widetilde{A}_{k,i} H_k^\sharp | \psi_k \rangle - \langle \widetilde{A}_{k,i} \rangle_{\psi_k} E^\sharp_{\psi_k,k} \Big) (G_k^+)_{ij} \\
&\qquad \times \Re\Big(\langle \psi_k | \widetilde{A}_{k,j} H_k^\sharp | \psi_k \rangle - \langle \widetilde{A}_{k,j} \rangle_{\psi_k} E^\sharp_{\psi_k,k} \Big), \\
\rho_{\mathrm{miss},k} &:= \frac{\epsilon^2_{\mathrm{proj},k}}{\mathrm{Var}_{\psi_k}(H_k^\sharp) + \varepsilon} \\
&= \frac{\langle \psi_k | (H_k^\sharp - E^\sharp_{\psi_k,k})^2 | \psi_k \rangle - \sum_{i,j=1}^{n_k} \Re\Big(\langle \psi_k | \widetilde{A}_{k,i} H_k^\sharp | \psi_k \rangle - \langle \widetilde{A}_{k,i} \rangle_{\psi_k} E^\sharp_{\psi_k,k} \Big) (G_k^+)_{ij} \Re\Big(\langle \psi_k | \widetilde{A}_{k,j} H_k^\sharp | \psi_k \rangle - \langle \widetilde{A}_{k,j} \rangle_{\psi_k} E^\sharp_{\psi_k,k} \Big)}{\langle \psi_k | (H_k^\sharp - E^\sharp_{\psi_k,k})^2 | \psi_k \rangle + \varepsilon}.
\end{aligned}
$$

After this fully expanded form one may use the shorthand $\rho_{\mathrm{miss},k}$ alone.

17A.4 Checkpoint-local decision surfaces and calmness gate

The projected differential law determines motion only while the scaffold is frozen; it does not decide whether the scaffold itself should change. The checkpoint law does. Here $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$ are coordinate vectors of possibly different lengths, $r := \min(m, n)$ is their overlap length, $\dot{\theta}^{\mathrm{acc}}_{k-1}$ is the previously accepted checkpoint velocity after zero padding to the current coordinate length, and $\tau^{\mathrm{cos}}_{\mathrm{calm}}, \tau^{\mathrm{rate}}_{\mathrm{calm}}, \tau_{\mathrm{miss}}, \tau_{\mathrm{pr}} > 0$ are the calmness and miss thresholds.

$$
\begin{aligned}
\|x - y\|^2_{\mathrm{ov}} &:= \sum_{j=1}^{r} (x_j - y_j)^2 + \sum_{j=r+1}^{m} x_j^2 + \sum_{j=r+1}^{n} y_j^2, \\
c_k^{\mathrm{dir}} &:= \frac{\langle \dot{\theta}_k, \dot{\theta}^{\mathrm{acc}}_{k-1} \rangle}{\max\{\|\dot{\theta}_k\|_2 \, \|\dot{\theta}^{\mathrm{acc}}_{k-1}\|_2, \varepsilon\}}, \qquad \eta_k^{\mathrm{rate}} := \frac{\|\dot{\theta}_k - \dot{\theta}^{\mathrm{acc}}_{k-1}\|_{\mathrm{ov}}}{\max\{\|\dot{\theta}_k\|_2, \|\dot{\theta}^{\mathrm{acc}}_{k-1}\|_2, \varepsilon\}}, \\
\mathfrak{C}_k &:= \mathbf{1}[c_k^{\mathrm{dir}} \geq \tau^{\mathrm{cos}}_{\mathrm{calm}}] \, \mathbf{1}[\eta_k^{\mathrm{rate}} \leq \tau^{\mathrm{rate}}_{\mathrm{calm}}], \\
\mathcal{U}_k^{\mathrm{stay}} &:= \{(\mathrm{stay}, p, a) : p \in \mathcal{P}_k^{\mathrm{stay}}, \ a \in \mathcal{A}_k\}, \qquad \mathcal{U}_k^{\mathrm{app}} := \{(\mathrm{append}, r, a) : r \in \mathcal{R}_k^{\mathrm{conf}}, \ a \in \mathcal{A}_k\}, \\
\nu_k^\star &:= \begin{cases} \arg \min_{u \in \mathcal{U}_k^{\mathrm{stay}} \cup \mathcal{U}_k^{\mathrm{app}}} \mathcal{J}_k^{\mathrm{forecast}}(u), & \rho_{\mathrm{miss},k} > \tau_{\mathrm{miss}}, \\ \text{best admissible prune of §17A.9 or stay}, & \rho_{\mathrm{miss},k} \leq \tau_{\mathrm{pr}} \text{ and } \mathfrak{C}_k = 1, \\ \mathrm{stay}, & \text{otherwise.} \end{cases}
\end{aligned}
$$

Thus the miss-dominant regime compares stay and append through the forecast score, the low-miss calm regime opens prune, and all other cases reduce to stay.

17A.5 Position-aware append geometry and confirm-grade gain

An append move is a record $r = (m, p)$ rather than a bare generator because the new block is inserted at a definite ordered position. Let m_r denote the number of new coordinates carried by the appended block, let $\eta = (\eta_1, \dots, \eta_{m_r})$ denote those new coordinates, let $\vartheta = (\theta, \eta)$ denote the augmented coordinate vector, and let $\eta = 0$ at the append instant. The physical state therefore does not jump at the checkpoint, but the tangent plane acquires new columns. The append gain comes from solving the augmented McLachlan normal system and eliminating the old coordinates exactly.

$$
\begin{aligned}
\bar{U}_{r,k} &:= \left[(\bar{U}_{r,k})_{sa}\right]_{s=1,\dots,d}^{a=1,\dots,m_r} = \left[(Q_{\psi_k}\partial_{\eta_a}|\psi(\theta_k,\eta;\mathcal{O}_k^+(r))\rangle|_{\eta=0})_s\right]_{s=1,\dots,d}^{a=1,\dots,m_r} \\
&= \begin{bmatrix} (\bar{u}_{r,1})_1 & \cdots & (\bar{u}_{r,m_r})_1 \\ \vdots & \ddots & \vdots \\ (\bar{u}_{r,1})_d & \cdots & (\bar{u}_{r,m_r})_d \end{bmatrix}, \\
B_{r,k} &:= \left[(B_{r,k})_{ia}\right]_{i=1,\dots,n_k}^{a=1,\dots,m_r} = \Re(\bar{T}_k^\dagger \bar{U}_{r,k}) = \left[\Re\left(\sum_{s=1}^{d}(\bar{T}_k)_{si}^*(\bar{U}_{r,k})_{sa}\right)\right]_{i=1,\dots,n_k}^{a=1,\dots,m_r}, \\
C_{r,k} &:= \left[(C_{r,k})_{ab}\right]_{a,b=1}^{m_r} = \Re(\bar{U}_{r,k}^\dagger \bar{U}_{r,k}) = \left[\Re\left(\sum_{s=1}^{d}(\bar{U}_{r,k})_{sa}^*(\bar{U}_{r,k})_{sb}\right)\right]_{a,b=1}^{m_r}, \\
q_{r,k} &:= \left[(q_{r,k})_a\right]_{a=1}^{m_r} = \Re(\bar{U}_{r,k}^\dagger \bar{b}_k) = \left[\Re\left(\sum_{s=1}^{d}(\bar{U}_{r,k})_{sa}^*(\bar{b}_k)_s\right)\right]_{a=1}^{m_r}, \\
\mathfrak{m}_{r,k}(v,\xi) &:= \|\bar{T}_k v + \bar{U}_{r,k}\xi - \bar{b}_k\|_2^2 + v^\top \Lambda_k v + \xi^\top \Lambda_r \xi \\
&= \begin{bmatrix} v \\ \xi \end{bmatrix}^\top \begin{bmatrix} K_k & B_{r,k} \\ B_{r,k}^\top & C_{r,k} + \Lambda_r \end{bmatrix} \begin{bmatrix} v \\ \xi \end{bmatrix} - 2\begin{bmatrix} v \\ \xi \end{bmatrix}^\top \begin{bmatrix} f_k \\ q_{r,k} \end{bmatrix} + \langle \psi_k | (H_k^\sharp - E_{\psi_k,k}^\sharp)^2 | \psi_k \rangle, \\
\begin{bmatrix} \dot{\theta}_{r,k}^{\mathrm{old}} \\ \dot{\eta}_{r,k} \end{bmatrix} &:= \begin{bmatrix} K_k & B_{r,k} \\ B_{r,k}^\top & C_{r,k} + \Lambda_r \end{bmatrix}^+ \begin{bmatrix} f_k \\ q_{r,k} \end{bmatrix}, \qquad v^\star(\xi) := K_k^+(f_k - B_{r,k}\xi), \\
\mathfrak{m}_{r,k}(v^\star(\xi),\xi) &= \langle \psi_k | (H_k^\sharp - E_{\psi_k,k}^\sharp)^2 | \psi_k \rangle - f_k^\top K_k^+ f_k + \xi^\top S_{r,k}\xi - 2\xi^\top w_{r,k}, \\
w_{r,k} &:= q_{r,k} - B_{r,k}^\top K_k^+ f_k \\
&= \left[\Re\left(\sum_{s=1}^{d}(\bar{U}_{r,k})_{sa}^*(\bar{b}_k)_s\right) - \sum_{i=1}^{n_k}(B_{r,k})_{ia}(K_k^+ f_k)_i\right]_{a=1}^{m_r}, \\
S_{r,k} &:= C_{r,k} + \Lambda_r - B_{r,k}^\top K_k^+ B_{r,k} \\
&= \left[(C_{r,k})_{ab} + (\Lambda_r)_{ab} - \sum_{i,j=1}^{n_k}(B_{r,k})_{ia}(K_k^+)_{ij}(B_{r,k})_{jb}\right]_{a,b=1}^{m_r}, \\
\Delta_{r,k}^{\mathrm{exact}} &:= w_{r,k}^\top S_{r,k}^+ w_{r,k} \\
&= (q_{r,k} - B_{r,k}^\top K_k^+ f_k)^\top (C_{r,k} + \Lambda_r - B_{r,k}^\top K_k^+ B_{r,k})^+ (q_{r,k} - B_{r,k}^\top K_k^+ f_k), \\
\rho_{r,k}^{\mathrm{gain}} &:= \frac{\Delta_{r,k}^{\mathrm{exact}}}{\langle \psi_k | (H_k^\sharp - E_{\psi_k,k}^\sharp)^2 | \psi_k \rangle + \varepsilon}.
\end{aligned}
$$

The confirm-grade compressed path keeps the same append object and resolves only the old-coordinate contribution on a retained spectral support. If $K_k = V_k \Sigma_k^2 V_k^\top$ is the supported factorization, $z_k := \Sigma_k^{-1} V_k^\top f_k$, and $\Gamma_{r,k} := \Sigma_k^{-1} V_k^\top B_{r,k}$, then $w_{r,k} = q_{r,k} - \Gamma_{r,k}^\top z_k$, the compressed Schur block on a retained mode set J_r is $\widetilde{S}_{r,k}(J_r) :=$

$C_{r,k} + \Lambda_r - \Gamma_{r,k;J_r}^\top \Gamma_{r,k;J_r}$, the compressed gain is $\widetilde{\Delta}_{r,k}(J_r) := (q_{r,k} - \Gamma_{r,k}^\top z_k)^\top \widetilde{S}_{r,k}(J_r)^+ (q_{r,k} - \Gamma_{r,k}^\top z_k)$, and with direction penalty $\delta_{r,k}^{\mathrm{dir}} := \|\dot{\vartheta}_{r,k}^{\mathrm{aug}} - \dot{\theta}_{k-1}^{\mathrm{acc}}\|_{\mathrm{ov}}$ the confirm score is

$$S_{r,k}^{\mathrm{conf}} := \begin{cases} \dfrac{(q_{r,k} - B_{r,k}^\top K_k^+ f_k)^\top (C_{r,k} + \Lambda_r - B_{r,k}^\top K_k^+ B_{r,k})^+ (q_{r,k} - B_{r,k}^\top K_k^+ f_k)}{\langle \psi_k | (H_k^\sharp - E_{\psi_k,k}^\sharp)^2 | \psi_k \rangle + \varepsilon} - \lambda_{\mathrm{dir}}\, \delta_{r,k}^{\mathrm{dir}} - \lambda_{\mathrm{meas}} g_{r,k}^{\mathrm{new}}, & \text{exact confirm,} \\ \dfrac{(q_{r,k} - \Gamma_{r,k}^\top z_k)^\top \widetilde{S}_{r,k}(J_r)^+ (q_{r,k} - \Gamma_{r,k}^\top z_k)}{\langle \psi_k | (H_k^\sharp - E_{\psi_k,k}^\sharp)^2 | \psi_k \rangle + \varepsilon} - \lambda_{\mathrm{dir}}\, \delta_{r,k}^{\mathrm{dir}} - \lambda_{\mathrm{meas}} g_{r,k}^{\mathrm{new}}, & \text{compressed confirm.} \end{cases}$$

Thus the confirmed append shortlist is $\mathcal{R}_k^{\mathrm{conf}} := \{r : \rho_{r,k}^{\mathrm{gain}} \geq \tau_{\mathrm{gain}},\ S_{r,k}^{\mathrm{conf}} \geq \tau_{\mathrm{conf}}\}$.

17A.6 Proposal family for within-scaffold motion

A stay action is a finite family of same-scaffold forecast proposals, not a single velocity. Here M_k is the proposal metric, $I_k^{\mathrm{drv}} \subseteq \{1, \dots, n_k\}$ is the drive-aligned coordinate block, $\mathcal{B}_k$ is the finite blend-weight set, and $\mathcal{A}_k = \{a_{k,1}, \dots, a_{k,m_k}\} \subset (0, \infty)$ is the finite step-scale set.

$$M_k := \begin{cases} G_k, & \text{when the tangent metric is used on its supported subspace,} \\ I, & \text{otherwise,} \end{cases} \qquad \|v\|_{M_k} := \sqrt{v^\top M_k v}, \qquad \langle x, y \rangle_{M_k} := x^\top M_k y,$$

$$v_k^{\mathrm{McL}} := \dot{\theta}_k,$$

$$v_{k,\mathrm{drv}}^{\mathrm{cur}} := \text{the unique vector with } v_{k,\mathrm{drv}}^{\mathrm{cur}}[I_k^{\mathrm{drv}}] = K_k[I_k^{\mathrm{drv}}, I_k^{\mathrm{drv}}]^+ f_k[I_k^{\mathrm{drv}}] \text{ and } v_{k,\mathrm{drv}}^{\mathrm{cur}}[(I_k^{\mathrm{drv}})^c] = 0,$$

$$v_{k,\mathrm{drv}}^{\mathrm{next}} := \text{the unique vector with } v_{k,\mathrm{drv}}^{\mathrm{next}}[I_k^{\mathrm{drv}}] = K_{k+1}^\sharp[I_k^{\mathrm{drv}}, I_k^{\mathrm{drv}}]^+ f_{k+1}^\sharp[I_k^{\mathrm{drv}}] \text{ and } v_{k,\mathrm{drv}}^{\mathrm{next}}[(I_k^{\mathrm{drv}})^c] = 0,$$

$$d_k^\perp := d_k - \frac{\langle v_k^{\mathrm{McL}}, d_k \rangle_{M_k}}{\|v_k^{\mathrm{McL}}\|_{M_k}^2} v_k^{\mathrm{McL}}, \qquad \tilde{d}_k^\perp := \sqrt{\frac{\|v_k^{\mathrm{McL}}\|_{M_k}^2}{\|d_k^\perp\|_{M_k}^2}}\, d_k^\perp \text{ when } d_k^\perp \neq 0,$$

$$v_{k,\beta}^{\mathrm{blend}} := \sqrt{\frac{\|v_k^{\mathrm{McL}}\|_{M_k}^2}{\|v_k^{\mathrm{McL}} + \beta \tilde{d}_k^\perp\|_{M_k}^2}}\, (v_k^{\mathrm{McL}} + \beta \tilde{d}_k^\perp), \qquad \beta \in \mathcal{B}_k,$$

$$\hat{v} := v / \|v\|_{M_k} \text{ for } v \neq 0, \qquad \mathcal{P}_k^{\mathrm{stay}} \subseteq \{\hat{v}_k^{\mathrm{McL}}, \hat{v}_{k,\mathrm{drv}}^{\mathrm{cur}}, \hat{v}_{k,\mathrm{drv}}^{\mathrm{next}}, \hat{v}_{k,\beta}^{\mathrm{blend}}, \hat{v}_k^{\mathrm{sec}}\},$$

$$\mathcal{U}_k^{\mathrm{stay}} := \{(\mathrm{stay}, p, a) : p \in \mathcal{P}_k^{\mathrm{stay}},\ a \in \mathcal{A}_k\}.$$

Thus every stay action is already written as an explicit proposal direction together with an explicit discrete scale.

17A.7 Tangent-secant forecast proposal and signed-energy-lead taper

The tangent-secant proposal is the tangent inverse image of the phase-aligned exact one-step secant displacement, followed by a metric trust clip and a signed-energy-lead taper. Here $R_k > 0$ is the trust radius and $\eta_E > 0$ is the relative energy-lead cap.

$$
\begin{aligned}
|\widetilde{\psi}^{\mathrm{ex}}_{k+1}\rangle &:= \exp(-i\arg\langle\psi_k,\psi^{\mathrm{ex}}_{k+1}\rangle)|\psi^{\mathrm{ex}}_{k+1}\rangle,\\
\delta^{\mathrm{sec}}_k &:= Q_{\psi_k}(|\widetilde{\psi}^{\mathrm{ex}}_{k+1}\rangle - |\psi_k\rangle) = |\widetilde{\psi}^{\mathrm{ex}}_{k+1}\rangle - |\psi_k\rangle - \langle\psi_k,\widetilde{\psi}^{\mathrm{ex}}_{k+1}-\psi_k\rangle|\psi_k\rangle,\\
\xi^{\mathrm{sec}}_k &:= \mathfrak{R}(\bar{T}_k^\dagger\delta^{\mathrm{sec}}_k) = \left[\mathfrak{R}\left(\sum_{a=1}^{d}(\bar{T}_k)^*_{aj}(\delta^{\mathrm{sec}}_k)_a\right)\right]_{j=1}^{n_k},\\
v^{\mathrm{sec}}_k &:= \frac{1}{\Delta\tau_k}K_k^+\xi^{\mathrm{sec}}_k, \qquad v^{\mathrm{sec}}_k \leftarrow \min\left\{1,\frac{R_k}{\|v^{\mathrm{sec}}_k\|_{M_k}}\right\}v^{\mathrm{sec}}_k,\\
b^E_k &:= E^{\mathrm{ctrl}}_k - E^{\mathrm{ex}}_k, \qquad \Delta E^{\mathrm{ex}}_k := E^{\mathrm{ex}}_{k+1} - E^{\mathrm{ex}}_k, \qquad \ell^E_k := [\mathrm{sgn}(\Delta E^{\mathrm{ex}}_k)b^E_k]_+,\\
\chi^E_k &:= \left[1-\frac{\ell^E_k}{\eta_E|\Delta E^{\mathrm{ex}}_k|+\varepsilon}\right]_+, \qquad v^{\mathrm{sec}}_k \leftarrow \chi^E_k v^{\mathrm{sec}}_k.
\end{aligned}
$$

After the fully expanded chain above, one may use $\widehat{v}^{\mathrm{sec}}_k := v^{\mathrm{sec}}_k/\|v^{\mathrm{sec}}_k\|_{M_k}$ as shorthand when it is inserted into $\mathcal{P}^{\mathrm{stay}}_k$.

17A.8 Exact-forecast horizon score and stay-append decision law

This subsection defines the forecast functional used when the miss ratio is too large to stay on the current scaffold blindly. The horizon index is $h \in \{1,\dots,H_k\}$, the horizon length is $H_k \geq 1$, the positive horizon weights are ω_h, and $W_k := \sum_{h=1}^{H_k}\omega_h$. The action u is either a stay action $u = (\mathrm{stay},p,a)$ with $p \in \mathcal{P}^{\mathrm{stay}}_k$ and $a \in \mathcal{A}_k$, or an append action $u = (\mathrm{append},r,a)$ with $r = (m,p_{\mathrm{ins}}) \in \mathcal{R}^{\mathrm{conf}}_k$ and $a \in \mathcal{A}_k$. The nonnegative weights $\lambda_F,\lambda_s,\lambda_D,\lambda_n,\lambda_E,\lambda_{\mathrm{slope}},\lambda_{\mathrm{curv}},\lambda_{\mathrm{under}},\lambda_{\mathrm{over}}$ determine the relative importance of fidelity, staggered density, doublon, site occupations, total energy, and the optional energy-shape and excursion penalties.

A stay forecast keeps the current scaffold and moves along a normalized within-scaffold proposal, while an append forecast lifts to the zero-initialized augmented scaffold and moves along the augmented append velocity. Thus

$$
\begin{aligned}
\theta^{(\mathrm{stay},p,a)}_{k+h|k} &:= \theta_k + h\,a\,\Delta\tau_k\,p, \qquad |\psi^{(\mathrm{stay},p,a)}_{k+h}\rangle := U(\theta^{(\mathrm{stay},p,a)}_{k+h|k};\mathcal{O}_k)|\psi_{\mathrm{ref}}\rangle,\\
\vartheta^{(r)}_k &:= (\theta_k,0), \qquad \vartheta^{(\mathrm{append},r,a)}_{k+h|k} := \vartheta^{(r)}_k + h\,a\,\Delta\tau_k\,\dot{\vartheta}^{\mathrm{aug}}_{r,k}, \qquad |\psi^{(\mathrm{append},r,a)}_{k+h}\rangle := U(\vartheta^{(\mathrm{append},r,a)}_{k+h|k};\mathcal{O}^+_k(r))|\psi_{\mathrm{ref}}\rangle,\\
|\psi^{\mathrm{ex}}_{k+h}\rangle &:= \text{exact driven reference state at the same future checkpoint time.}
\end{aligned}
$$

The tracked observables are the site occupations, the doublon, the staggered density, and the total energy. If $|\phi\rangle = \sum_x \phi_x|x\rangle$, then $b_{j,\sigma}(x) \in \{0,1\}$ is the occupation bit of basis state $|x\rangle$ at site-spin pair (j,σ), and

$$
\begin{aligned}
&\hat{n}_j := \hat{n}_{j,\uparrow} + \hat{n}_{j,\downarrow}, \qquad n_{j,\sigma}[\phi] := \langle\phi|\hat{n}_{j,\sigma}|\phi\rangle = \sum_x |\phi_x|^2 b_{j,\sigma}(x), \qquad n_j[\phi] := \langle\phi|\hat{n}_j|\phi\rangle = \sum_x |\phi_x|^2(b_{j,\uparrow}(x)+b_{j,\downarrow}(x)),\\
&\hat{D} := \sum_{j=1}^{L}\hat{n}_{j,\uparrow}\hat{n}_{j,\downarrow}, \qquad D[\phi] := \langle\phi|\hat{D}|\phi\rangle = \sum_x |\phi_x|^2\sum_{j=1}^{L} b_{j,\uparrow}(x)b_{j,\downarrow}(x),\\
&\hat{S} := \frac{1}{L}\sum_{j=1}^{L}(-1)^{j-1}\hat{n}_j, \qquad s[\phi] := \langle\phi|\hat{S}|\phi\rangle = \frac{1}{L}\sum_{j=1}^{L}(-1)^{j-1}n_j[\phi],\\
&E^{\mathrm{ctrl}}_{k+h}(u) := \langle\psi^{(u)}_{k+h}|H^\sharp_{k+h}|\psi^{(u)}_{k+h}\rangle, \qquad E^{\mathrm{ex}}_{k+h} := \langle\psi^{\mathrm{ex}}_{k+h}|H^\sharp_{k+h}|\psi^{\mathrm{ex}}_{k+h}\rangle, \qquad H^\sharp_{k+h} := H(t^\sharp_{k+h}).
\end{aligned}
$$

From these primitives the first-order forecast discrepancies are defined once by

$$\begin{aligned}
\varepsilon^{F}_{k+h}(u) &:= 1 - \big|\langle \psi^{\mathrm{ex}}_{k+h}, \psi^{(u)}_{k+h}\rangle\big|^2,\\
\varepsilon^{s}_{k+h}(u) &:= \left|\frac{1}{L}\sum_{j=1}^{L}(-1)^{j-1}\big(n_j[\psi^{(u)}_{k+h}] - n_j[\psi^{\mathrm{ex}}_{k+h}]\big)\right|,\\
\varepsilon^{D}_{k+h}(u) &:= \left|\sum_{j=1}^{L}\Big(\langle \psi^{(u)}_{k+h}|\hat{n}_{j,\uparrow}\hat{n}_{j,\downarrow}|\psi^{(u)}_{k+h}\rangle - \langle \psi^{\mathrm{ex}}_{k+h}|\hat{n}_{j,\uparrow}\hat{n}_{j,\downarrow}|\psi^{\mathrm{ex}}_{k+h}\rangle\Big)\right|,\\
\varepsilon^{n}_{k+h}(u) &:= \max_{1\le j\le L}\big|n_j[\psi^{(u)}_{k+h}] - n_j[\psi^{\mathrm{ex}}_{k+h}]\big|,\\
\varepsilon^{E}_{k+h}(u) &:= \big|\langle \psi^{(u)}_{k+h}|H^{\sharp}_{k+h}|\psi^{(u)}_{k+h}\rangle - \langle \psi^{\mathrm{ex}}_{k+h}|H^{\sharp}_{k+h}|\psi^{\mathrm{ex}}_{k+h}\rangle\big|.
\end{aligned}$$

The energy-shape corrections compare first and second forward differences of the predicted and exact energy paths, while the excursion corrections compare the signed forecast excursion against a relative exact band. Writing $\varpi^{(1)}_h := (\omega_h + \omega_{h+1})/2$, $\varpi^{(2)}_h := (\omega_h + \omega_{h+1} + \omega_{h+2})/3$, and $\varrho_E \in [0,1)$,

$$\begin{aligned}
\Delta E^{\mathrm{ctrl}}_{k+h}(u) &:= E^{\mathrm{ctrl}}_{k+h+1}(u) - E^{\mathrm{ctrl}}_{k+h}(u), \qquad \Delta E^{\mathrm{ex}}_{k+h} := E^{\mathrm{ex}}_{k+h+1} - E^{\mathrm{ex}}_{k+h},\\
\Delta^2 E^{\mathrm{ctrl}}_{k+h}(u) &:= E^{\mathrm{ctrl}}_{k+h+2}(u) - 2E^{\mathrm{ctrl}}_{k+h+1}(u) + E^{\mathrm{ctrl}}_{k+h}(u), \qquad \Delta^2 E^{\mathrm{ex}}_{k+h} := E^{\mathrm{ex}}_{k+h+2} - 2E^{\mathrm{ex}}_{k+h+1} + E^{\mathrm{ex}}_{k+h},\\
\Xi^{\mathrm{shape,slope}}_k(u) &:= \frac{1}{\sum_h \varpi^{(1)}_h}\sum_h \varpi^{(1)}_h\big|\Delta E^{\mathrm{ctrl}}_{k+h}(u) - \Delta E^{\mathrm{ex}}_{k+h}\big|,\\
\Xi^{\mathrm{shape,curv}}_k(u) &:= \frac{1}{\sum_h \varpi^{(2)}_h}\sum_h \varpi^{(2)}_h\big|\Delta^2 E^{\mathrm{ctrl}}_{k+h}(u) - \Delta^2 E^{\mathrm{ex}}_{k+h}\big|,\\
x^{\mathrm{ctrl}}_{k+h}(u) &:= E^{\mathrm{ctrl}}_{k+h}(u) - E^{\mathrm{ctrl}}_k, \qquad x^{\mathrm{ex}}_{k+h} := E^{\mathrm{ex}}_{k+h} - E^{\mathrm{ex}}_k, \qquad \Pi^{\mathrm{ctrl}}_{k+h}(u) := \mathrm{sgn}(x^{\mathrm{ex}}_{k+h})x^{\mathrm{ctrl}}_{k+h}(u),\\
L^{\mathrm{ex}}_{k+h} &:= (1-\varrho_E)|x^{\mathrm{ex}}_{k+h}|, \qquad U^{\mathrm{ex}}_{k+h} := (1+\varrho_E)|x^{\mathrm{ex}}_{k+h}|,\\
\Xi^{\mathrm{exc,under}}_k(u) &:= \frac{1}{W_k}\sum_{h=1}^{H_k}\omega_h\big[L^{\mathrm{ex}}_{k+h} - \Pi^{\mathrm{ctrl}}_{k+h}(u)\big]_+, \qquad \Xi^{\mathrm{exc,over}}_k(u) := \frac{1}{W_k}\sum_{h=1}^{H_k}\omega_h\big[\Pi^{\mathrm{ctrl}}_{k+h}(u) - U^{\mathrm{ex}}_{k+h}\big]_+.
\end{aligned}$$

The final quantity of interest in the miss-dominant regime is the full horizon score itself, so it is written first in fully expanded primitive form:

$$\begin{aligned}
\mathcal{J}^{\mathrm{forecast}}_k(u) := \frac{1}{W_k}\sum_{h=1}^{H_k}\omega_h\Bigg[&\lambda_F\Big(1 - \big|\langle \psi^{\mathrm{ex}}_{k+h}, \psi^{(u)}_{k+h}\rangle\big|^2\Big) + \lambda_s\left|\frac{1}{L}\sum_{j=1}^{L}(-1)^{j-1}\big(n_j[\psi^{(u)}_{k+h}] - n_j[\psi^{\mathrm{ex}}_{k+h}]\big)\right|\\
&+ \lambda_D\left|\sum_{j=1}^{L}\Big(\langle \psi^{(u)}_{k+h}|\hat{n}_{j,\uparrow}\hat{n}_{j,\downarrow}|\psi^{(u)}_{k+h}\rangle - \langle \psi^{\mathrm{ex}}_{k+h}|\hat{n}_{j,\uparrow}\hat{n}_{j,\downarrow}|\psi^{\mathrm{ex}}_{k+h}\rangle\Big)\right|\\
&+ \lambda_n \max_{1\le j\le L}\big|n_j[\psi^{(u)}_{k+h}] - n_j[\psi^{\mathrm{ex}}_{k+h}]\big| + \lambda_E\big|\langle \psi^{(u)}_{k+h}|H^{\sharp}_{k+h}|\psi^{(u)}_{k+h}\rangle - \langle \psi^{\mathrm{ex}}_{k+h}|H^{\sharp}_{k+h}|\psi^{\mathrm{ex}}_{k+h}\rangle\big|\\
&+ \lambda_{\mathrm{under}}\big[(1-\varrho_E)|E^{\mathrm{ex}}_{k+h} - E^{\mathrm{ex}}_k| - \mathrm{sgn}(E^{\mathrm{ex}}_{k+h} - E^{\mathrm{ex}}_k)(E^{\mathrm{ctrl}}_{k+h}(u) - E^{\mathrm{ctrl}}_k)\big]_+\\
&+ \lambda_{\mathrm{over}}\big[\mathrm{sgn}(E^{\mathrm{ex}}_{k+h} - E^{\mathrm{ex}}_k)(E^{\mathrm{ctrl}}_{k+h}(u) - E^{\mathrm{ctrl}}_k) - (1+\varrho_E)|E^{\mathrm{ex}}_{k+h} - E^{\mathrm{ex}}_k|\big]_+\Bigg]\\
&+ \lambda_{\mathrm{slope}}\frac{1}{\sum_h \varpi^{(1)}_h}\sum_h \varpi^{(1)}_h\big|\Delta E^{\mathrm{ctrl}}_{k+h}(u) - \Delta E^{\mathrm{ex}}_{k+h}\big| + \lambda_{\mathrm{curv}}\frac{1}{\sum_h \varpi^{(2)}_h}\sum_h \varpi^{(2)}_h\big|\Delta^2 E^{\mathrm{ctrl}}_{k+h}(u) - \Delta^2 E^{\mathrm{ex}}_{k+h}\big|.
\end{aligned}$$

For shorthand only, one may now write $\mathcal{J}^{\mathrm{forecast}}_k = \mathcal{J}^{\mathrm{trk}}_k + \lambda_{\mathrm{slope}}\Xi^{\mathrm{shape,slope}}_k + \lambda_{\mathrm{curv}}\Xi^{\mathrm{shape,curv}}_k + \lambda_{\mathrm{under}}\Xi^{\mathrm{exc,under}}_k + \lambda_{\mathrm{over}}\Xi^{\mathrm{exc,over}}_k$.

The stay-versus-append comparison is then defined once by

$$\nu^\star_{k,\mathrm{stay}} \in \arg\min_{u\in\mathcal{U}_k^{\mathrm{stay}}} \mathcal{J}_k^{\mathrm{forecast}}(u), \qquad \nu^\star_{k,\mathrm{app}} \in \arg\min_{u\in\mathcal{U}_k^{\mathrm{app}}} \mathcal{J}_k^{\mathrm{forecast}}(u),$$

$$\Delta F_k := \left|\langle \psi^{\mathrm{ex}}_{k+1}, \psi^{(\nu^\star_{k,\mathrm{app}})}_{k+1}\rangle\right|^2 - \left|\langle \psi^{\mathrm{ex}}_{k+1}, \psi^{(\nu^\star_{k,\mathrm{stay}})}_{k+1}\rangle\right|^2, \qquad \Delta\varepsilon^E_k := \varepsilon^E_{k+1}(\nu^\star_{k,\mathrm{app}}) - \varepsilon^E_{k+1}(\nu^\star_{k,\mathrm{stay}}),$$

$$\nu^\star_k := \begin{cases} \nu^\star_{k,\mathrm{stay}}, & 1-\left|\langle \psi^{\mathrm{ex}}_{k+1}, \psi^{(\nu^\star_{k,\mathrm{stay}})}_{k+1}\rangle\right|^2 \le 10^{-3},\ \varepsilon^s_{k+1}(\nu^\star_{k,\mathrm{stay}}) \le 2\times 10^{-2},\ \varepsilon^D_{k+1}(\nu^\star_{k,\mathrm{stay}}) \le 2\times 10^{-3},\ \varepsilon^n_{k+1}(\nu^\star_{k,\mathrm{stay}}) \le 2\times 10^{-2}, \\ \nu^\star_{k,\mathrm{stay}}, & \Delta F_k < -\tau_F \text{ and } \Delta\varepsilon^E_k > \tau_E, \\ \nu^\star_{k,\mathrm{app}}, & \mathcal{J}_k^{\mathrm{forecast}}(\nu^\star_{k,\mathrm{app}}) < \mathcal{J}_k^{\mathrm{forecast}}(\nu^\star_{k,\mathrm{stay}}), \\ \nu^\star_{k,\mathrm{stay}}, & \text{otherwise.} \end{cases}$$

17A.9 Exact local prune law in the calm regime

The prune law deletes one old block only in the calm low-miss regime. Here j is the logical block index, I_j is the runtime coordinate block occupied by block j, I_j^c is its complement, $m^{\mathrm{mov}}_{j,k}$ and $m^{\mathrm{fit}}_{j,k}$ are the recent normalized motion and fit-activity indicators, $\alpha_{\mathrm{pr}} \in [0,1]$ is the stagnation mixing weight, N_{prot} is the protection age, and $\tau_{\mathrm{stale}}, \tau^{\mathrm{pr}}_{\mathrm{loss}}, \tau^{\mathrm{pr}}_\theta, \tau^{\mathrm{pr}}_\psi, \tau^{\mathrm{pr}}_\rho > 0$ are the prune thresholds.

$$\sigma^{\mathrm{stag}}_{j,k} := \alpha_{\mathrm{pr}}(1-m^{\mathrm{mov}}_{j,k}) + (1-\alpha_{\mathrm{pr}})(1-m^{\mathrm{fit}}_{j,k}),$$

$$j \in \mathcal{D}^{\mathrm{pr}}_k \iff \mathrm{age}_{j,k} \ge N_{\mathrm{prot}},\ \mathrm{cooldown}_{j,k} = 0,\ \sigma^{\mathrm{stag}}_{j,k} \ge \tau_{\mathrm{stale}},\ \mathfrak{C}_k = 1,$$

$$G_{k,-j} := G_k[I_j^c, I_j^c], \qquad f_{k,-j} := f_k[I_j^c], \qquad \dot\theta^{\mathrm{proj}}_{k,-j} := G^+_{k,-j} f_{k,-j},$$

$$L^{\mathrm{cache}}_{j,k} := \frac{[f_k^\top G_k^+ f_k - f_{k,-j}^\top G^+_{k,-j} f_{k,-j}]_+}{\langle \psi_k | (H_k^\sharp - E^\sharp_{\psi_k,k})^2 | \psi_k\rangle + \varepsilon},$$

$$|\psi_k^{(-j)}\rangle := U(\theta_k^{(-j)}; \mathcal{O}_k^{(-j)})|\psi_{\mathrm{ref}}\rangle, \qquad \Theta^{\mathrm{blk}}_{j,k} := \|\theta_{k,I_j}\|_2, \qquad \Delta\psi^{\mathrm{pr}}_{j,k} := \|U(\theta_k^{(-j)}; \mathcal{O}_k^{(-j)})|\psi_{\mathrm{ref}}\rangle - U(\theta_k; \mathcal{O}_k)|\psi_{\mathrm{ref}}\rangle\|_2,$$

$$\rho^{(-j)}_{\mathrm{miss},k} := \frac{\langle \psi_k^{(-j)} | (H_k^\sharp - E^\sharp_{\psi_k^{(-j)},k})^2 | \psi_k^{(-j)}\rangle - f_{k,-j}^\top G^+_{k,-j} f_{k,-j}}{\langle \psi_k^{(-j)} | (H_k^\sharp - E^\sharp_{\psi_k^{(-j)},k})^2 | \psi_k^{(-j)}\rangle + \varepsilon}, \qquad \Delta\rho^{\mathrm{pr}}_{j,k} := \rho^{(-j)}_{\mathrm{miss},k} - \rho_{\mathrm{miss},k},$$

$$\mathrm{accept_prune}(j) \iff L^{\mathrm{cache}}_{j,k} \le \tau^{\mathrm{pr}}_{\mathrm{loss}},\ \Theta^{\mathrm{blk}}_{j,k} \le \tau^{\mathrm{pr}}_\theta,\ \Delta\psi^{\mathrm{pr}}_{j,k} \le \tau^{\mathrm{pr}}_\psi,\ \Delta\rho^{\mathrm{pr}}_{j,k} \le \tau^{\mathrm{pr}}_\rho.$$

Thus a block is deleted only if the cached projective loss, deleted-block norm, exact reduced-state jump, and reduced miss-ratio increase are all simultaneously small.

17A.10 Canonical piecewise-smooth law

We now assemble the previous pieces into the final law. Here $\mathcal{O}(\tau)$ is the active scaffold at physical time τ, $\theta(\tau)$ is its coordinate vector on the open checkpoint interval, and the checkpoint times satisfy $\tau_0 < \tau_1 < \tau_2 < \cdots$. The full law is written first in expanded form and only then summarized.

$$
\begin{aligned}
|\psi(\tau)\rangle &:= U(\theta(\tau);\mathcal{O}_k)|\psi_{\mathrm{ref}}\rangle, \qquad \tau \in [\tau_k, \tau_{k+1}),\\
\dot\theta(\tau) &:= \left(\left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}\widetilde{A}_{k,j}|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}\langle\widetilde{A}_{k,j}\rangle_{\psi_k}\Big) + (\Lambda_k)_{ij}\right]_{i,j=1}^{n_k}\right)^{+} \left[\Re\Big(\langle\psi_k|\widetilde{A}_{k,i}H_k^{\sharp}|\psi_k\rangle - \langle\widetilde{A}_{k,i}\rangle_{\psi_k}E^{\sharp}_{\psi_k,k}\Big)\right]_{i=1}^{n_k},\\
\nu_k^\star &:= \begin{cases} \arg\min\limits_{u\in\mathcal{U}_k^{\mathrm{stay}}\cup\mathcal{U}_k^{\mathrm{app}}} \mathcal{J}_k^{\mathrm{forecast}}(u), & \rho_{\mathrm{miss},k} > \tau_{\mathrm{miss}},\\ \text{the best } j\in\mathcal{D}_k^{\mathrm{pr}} \text{ satisfying accept_prune}(j), & \rho_{\mathrm{miss},k} \le \tau_{\mathrm{pr}} \text{ and } \mathfrak{C}_k = 1,\\ \text{stay}, & \text{otherwise}, \end{cases}\\
(\mathcal{O}_{k+1},\theta_{k+1}) &:= \begin{cases} (\mathcal{O}_k,\theta_{k+1}^{\mathrm{stay}}), & \nu_k^\star = \nu_{k,\mathrm{stay}}^\star,\\ (\mathcal{O}_k^{+}(r_k^\star),\vartheta_{k+1}^{\mathrm{app}}), & \nu_k^\star = \nu_{k,\mathrm{app}}^\star = (\mathrm{append}, r_k^\star, a_k^\star),\\ (\mathcal{O}_k^{(-j_k^\star)},\theta_{k+1}^{(-j_k^\star)}), & \nu_k^\star = \mathrm{prune}(j_k^\star). \end{cases}
\end{aligned}
$$

In condensed words only after the expanded law: project the sampled Schrödinger drift onto the current projective tangent space on each checkpoint interval; when the intrinsic miss ratio is large, choose between stay and append by the fully expanded horizon score; when the miss ratio is small and the motion is calm, choose deletion only by the fully expanded exact local prune law; then restart the next interval on the resulting scaffold.

17B. Projective McLachlan Real-Time Dynamics and Checkpoint-Adaptive Snake Control

This chapter is the real-time analogue of §11. The governing surface is the same staged-controller picture: fixed local geometry on the current scaffold, checkpoint-local structural search over candidate-position records $r = (m, p)$, scout/confirm/commit screening for append actions, and checkpoint-local scaffold pruning to keep the tangent plane small. The intended snake doctrine is:

$$\text{McLachlan} = \text{continuous within-scaffold motion law}, \qquad \text{controller} = \text{discrete append/prune law at checkpoints}.$$

Thus the scaffold is piecewise constant in physical time, and may change only at checkpoint events. Ordinary McLachlan dynamics is a local projection law on a fixed scaffold: at the current state it asks for the parameter-velocity vector whose tangent combination best matches the physical Schr"odinger drift after removal of the physically irrelevant global-phase direction. Equivalently,

$$\dot\theta_\lambda \in \arg\min_{v\in\mathbb{R}^{n_\theta}} \Big\{\|\bar T v - \bar b\|^2 + v^\top \Lambda v\Big\}.$$

Its practical outputs are the tangent metric $\bar G$, the force vector $\bar f$, and the projected residual $\bar r$; this residual is the correct local inadequacy signal for the current scaffold, and its natural normalization scale is the energy variance $\mathrm{Var}_\psi(H)$. Therefore McLachlan is treated here as the continuous within-scaffold motion oracle, while the checkpoint controller alone decides when the scaffold should expand or contract. The live controller-defining expressions are

$$\rho_{\mathrm{miss},k}, \qquad \Gamma_k^{\mathrm{live}}, \qquad \Sigma_{\mathrm{scout}}(r;k), \qquad \Delta_{\mathrm{McL}}(r;\tau_k), \qquad \Sigma_{\mathrm{conf}}(r;k), \qquad P_j^{\mathrm{pr}}(k),$$

and the displays below isolate these quantities in terms of the primitive projective objects Q_ψ, $\bar b$, $\bar T$, K, and $\bar r$. The older branch-level rollout and beam doctrine is retained only as a future extension, not as the canonical live runtime surface.

17B.1 Fixed-scaffold projective geometry

Fix a scaffold $\mathcal{O}$, state family $|\psi(\theta;\mathcal{O})\rangle$ with $\theta\in\mathbb{R}^{n_\theta}$, and Hamiltonian $H(\tau)$. At the current state, let $d := \dim\mathcal{H}$, let

$$Q_\psi := I - |\psi\rangle\langle\psi|, \qquad E_\psi := \langle\psi|H|\psi\rangle,$$

and define the projected Schr"odinger drift vector by

$$\bar{b} := Q_\psi(-iH\psi) = \begin{bmatrix} (Q_\psi(-iH\psi))_1 \\ \vdots \\ (Q_\psi(-iH\psi))_d \end{bmatrix} = \begin{bmatrix} (-i(H-E_\psi)\psi)_1 \\ \vdots \\ (-i(H-E_\psi)\psi)_d \end{bmatrix} \in \mathbb{C}^d.$$

Define the projected tangent matrix by

$$\bar{T} := \begin{bmatrix} (Q_\psi\, \partial_{\theta_1}|\psi(\theta;\mathcal{O})\rangle)_1 & \cdots & (Q_\psi\, \partial_{\theta_{n_\theta}}|\psi(\theta;\mathcal{O})\rangle)_1 \\ \vdots & \ddots & \vdots \\ (Q_\psi\, \partial_{\theta_1}|\psi(\theta;\mathcal{O})\rangle)_d & \cdots & (Q_\psi\, \partial_{\theta_{n_\theta}}|\psi(\theta;\mathcal{O})\rangle)_d \end{bmatrix} \in \mathbb{C}^{d\times n_\theta}.$$

Define the regularizer by

$$\Lambda := \begin{bmatrix} \Lambda_{11} & \cdots & \Lambda_{1n_\theta} \\ \vdots & \ddots & \vdots \\ \Lambda_{n_\theta 1} & \cdots & \Lambda_{n_\theta n_\theta} \end{bmatrix} \in \mathbb{R}^{n_\theta\times n_\theta}, \qquad \Lambda^\top = \Lambda, \qquad \Lambda \succeq 0,$$

with simplest default $\Lambda = \lambda_{\text{ridge}} I$. Then the real McLachlan metric, force, and regularized system are

$$\bar{G} := \Re(\bar{T}^\dagger \bar{T}) = \begin{bmatrix} \Re(\sum_{a=1}^d (\bar{T})^*_{a1}(\bar{T})_{a1}) & \cdots & \Re(\sum_{a=1}^d (\bar{T})^*_{a1}(\bar{T})_{an_\theta}) \\ \vdots & \ddots & \vdots \\ \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta}(\bar{T})_{a1}) & \cdots & \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta}(\bar{T})_{an_\theta}) \end{bmatrix},$$

$$\bar{f} := \Re(\bar{T}^\dagger \bar{b}) = \begin{bmatrix} \Re(\sum_{a=1}^d (\bar{T})^*_{a1}\, \bar{b}_a) \\ \vdots \\ \Re(\sum_{a=1}^d (\bar{T})^*_{an_\theta}\, \bar{b}_a) \end{bmatrix}, \qquad K := \bar{G} + \Lambda, \qquad K\dot{\theta}_\lambda = \bar{f}.$$

Equivalently, on the fixed scaffold $\mathcal{O}$, ordinary McLachlan dynamics solves

$$\dot{\theta}_\lambda \in \arg\min_{v\in\mathbb{R}^{n_\theta}} \Big\{ \|\bar{T}v - \bar{b}\|^2 + v^\top \Lambda v \Big\},$$

so the only role of the current scaffold is to define the available tangent plane. The solve itself is therefore a within-scaffold motion law, not a scaffold-construction law. On a fixed scaffold $\mathcal{O}_k$, the runtime first solves $K\dot{\theta}_\lambda = \bar{f}$ and propagates $\theta(\tau)$ over the next checkpoint segment; it then measures $\rho_{\text{miss},k}$, and only if that miss is too large does the discrete checkpoint controller open the structural action set $\{\text{stay}, \text{append}(r), \text{prune}(j)\}$. The baseline miss telemetry is

$$\epsilon^2_{\text{proj}} := \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}, \qquad \epsilon^2_{\text{step}} := \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2, \qquad \rho_{\text{miss}} := \frac{\epsilon^2_{\text{proj}}}{\|\bar{b}\|^2 + \varepsilon} = \frac{\epsilon^2_{\text{proj}}}{\text{Var}_\psi(H) + \varepsilon}.$$

So the miss statistic itself may be read as

$$\begin{aligned} \epsilon^2_{\text{proj}} &= \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f} \\ &= \|Q_\psi(-iH\psi)\|^2 - \big(\Re(\bar{T}^\dagger\bar{b})\big)^\top \big(\Re(\bar{T}^\dagger\bar{T})\big)^+ \Re(\bar{T}^\dagger\bar{b}), \\ \rho_{\text{miss}} &= \frac{\epsilon^2_{\text{proj}}}{\text{Var}_\psi(H) + \varepsilon}. \end{aligned}$$

For any admissible checkpoint-local tangent baseline A with basis matrix $\bar{T}_{k,A}$ (for example a coordinate subset of the current scaffold tangents, or any chosen real tangent subspace with an explicit basis), define

$$\bar{G}_{k,A} := \Re(\bar{T}^\dagger_{k,A}\bar{T}_{k,A}), \qquad \bar{f}_{k,A} := \Re(\bar{T}^\dagger_{k,A}\bar{b}_k),$$

and the intrinsic reduced-space projective miss

$$\epsilon^2_{\text{proj},k}(A) := \|\bar{b}_k\|^2 - \bar{f}^\top_{k,A}\bar{G}^+_{k,A}\bar{f}_{k,A}, \qquad \rho_{\text{miss},k}(A) := \frac{\epsilon^2_{\text{proj},k}(A)}{\|\bar{b}_k\|^2 + \varepsilon}.$$

This quantity is exact for the reduced tangent space $\mathrm{span}_{\mathbb{R}}(\bar{T}_{k,A})$. If $A \subseteq A'$ as tangent subspaces, then

$$\epsilon^2_{\mathrm{proj},k}(A') \le \epsilon^2_{\mathrm{proj},k}(A), \qquad \rho_{\mathrm{miss},k}(A') \le \rho_{\mathrm{miss},k}(A),$$

because enlarging the tangent baseline can only decrease the orthogonal projection distance of $\bar{b}_k$. In particular, if $A^{\mathrm{sc}}_k \subseteq A^{\mathrm{cm}}_k$, then

$$\rho_{\mathrm{miss},k}(A^{\mathrm{cm}}_k) \le \rho_{\mathrm{miss},k}(A^{\mathrm{sc}}_k),$$

so the scout-baseline miss is conservative for the larger shared commit baseline. The projective and raw residuals are

$$\bar{r} := \bar{T}\dot{\theta}_\lambda - \bar{b}, \qquad r_{\mathrm{raw}} := T\dot{\theta}_\lambda + iH\psi, \qquad \bar{r} = Q_\psi r_{\mathrm{raw}}.$$

Thus $\bar{r}$ is the local defect after the best available tangent velocity has been chosen on the current scaffold, and ρ_{miss} is the corresponding local inadequacy ratio relative to exact projective drift. So, for $\tau \in [\tau_k, \tau_{k+1})$, the live controller holds $\mathcal{O}(\tau) = \mathcal{O}_k$ fixed and uses McLachlan only to evolve $\theta(\tau)$ on that scaffold.

17B.2 Checkpoint miss test and telemetry-driven controller outputs

At checkpoint k, with physical time τ_k, the controller baseline summary is

$$\Xi^{\mathrm{base}}_k := (E_k,\, \mathrm{Var}_k,\, \epsilon^2_{\mathrm{proj},k},\, \epsilon^2_{\mathrm{step},k},\, \rho_{\mathrm{miss},k},\, \|\dot{\theta}_{\lambda,k}\|_2,\, \mathrm{rank}(K_k),\, \mathrm{cond}(K_k)).$$

Let

$$A^{\mathrm{sc}}_k \subseteq A^{\mathrm{cm}}_k \subseteq \{1, \dots, n_{\theta,k}\}$$

denote the scout and shared commit baselines at checkpoint k. Unless decorated otherwise, the live checkpoint shorthand is

$$\epsilon^2_{\mathrm{proj},k} := \epsilon^2_{\mathrm{proj},k}(A^{\mathrm{sc}}_k), \qquad \rho_{\mathrm{miss},k} := \rho_{\mathrm{miss},k}(A^{\mathrm{sc}}_k).$$

The clean rule is

$$\text{stay on } \mathcal{O}_k \text{ if } \rho_{\mathrm{miss},k} \le \tau_{\mathrm{miss}}, \qquad \text{open structural search otherwise.}$$

The `fullclone_3` runtime enriches this baseline with checkpoint cadence and telemetry-driven controller outputs. If $\dot{\theta}_k, \dot{\theta}_{k-1}, \dot{\theta}_{k-2}$ denote the aligned recent solved steps, define

$$\delta_k := \dot{\theta}_k - \dot{\theta}_{k-1}, \qquad \alpha_k := \delta_k - \delta_{k-1}, \qquad \rho^{\mathrm{rate}}_k := \frac{\|\delta_k\|_2}{\max(\|\dot{\theta}_k\|_2, \|\dot{\theta}_{k-1}\|_2, \varepsilon)},$$

$$c^{\mathrm{dir}}_k := \cos(\dot{\theta}_k, \dot{\theta}_{k-1}), \qquad c^{\mathrm{curv}}_k := \cos(\alpha_k, \alpha_{k-1}), \qquad d^{\mathrm{pred}}_k := \text{predicted displacement at checkpoint } k.$$

Then the controller uses the indicators

$$\mathbf{1}^{\mathrm{rev}}_k := \mathbf{1}[c^{\mathrm{dir}}_k \le \tau_{\mathrm{rev}} \ \wedge\ \rho^{\mathrm{rate}}_k \ge \tau^{\mathrm{rate}}_{\mathrm{calm}} \ \wedge\ \|\delta_k\|_2 \ge \tau_{\mathrm{acc}}],$$

$$\mathbf{1}^{\mathrm{flip}}_k := \mathbf{1}[c^{\mathrm{curv}}_k \le \tau_{\mathrm{flip}} \ \wedge\ \|\alpha_k\|_2 \ge \tau_{\mathrm{acc}} \ \wedge\ \|\alpha_{k-1}\|_2 \ge \tau_{\mathrm{acc}}],$$

$$\mathbf{1}^{\mathrm{kink}}_k := \mathbf{1}[\mathbf{1}^{\mathrm{rev}}_k = 1 \ \vee\ \mathbf{1}^{\mathrm{flip}}_k = 1 \ \vee\ (\|\delta_k\|_2 \ge \tau_{\mathrm{acc}} \ \wedge\ \rho^{\mathrm{rate}}_k \ge \tau^{\mathrm{rate}}_{\mathrm{kink}})],$$

$$\mathbf{1}^{\mathrm{calm}}_k := \mathbf{1}[k > 0 \ \wedge\ c^{\mathrm{dir}}_k \ge \tau^{\mathrm{dir}}_{\mathrm{calm}} \ \wedge\ \rho^{\mathrm{rate}}_k \le \tau^{\mathrm{rate}}_{\mathrm{calm}} \ \wedge\ \mathbf{1}^{\mathrm{rev}}_k = 0 \ \wedge\ \mathbf{1}^{\mathrm{flip}}_k = 0 \ \wedge\ d^{\mathrm{pred}}_k \le 0.05].$$

If N^0_{short} and f^0_{short} are the base shortlist size and fraction, then

$$N_{\mathrm{short}}(k) = \begin{cases} \max(1, N^0_{\mathrm{short}} + b^{\mathrm{short}}_{\mathrm{kink}}), & \mathbf{1}^{\mathrm{kink}}_k = 1, \\ \max(1, \lceil s^{\mathrm{short}}_{\mathrm{calm}} N^0_{\mathrm{short}} \rceil), & \mathbf{1}^{\mathrm{calm}}_k = 1, \\ N^0_{\mathrm{short}}, & \text{otherwise}, \end{cases}$$

$$f_{\mathrm{short}}(k) = \begin{cases} \min(1, 1.5\, f^0_{\mathrm{short}}), & \mathbf{1}^{\mathrm{kink}}_k = 1, \\ \max(0.05, s^{\mathrm{short}}_{\mathrm{calm}} f^0_{\mathrm{short}}), & \mathbf{1}^{\mathrm{calm}}_k = 1, \\ f^0_{\mathrm{short}}, & \text{otherwise}, \end{cases}$$

while the effective refresh pressure is

$$\pi_k := \max_{\text{ord}}\big(\pi_k^{\text{base}}, \pi_k^{\text{tele}}\big), \qquad \pi_k^{\text{tele}} = \begin{cases} \text{medium}, & k = 0, \\ \text{high}, & \mathbf{1}_k^{\text{kink}} = 1, \\ \text{low}, & \text{otherwise.} \end{cases}$$

The oracle-budget multiplier is then

$$\beta_{\text{orc}}(k) = \begin{cases} \max\big(1, s_{\text{kink}}^{\text{orc}}\big), & \mathbf{1}_k^{\text{kink}} = 1 \ \vee \ \pi_k = \text{high}, \\ \max\big(0.25, s_{\text{calm}}^{\text{orc}}\big), & \mathbf{1}_k^{\text{calm}} = 1 \ \wedge \ \pi_k = \text{low}, \\ 1, & \text{otherwise.} \end{cases}$$

17B.3 Live checkpoint controller: stay, append, and prune

At each checkpoint, the discrete structural controller applies

$$\Gamma_k^{\text{live}} : (\mathcal{O}_k, \theta_k^-, \Xi_k^{\text{base}}) \longmapsto (\mathcal{O}_{k+1}, \theta_{k+1}^+),$$

with action set

$$\mathcal{U}_k^{\text{live}} := \{\text{stay}\} \cup \{\text{append}(r) : r \in \mathcal{R}_k^{\text{probe}}\} \cup \{\text{prune}(j) : j \in \mathcal{D}_k^{\text{prune}}\}.$$

Thus the live controller separates the continuous within-scaffold McLachlan motion from the discrete checkpoint-local scaffold edits. The append surface is the candidate-position record set from the current checkpoint, while the prune surface is the mature scaffold-coordinate set selected by the local prune controller. Only one discrete action is committed per checkpoint. The controller is lane-ordered rather than symmetric:

$$\text{lane}_k := \begin{cases} \text{append}, & \rho_{\text{miss},k} > \tau_{\text{miss}}, \\ \text{prune}, & \rho_{\text{miss},k} \le \tau_{\text{miss}} \ \wedge \ \Gamma_k^{\text{prune,perm}} = 1, \\ \text{stay}, & \text{otherwise.} \end{cases}$$

Hence a large miss opens the append lane, pruning is visible only in the calm low-miss regime, and failed append confirmation falls back to stay, not to prune at the same checkpoint.

17B.4 Append broad screen over candidate-position records

Candidate append is position-aware:

$$r = (m, p), \qquad \mathcal{O}_r^+ := \mathcal{O}_k \oplus_p m, \qquad \theta_r^+ := \theta_k \oplus_p 0.$$

The probed positions are restricted to a capped active set built from a fixed active suffix window,

$$W_k^{\text{act}} := \{\max(0, n_{\theta,k} - \omega_{\text{act}}), \dots, n_{\theta,k} - 1\},$$

$$\mathcal{P}_k := \text{AllowedPositions}\big(n_{\theta,k}, \text{append} = n_{\theta,k}, W_k^{\text{act}}, N_{\text{probe}}^{\max}\big) \subseteq \{0, \dots, n_{\theta,k}\}, \qquad |\mathcal{P}_k| \le N_{\text{probe}}^{\max}.$$

At zero initialization,

$$|\psi_r^+(\theta_k, 0)\rangle = |\psi_k\rangle, \qquad Q_{\psi_r^+} = Q_{\psi_k},$$

and the old tangent columns transport exactly. If $r = (m, p)$ inserts a q_r-parameter candidate block at position p, define the old-index transport

$$I_{r,p}(j) := \begin{cases} j, & j \le p, \\ j + q_r, & j > p. \end{cases}$$

Then

$$\bar{\tau}_{k,r,I_{r,p}(j)}^+ = \bar{\tau}_{k,j},$$

hence for every old tangent baseline A,

$$\bar{G}_{k,I_{r,p}(A),I_{r,p}(A)}^+ = \bar{G}_{k,A}, \qquad \bar{f}_{k,I_{r,p}(A)}^+ = \bar{f}_{k,A}, \qquad K_{k,I_{r,p}(A),I_{r,p}(A)}^+ = K_{k,A}.$$

Thus zero-init append transports the full old-old geometry exactly and requires no remeasurement of old-old blocks.

At checkpoint k, let $x_k \in \mathbb{R}^{M_k}$ collect the measurement bank already available on the current scaffold. Then every current old-old block is a fixed linear image of x_k:

$$\mathrm{vec}(\bar{G}_{k,A}) = L_{G,A} x_k, \qquad \bar{f}_{k,A} = L_{f,A} x_k, \qquad \|\bar{b}_k\|^2 = \ell_{b,k}^\top x_k.$$

For a candidate r, let

$$x_{k,r}^{\mathrm{aug}} := \begin{bmatrix} x_k \\ y_{k,r} \end{bmatrix}$$

augment the bank only by those candidate-local observables not already present. Then scout, confirm, and commit are nested linear images of this common candidate-local augmentation bank:

$$w_{r,k}^{\mathrm{cm}} = L_{w,r}^{\mathrm{sc}} x_{k,r}^{\mathrm{aug}}, \qquad \mathrm{vec}(B_{r,k}^{\mathrm{cm}}) = L_{B,r}^{\mathrm{cm}} x_{k,r}^{\mathrm{aug}}, \qquad \mathrm{vec}(C_{r,k}) = L_{C,r} x_{k,r}^{\mathrm{aug}}.$$

Scout therefore needs only $w_{r,k}^{\mathrm{cm}}$ and $C_{r,k}$; confirm requests only selected K_k^{cm}-whitened rows of $B_{r,k}^{\mathrm{cm}}$; commit may materialize the full old-new block.

Now fix the shared commit baseline A_k^{cm}. Write

$$\bar{T}_k^{\mathrm{cm}} := \bar{T}_{k,A_k^{\mathrm{cm}}}, \qquad \bar{G}_k^{\mathrm{cm}} := \bar{G}_{k,A_k^{\mathrm{cm}}}, \qquad \bar{f}_k^{\mathrm{cm}} := \bar{f}_{k,A_k^{\mathrm{cm}}}, \qquad K_k^{\mathrm{cm}} := \bar{G}_k^{\mathrm{cm}} + \Lambda_k^{\mathrm{cm}},$$

and let

$$K_k^{\mathrm{cm}} \dot{\theta}_{k,\lambda}^{\mathrm{cm}} = \bar{f}_k^{\mathrm{cm}}, \qquad \bar{r}_k^{\mathrm{cm}} := \bar{T}_k^{\mathrm{cm}} \dot{\theta}_{k,\lambda}^{\mathrm{cm}} - \bar{b}_k.$$

The tangent space still gains a new column

$$u_r := \partial_{\vartheta_r} |\psi_r^+(\theta_k, \vartheta_r)\rangle\big|_{\vartheta_r=0}, \qquad \bar{u}_r := Q_\psi u_r.$$

For the candidate block $\bar{U}_r \in \mathbb{C}^{d \times q_r}$, define

$$B_{r,k}^{\mathrm{cm}} := \mathfrak{R}\big((\bar{T}_k^{\mathrm{cm}})^\dagger \bar{U}_r\big), \qquad C_{r,k} := \mathfrak{R}(\bar{U}_r^\dagger \bar{U}_r), \qquad q_{r,k} := \mathfrak{R}(\bar{U}_r^\dagger \bar{b}_k),$$

and the residualized force on the shared commit baseline

$$w_{r,k}^{\mathrm{cm}} := q_{r,k} - (B_{r,k}^{\mathrm{cm}})^\top (K_k^{\mathrm{cm}})^{-1} \bar{f}_k^{\mathrm{cm}} = -\mathfrak{R}(\bar{U}_r^\dagger \bar{r}_k^{\mathrm{cm}}).$$

The scout-safe lower-bound Schur gain is

$$\underline{\Delta}_{\mathrm{McL}}(r; \tau_k) := (w_{r,k}^{\mathrm{cm}})^\top (C_{r,k} + \Lambda_r)^+ w_{r,k}^{\mathrm{cm}} = \max_{\eta \in \mathbb{R}^{q_r}} \left\{ 2\eta^\top w_{r,k}^{\mathrm{cm}} - \eta^\top (C_{r,k} + \Lambda_r)\eta \right\}.$$

This is a conservative lower bound to the exact commit-grade gain because

$$S_{r,k}^{\mathrm{cm}} := C_{r,k} + \Lambda_r - (B_{r,k}^{\mathrm{cm}})^\top (K_k^{\mathrm{cm}})^{-1} B_{r,k}^{\mathrm{cm}} \preceq C_{r,k} + \Lambda_r.$$

The normalized scout score is therefore

$$\underline{\rho}_{\mathrm{gain}}(r; k) := \frac{\underline{\Delta}_{\mathrm{McL}}(r; \tau_k)}{\|\bar{b}_k\|^2 + \varepsilon},$$

$$\Sigma_{\mathrm{scout}}(r; k) := \underline{\rho}_{\mathrm{gain}}(r; k) + b_{\mathrm{temp}}(r; k) - \beta_{\mathrm{comp}}\, C_{\mathrm{comp}}(r; k) - \beta_{\mathrm{meas}}\, G_{\mathrm{new}}(r; k) - \beta_{\mathrm{jump}}\, J_{\mathrm{jump}}(r; k).$$

Hence scout is already written on the same McLachlan geometry as commit, but uses only the residualized force and the candidate self-block rather than the full old-new Schur data.

17B.5 Append confirm, compressed rerank, and exact commit

The runtime remains organized in three tiers,

$$\text{scout}, \qquad \text{confirm}, \qquad \text{commit},$$

but confirm now enriches the same candidate on the shared commit baseline by a monotone family of K_k^{cm}-whitened old-mode responses.

Let

$$K_k^{\mathrm{cm}} = V_k \Sigma_k^2 V_k^\top$$

be the spectral factorization of the shared commit baseline on its numerical support, and define the corresponding K_k^{cm}-whitened coordinates

$$z_k := \Sigma_k^{-1} V_k^\top \bar{f}_k^{\mathrm{cm}}, \qquad \Gamma_{r,k} := \Sigma_k^{-1} V_k^\top B_{r,k}^{\mathrm{cm}}.$$

Then

$$w_{r,k}^{\mathrm{cm}} = q_{r,k} - \Gamma_{r,k}^\top z_k, \qquad S_{r,k}^{\mathrm{cm}} = C_{r,k} + \Lambda_r - \Gamma_{r,k}^\top \Gamma_{r,k}.$$

Let

$$J_r^{(0)} \subseteq J_r^{(1)} \subseteq \cdots \subseteq J_r^{(m_{\max})} \subseteq \{1, \dots, \mathrm{rank}(K_k^{\mathrm{cm}})\}$$

be any nested candidate-local family of whitened baseline mode sets, and let $\Pi_{J_r^{(m)}}$ denote the coordinate projector onto $J_r^{(m)}$. A deterministic implementation may, for example, order the modes by decreasing row norm of $\Gamma_{r,k}$, but the ladder below does not depend on the chosen ordering. Define the compressed Schur family

$$\widetilde{S}_{r,k}^{(m)} := C_{r,k} + \Lambda_r - \Gamma_{r,k}^\top \Pi_{J_r^{(m)}} \Gamma_{r,k},$$

and the corresponding confirm-grade lower-bound gains

$$\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r; \tau_k, J_r) := (w_{r,k}^{\mathrm{cm}})^\top (\widetilde{S}_{r,k}^{(m)})^+ w_{r,k}^{\mathrm{cm}}.$$

Equivalently,

$$\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r; \tau_k, J_r) = \max_{\eta \in \mathbb{R}^{q_r},\, \zeta \in \mathrm{range}(\Pi_{J_r^{(m)}})} \left\{ 2\eta^\top w_{r,k}^{\mathrm{cm}} - \eta^\top (C_{r,k} + \Lambda_r)\eta - 2\zeta^\top \Gamma_{r,k}\eta - \|\zeta\|_2^2 \right\}.$$

Thus the confirm ladder is monotone:

$$\underline{\Delta}_{\mathrm{McL}}(r; \tau_k) = \widetilde{\Delta}_{\mathrm{McL}}^{(0)}(r; \tau_k, J_r) \le \widetilde{\Delta}_{\mathrm{McL}}^{(1)}(r; \tau_k, J_r) \le \cdots \le \widetilde{\Delta}_{\mathrm{McL}}^{(m_{\max})}(r; \tau_k, J_r) \le \Delta_{\mathrm{McL}}(r; \tau_k, A_k^{\mathrm{cm}}),$$

where the exact commit-grade gain on the shared baseline is

$$\Delta_{\mathrm{McL}}(r; \tau_k, A_k^{\mathrm{cm}}) := (w_{r,k}^{\mathrm{cm}})^\top (S_{r,k}^{\mathrm{cm}})^+ w_{r,k}^{\mathrm{cm}}.$$

The normalized confirm and commit gains are

$$\tilde{\rho}_{\mathrm{gain}}^{(m)}(r; k) := \frac{\widetilde{\Delta}_{\mathrm{McL}}^{(m)}(r; \tau_k, J_r)}{\|\bar{b}_k\|^2 + \varepsilon}, \qquad \rho_{\mathrm{gain}}^{\mathrm{cm}}(r; k) := \frac{\Delta_{\mathrm{McL}}(r; \tau_k, A_k^{\mathrm{cm}})}{\|\bar{b}_k\|^2 + \varepsilon}.$$

Scout keeps

$$\mathcal{S}_k^{\mathrm{scout}} := \mathrm{Top}_{N_{\mathrm{short}}(k)}(\Sigma_{\mathrm{scout}}(r; k);\, r \in \mathcal{R}_k^{\mathrm{probe}}).$$

Confirm evaluates one chosen compressed level m_r for each scout survivor and ranks with

$$\Sigma_{\mathrm{conf}}(r; k) := \tilde{\rho}_{\mathrm{gain}}^{(m_r)}(r; k) - \beta_{\mathrm{meas}}\, G_{\mathrm{new}}(r; k) - \beta_{\mathrm{dir}}\, D_{\mathrm{dir}}(r; k).$$

The confirm shortlist is therefore

$$\mathcal{S}_k^{\mathrm{conf}} := \mathrm{Top}_{\max\{1, \lceil f_{\mathrm{short}}(k)\, |\mathcal{S}_k^{\mathrm{scout}}| \rceil\}}(\Sigma_{\mathrm{conf}}(r; k);\, r \in \mathcal{S}_k^{\mathrm{scout}}).$$

The selected append record is

$$r_k^\star := \arg\max_{r \in \mathcal{S}_k^{\mathrm{conf}}} \Sigma_{\mathrm{conf}}(r; k).$$

The clean commit rule is

$$\text{append } r_k^\star \iff \rho_{\mathrm{miss},k}(A_k^{\mathrm{cm}}) > \tau_{\mathrm{miss}}, \quad \rho_{\mathrm{gain}}^{\mathrm{cm}}(r_k^\star; k) \ge \tau_{\mathrm{gain}}, \quad \Delta_{\mathrm{McL}}(r_k^\star; \tau_k, A_k^{\mathrm{cm}}) \ge \delta_{\mathrm{append}}.$$

Otherwise the controller stays on $\mathcal{O}_k$. Thus scout, confirm, and commit all remain inside the same McLachlan geometry, but use a staged Schur family that narrows measurement effort from residualized force, to selected old-mode responses, to the exact shared-baseline Schur complement.

17B.6 Local scaffold pruning and rollback

Pruning acts on scaffold coordinates, not on branch objects. Let the current scaffold be $\mathcal{O}_k = (m_1, \dots, m_{n_k})$ with $n_k := |\mathcal{O}_k|$, and let each coordinate carry the metadata (ι_j, κ_j, c_j), where ι_j is the checkpoint at which coordinate j was appended, κ_j is its stored append-time burden, and $c_j(k) \in \{0, 1, \dots, c_{\text{cool}}\}$ is its prune-cooldown counter. The hard protection set is

$$\mathcal{P}_{\text{protect}}(k) := \{\, j : \; k - \iota_j < \ell_{\text{protect}} \,\},$$

so coordinates inside $\mathcal{P}_{\text{protect}}(k)$ are never considered for pruning. For the recent-history stagnation window, let

$$\mathcal{J}_j^{\text{stag}}(k) := \{\, \ell : \max(\iota_j + 1, k - L_{\text{stag}} + 1) \le \ell \le k \,\}, \qquad \Delta\theta_{j,\ell} := \theta_j(\tau_\ell^-) - \theta_j(\tau_{\ell-1}^+), \qquad \bar{f}_{j,\ell} := (\bar{f}_\ell)_j.$$

Then the normalized recent-motion and fit-contribution summaries are

$$m_j^{\text{mov}}(k) := \frac{1}{|\mathcal{J}_j^{\text{stag}}(k)|} \sum_{\ell \in \mathcal{J}_j^{\text{stag}}(k)} \frac{|\Delta\theta_{j,\ell}|}{\|\Delta\theta_\ell\|_\infty + \varepsilon_\theta},$$

$$m_j^{\text{fit}}(k) := \frac{1}{|\mathcal{J}_j^{\text{stag}}(k)|} \sum_{\ell \in \mathcal{J}_j^{\text{stag}}(k)} \frac{|\Delta\theta_{j,\ell}|\,|\bar{f}_{j,\ell}|}{\sum_i |\Delta\theta_{i,\ell}|\,|\bar{f}_{i,\ell}| + \varepsilon_f}.$$

The hybrid checkpoint-local stagnation score is

$$s_j^{\text{stag}}(k) := \alpha_{\text{stag}}\Big(1 - \text{clip}(m_j^{\text{mov}}(k), 0, 1)\Big) + (1 - \alpha_{\text{stag}})\Big(1 - \text{clip}(m_j^{\text{fit}}(k), 0, 1)\Big), \qquad 0 \le \alpha_{\text{stag}} \le 1,$$

so larger values mean less recent motion and less recent contribution to local drift tracking. For coordinates outside the hard protection window, the optional soft early-removal penalty is

$$\Pi_j^{\text{age}}(k) := 1 + \lambda_{\text{age}}^{\text{pr}} \exp\left(-\frac{k - \iota_j - \ell_{\text{protect}}}{\ell_{\text{age}}}\right), \qquad j \notin \mathcal{P}_{\text{protect}}(k).$$

The mature eligible set is

$$\mathcal{E}_k^{\text{prune}} := \{\, j : \; j \notin \mathcal{P}_{\text{protect}}(k),\; c_j(k) = 0,\; |\mathcal{J}_j^{\text{stag}}(k)| \ge 1,\; s_j^{\text{stag}}(k) \ge s_{\text{stale}} \,\}.$$

Pruning is opened only when the current scaffold is already tracking the drift adequately. The permissibility gate is

$$\Gamma_k^{\text{prune,perm}} := \mathbf{1}[\rho_{\text{miss},k} \le \tau_{\text{prune}}^{\text{miss}}]\, \mathbf{1}[c_k^{\text{dir}} \ge \tau_{\text{calm}}^{\text{dir}}]\, \mathbf{1}[\rho_k^{\text{rate}} \le \tau_{\text{calm}}^{\text{rate}}]\, \mathbf{1}[n_k \ge 2], \qquad \tau_{\text{prune}}^{\text{miss}} \le \tau_{\text{miss}}.$$

Once the current baseline geometry $(\bar{G}_k^{\text{cm}}, \bar{f}_k^{\text{cm}})$ is cached, broad prune nomination is shot-free. For each $j \in \mathcal{E}_k^{\text{prune}} \cap A_k^{\text{cm}}$, let

$$A_{k,-j}^{\text{cm}} := A_k^{\text{cm}} \quad \{j\}, \qquad \bar{G}_{k,-j} := \bar{G}_{k,A_{k,-j}^{\text{cm}}}, \qquad \bar{f}_{k,-j} := \bar{f}_{k,A_{k,-j}^{\text{cm}}}.$$

Define the cached frozen-ablation miss increase

$$L_{j,\text{frz}}^{\text{cache}}(k) := \left[\frac{\epsilon_{\text{proj},k}^2(A_{k,-j}^{\text{cm}}) - \epsilon_{\text{proj},k}^2(A_k^{\text{cm}})}{\|\bar{b}_k\|^2 + \varepsilon}\right]_+ = \left[\frac{(\bar{f}_k^{\text{cm}})^\top (\bar{G}_k^{\text{cm}})^+ \bar{f}_k^{\text{cm}} - \bar{f}_{k,-j}^\top \bar{G}_{k,-j}^+ \bar{f}_{k,-j}}{\|\bar{b}_k\|^2 + \varepsilon}\right]_+.$$

This quantity depends only on cached submatrices and subvectors of the current geometry and therefore requires no new shots. The local prune probe set is then

$$\mathcal{D}_k^{\text{prune}} := \begin{cases} \text{Top}_{K_{\text{pr}}}\Big(-L_{j,\text{frz}}^{\text{cache}}(k);\, j \in \mathcal{E}_k^{\text{prune}} \cap A_k^{\text{cm}}\Big), & \Gamma_k^{\text{prune,perm}} = 1, \\ \emptyset, & \Gamma_k^{\text{prune,perm}} = 0. \end{cases}$$

When cached losses tie numerically, ties are broken in favor of larger stored burden κ_j. Inside the nominated set, use the cheap shot-free prune score

$$P_j^{\text{pr}}(k) := L_{j,\text{frz}}^{\text{cache}}(k), \qquad j_k^\star := \arg\min_{j \in \mathcal{D}_k^{\text{prune}}} P_j^{\text{pr}}(k).$$

Thus broad prune ranking is geometry-exact on the cached baseline and requires no new measurements; the later remove-and-reintegrate test remains the authoritative prune accept/reject stage. The short local prune horizon is fixed as

$$\Delta\tau_{\mathrm{pr}} := \eta_{\mathrm{pr}}(\tau_{k+1} - \tau_k), \qquad 0 < \eta_{\mathrm{pr}} \le 1.$$

The authoritative prune test removes $j_k^\star$, resolves McLachlan on the reduced scaffold over the short local probe segment $[\tau_k, \tau_k + \Delta\tau_{\mathrm{pr}}]$, and compares the reduced run against the incumbent by

$$J_{\mathrm{seg},k} := \int_{\tau_k}^{\tau_k+\Delta\tau_{\mathrm{pr}}} \|\bar{r}(\tau)\|^2 \, d\tau,$$

$$\Delta\rho_j^{\mathrm{pr}}(k) := \rho_{\mathrm{miss},k}^{(-j)} - \rho_{\mathrm{miss},k}, \qquad \Delta J_j^{\mathrm{pr}}(k) := \int_{\tau_k}^{\tau_k+\Delta\tau_{\mathrm{pr}}} \Big(\|\bar{r}^{(-j)}(\tau)\|^2 - \|\bar{r}(\tau)\|^2 \Big) \, d\tau.$$

Accept the prune iff

$$\Delta\rho_{j_k^\star}^{\mathrm{pr}}(k) \le \Delta_{\mathrm{safe}}^{\mathrm{miss}}, \qquad \Delta J_{j_k^\star}^{\mathrm{pr}}(k) \le \Delta_{\mathrm{safe}}^{\mathrm{seg}}.$$

Otherwise rollback to the pre-prune scaffold and set $c_{j_k^\star}(k^+) \leftarrow c_{\mathrm{cool}}$. Thus the live checkpoint action is

$$\Gamma_k^{\mathrm{live}} \in \{\mathrm{stay}, \mathrm{append}(r_k^\star), \mathrm{prune}(j_k^\star)\},$$

where append is selected by the shortlist/confirm/commit controller above, and prune is selected by the local remove-and-reintegrate controller above. Between checkpoints, the scaffold is fixed; at checkpoints, only one discrete append or prune action is committed.

17B.7 Future branch-level rollout and beam extension

If one later lifts the local stay/append/prune controller to a branch controller, a natural future extension is to compare short rollout alternatives by segment residual cost. On a fixed scaffold over a segment $[\tau_a, \tau_b]$, the raw and projective segment costs are

$$J_{\mathrm{seg}}^{\mathrm{raw}} := \int_{\tau_a}^{\tau_b} \|r_{\mathrm{raw}}(\tau)\|^2 \, d\tau, \qquad J_{\mathrm{seg}}^{\mathrm{proj}} := \int_{\tau_a}^{\tau_b} \|\bar{r}(\tau)\|^2 \, d\tau.$$

A clean scalar branch objective is

$$J_{\mathrm{dyn}}(\mathfrak{b}) := \sum_{k=0}^{K-1} J_{\mathrm{seg},k} + \beta \, C_{\mathrm{hw}}(\mathfrak{b}) + \gamma \, N_{\mathrm{adapt}}(\mathfrak{b}).$$

The associated residual ledgers may be summarized by

$$\Delta J^{\epsilon} := \int \epsilon_{\mathrm{step}}^2, \qquad \Delta J^{\mathrm{model}} := \int \epsilon_{\mathrm{proj}}^2, \qquad \Delta J^{\mathrm{damp}} := \int \max(\epsilon_{\mathrm{step}}^2 - \epsilon_{\mathrm{proj}}^2, 0),$$

then culls rollout alternatives lexicographically after deduplicating by checkpoint index, scaffold labels, and rounded parameters. Thus the branch objective may be read as the integrated residual ledger plus explicit hardware and adaptation penalties:

$$\begin{aligned}
J_{\mathrm{dyn}}(\mathfrak{b}) &= \sum_{k=0}^{K-1} J_{\mathrm{seg},k} + \beta \, C_{\mathrm{hw}}(\mathfrak{b}) + \gamma \, N_{\mathrm{adapt}}(\mathfrak{b}), \\
J_{\mathrm{seg},k} &\sim \int_{\tau_k}^{\tau_{k+1}} \|\bar{r}(\tau)\|^2 \, d\tau, \\
\Delta J^{\epsilon} &= \int \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2, \\
\Delta J^{\mathrm{model}} &= \int (\|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}), \\
\Delta J^{\mathrm{damp}} &= \int \max(\epsilon_{\mathrm{step}}^2 - \epsilon_{\mathrm{proj}}^2, 0).
\end{aligned}$$

This branch-level rollout and beam surface is retained only as a future extension. It is not the canonical live runtime surface of `fullclone_3`.

Implementation status. The repo surface in `fullclone_3` is already more structured than the original long-form derivation. It already implements checkpoint-local geometry caches, three measurement tiers, active-window and probe-position caps, position-aware zero-initialized single-parameter growth, and compile/measurement/directional/temporal-prior controller modifiers. The condensed doctrine is therefore:

> Propagate on the current scaffold by projective McLachlan. Measure miss at each checkpoint. If the miss is large, open a position-aware append search over records $r = (m, p)$. Scout by residual overlap and hardware penalties, confirm by local Schur-complement gain, and append only if the miss and gain thresholds are passed. If the scaffold is already calm and well fit, prune stale scaffold generators by a local remove-and-reintegrate test with rollback. Treat branch-level rollout and beam competition only as a future extension.

17C. Projective McLachlan Real-Time Geometry and Piecewise Beam-Adaptive Control

Below is the cleanest mathematical story for what the projected real-time implementation is doing. It is most naturally written as a **piecewise-smooth control problem on a family of variational manifolds**, with $\hbar = 1$, normalized pure states, and real parameters. The two geometric facts underneath everything are:

1. the McLachlan principle projects the exact Schrödinger velocity onto a **real-linear** tangent space using the real inner product $g(u, v) = \Re\langle u, v\rangle$, and
2. physical pure states are rays, so the natural state space is **projective Hilbert space**, which means global-phase directions should be removed from the local scoring geometry.

17C.1 What problem is being solved?

Fix a Hilbert space $\mathcal{H} \cong \mathbb{C}^d$, a Hamiltonian $H(\tau)$, and a current ansatz scaffold $\mathcal{O}$. That scaffold defines a parameterized family of states

$$|\psi(\theta; \mathcal{O})\rangle, \qquad \theta \in \mathbb{R}^{n_\theta}.$$

At physical time τ, the exact Schrödinger equation asks for the state velocity

$$\frac{d}{d\tau}|\psi_{\text{ex}}(\tau)\rangle = -iH(\tau)|\psi_{\text{ex}}(\tau)\rangle.$$

The local projected-dynamics question is therefore:

> **At the current state, how closely can the current ansatz manifold reproduce the exact instantaneous quantum velocity?**

This is the local question answered by real-time McLachlan dynamics. It is also the quantity that drives adaptive manifold growth: enlarge the ansatz when the instantaneous projected defect becomes too large.

17C.2 Why projective Hilbert space is the right setting

A normalized ket $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$ represent the same physical state. So the physical state space is projective Hilbert space $\mathbb{P}(\mathcal{H})$ rather than the unit sphere itself. At a normalized state $|\psi\rangle$, define the horizontal projector

$$Q_\psi := I - |\psi\rangle\langle\psi|.$$

For each parameter θ_j, the raw tangent vector is

$$\tau_j := \partial_{\theta_j}|\psi(\theta; \mathcal{O})\rangle,$$

and the horizontal tangent is

$$\bar{\tau}_j := Q_\psi \tau_j.$$

Collect the horizontal tangents as columns:

$$\bar{T} := [\bar{\tau}_1, \dots, \bar{\tau}_{n_\theta}] \in \mathbb{C}^{d \times n_\theta}.$$

The exact Schrödinger drift is

$$b := -iH\psi,$$

but the physically relevant local velocity is its horizontal part

$$\bar{b} := Q_\psi b = -i(H - E_\psi)\psi, \qquad E_\psi := \langle \psi | H | \psi \rangle.$$

Its squared norm is

$$\|\bar{b}\|^2 = \langle \psi | (H - E_\psi)^2 | \psi \rangle = \mathrm{Var}_\psi(H).$$

So the squared norm of the exact projective Schrödinger velocity is the energy variance. This is why $\mathrm{Var}_\psi(H)$ is the natural normalization scale for projected McLachlan defects.

17C.3 The current manifold and its allowed velocities

Because the parameters are real, the attainable instantaneous velocities form the **real-linear** tangent cone

$$T_\psi \mathcal{M}_{\mathcal{O}} = \left\{ \bar{T}\dot{\theta} : \dot{\theta} \in \mathbb{R}^{n_\theta} \right\}.$$

This is not a complex-linear subspace in general. So the correct local geometry is built from the real inner product

$$g(u, v) := \Re\langle u, v \rangle.$$

The variational velocity associated with parameter rates $\dot{\theta}$ is

$$v(\dot{\theta}) = \bar{T}\dot{\theta}.$$

17C.4 The fixed-structure McLachlan step

The fixed-structure local problem is

$$\dot{\theta}_0 = \arg\min_{\dot{\theta} \in \mathbb{R}^{n_\theta}} \|\bar{T}\dot{\theta} - \bar{b}\|^2.$$

Its quadratic form, projective metric, force vector, and solved stationarity system close to

$$\begin{aligned}
\|\bar{T}\dot{\theta} - \bar{b}\|^2 &= \|\bar{b}\|^2 - 2\Re\langle \bar{T}\dot{\theta}, \bar{b} \rangle + \dot{\theta}^\top \Re(\bar{T}^\dagger \bar{T})\dot{\theta}, \\
(\bar{G})_{ij} &= \Re\langle \bar{\tau}_i, \bar{\tau}_j \rangle = \Re\langle \partial_{\theta_i}\psi | Q_\psi | \partial_{\theta_j}\psi \rangle, \\
\bar{f}_i &= \Re\langle \bar{\tau}_i, \bar{b} \rangle = \Re\langle Q_\psi \partial_{\theta_i}\psi, -i(H - E_\psi)|\psi \rangle, \\
\mathcal{L}_0(\dot{\theta}) &= \|\bar{b}\|^2 - 2\bar{f}^\top \dot{\theta} + \dot{\theta}^\top \bar{G}\dot{\theta}, \\
\bar{G}\dot{\theta}_0 &= \bar{f}, \\
\dot{\theta}_0 &= \bar{G}^+ \bar{f} \qquad \text{if } \bar{G} \text{ is singular or rank-deficient}, \\
0 &= \Re\langle \bar{\tau}_i, \bar{T}\dot{\theta}_0 - \bar{b} \rangle \qquad \forall i.
\end{aligned}$$

17C.5 What regularization means

A stabilized runtime step solves

$$\dot{\theta}_\lambda = \arg\min_{\dot{\theta} \in \mathbb{R}^{n_\theta}} \left(\|\bar{T}\dot{\theta} - \bar{b}\|^2 + \dot{\theta}^\top \Lambda \dot{\theta} \right), \qquad \Lambda \succeq 0.$$

With

$$K := \bar{G} + \Lambda,$$

the regularized geometry closes to

$$\begin{aligned}
K\dot{\theta}_\lambda &= \bar{f}, \\
\epsilon^2_{\mathrm{proj}} &:= \min_{\dot{\theta}} \|\bar{T}\dot{\theta} - \bar{b}\|^2 = \|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}, \\
\epsilon^2_{\mathrm{step}} &:= \|\bar{T}\dot{\theta}_\lambda - \bar{b}\|^2, \\
\mathcal{L}^\star_\lambda &:= \min_{\dot{\theta}} \left(\|\bar{T}\dot{\theta} - \bar{b}\|^2 + \dot{\theta}^\top \Lambda \dot{\theta} \right) = \|\bar{b}\|^2 - \bar{f}^\top K^{-1} \bar{f} \qquad (K \text{ invertible}), \\
\mathcal{L}^\star_\lambda &= \epsilon^2_{\mathrm{step}} + \dot{\theta}_\lambda^\top \Lambda \dot{\theta}_\lambda.
\end{aligned}$$

So ϵ^2_{proj} measures intrinsic model inadequacy, ϵ^2_{step} measures the actual mismatch of the damped runtime step, and $\mathcal{L}^\star_\lambda$ is the stabilized solver objective.

17C.6 Why the residual is the right local error quantity

Choose a local gauge in which the representatives are horizontal at the current physical time τ. Then the exact and variational projective velocities are

$$\bar{b} \qquad \text{and} \qquad \bar{T}\dot{\theta}.$$

Define the horizontal residual

$$\bar{r} := \bar{T}\dot{\theta} - \bar{b}.$$

Then, for a short step h,

$$|\psi_{\text{ex}}(\tau + h)\rangle = |\psi\rangle + h\,\bar{b} + O(h^2),$$
$$|\psi_{\text{var}}(\tau + h)\rangle = |\psi\rangle + h\,\bar{T}\dot{\theta} + O(h^2),$$

so

$$|\psi_{\text{var}}(\tau + h)\rangle - |\psi_{\text{ex}}(\tau + h)\rangle = h\,\bar{r} + O(h^2).$$

So the correct local score is an **instantaneous derivative-mismatch score**. Static energy scores or novelty scores can help in secondary ranking, but they are not the primary real-time control quantity.

17C.7 The geometric projector behind the formulas

Because the tangent space is real-linear, the correct projector is

$$P^{\mathbb{R}}_{\bar{T}} v = \bar{T}\,\bar{G}^{+}\,\mathfrak{R}(\bar{T}^\dagger v).$$

The unregularized optimal residual is therefore

$$\bar{r}_0 = \bar{T}\dot{\theta}_0 - \bar{b} = \big(P^{\mathbb{R}}_{\bar{T}} - I\big)\,\bar{b}.$$

So the ansatz follows the component of the exact projective Schrödinger vector field that lies in the current real tangent plane, and the residual is the orthogonal complement.

17C.8 What a candidate generator really is

Suppose the current scaffold is an ordered product ansatz

$$|\psi(\theta; \mathcal{O})\rangle = e^{-i\theta_{n_\theta} a_{n_\theta}} \cdots e^{-i\theta_1 a_1} |\phi_0\rangle.$$

Now fix a candidate-position record $r = (m, p)$, insert the candidate generator A_m at position p, and let the new parameter ϑ_r be initialized at zero:

$$|\psi^+_r(\theta, \vartheta_r)\rangle = e^{-i\theta_{n_\theta} a_{n_\theta}} \cdots e^{-i\theta_p a_p} e^{-i\vartheta_r A_m} e^{-i\theta_{p-1} a_{p-1}} \cdots e^{-i\theta_1 a_1} |\phi_0\rangle.$$

At $\vartheta_r = 0$,

$$|\psi^+_r(\theta, 0)\rangle = |\psi(\theta)\rangle.$$

So the state is continuous, but the tangent space changes immediately. The new tangent column is

$$u_r := \partial_{\vartheta_r} |\psi^+_r(\theta, \vartheta_r)\rangle\Big|_{\vartheta_r = 0}, \qquad \bar{u}_r := Q_\psi u_r.$$

So a candidate is not merely "a promising operator." It is a **new tangent direction that becomes available at the current state without moving the state**. For a block candidate carrying q new real parameters, collect the new columns into

$$\bar{U}_r \in \mathbb{C}^{d \times q}.$$

17C.9 The exact local gain formula

Now augment the local problem:

$$\min_{\dot{\theta}, \eta} \Big(\|\bar{T}\dot{\theta} + \bar{U}_r \eta - \bar{b}\|^2 + \dot{\theta}^\top \Lambda \dot{\theta} + \eta^\top \Lambda_r \eta \Big), \qquad \Lambda_r \succeq 0.$$

With the mixed overlaps

$$
\begin{aligned}
B_{r,ia} &= \Re\langle \bar{\tau}_i, \bar{u}_{r,a}\rangle, & B_r &= \Re(\bar{T}^\dagger \bar{U}_r) \in \mathbb{R}^{n_\theta \times q},\\
C_{r,ab} &= \Re\langle \bar{u}_{r,a}, \bar{u}_{r,b}\rangle, & C_r &= \Re(\bar{U}_r^\dagger \bar{U}_r) \in \mathbb{R}^{q\times q},\\
q_{r,a} &= \Re\langle \bar{u}_{r,a}, \bar{b}\rangle, & q_r &= \Re(\bar{U}_r^\dagger \bar{b}) \in \mathbb{R}^{q},
\end{aligned}
$$

the augmented normal equations, Schur complement, and gain reduce to

$$
\begin{aligned}
\begin{bmatrix} K & B_r \\ B_r^\top & C_r + \Lambda_r \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \eta \end{bmatrix} &= \begin{bmatrix} \bar{f} \\ q_r \end{bmatrix},\\
\dot{\theta}(\eta) &= K^{-1}(\bar{f} - B_r \eta),\\
\mathcal{L}^\star_{\mathrm{aug},\lambda}(\eta) &= \mathcal{L}^\star_\lambda - 2\eta^\top w_r + \eta^\top S_r \eta,\\
S_r &= C_r + \Lambda_r - B_r^\top K^{-1} B_r,\\
w_r &= q_r - B_r^\top K^{-1} \bar{f},\\
\eta_\star &= S_r^+ w_r,\\
\Delta_{\mathrm{McL}}(r;\tau) &:= \mathcal{L}^\star_\lambda - \mathcal{L}^\star_{\mathrm{aug},\lambda} = w_r^\top S_r^+ w_r\\
&= \left(q_r - B_r^\top K^{-1}\bar{f}\right)^\top \left(C_r + \Lambda_r - B_r^\top K^{-1} B_r\right)^+ \left(q_r - B_r^\top K^{-1}\bar{f}\right).
\end{aligned}
$$

For a single real candidate direction ($q = 1$), this reduces to

$$
\Delta_{\mathrm{McL}}(r;\tau) = \frac{w_r^2}{S_r}.
$$

17C.10 What S and w mean

The Schur-complement pair and resulting score may be read in one chain:

$$
\begin{aligned}
S_r &= C_r + \Lambda_r - B_r^\top K^{-1} B_r,\\
w_r &= q_r - B_r^\top K^{-1} \bar{f},\\
\Delta_{\mathrm{McL}}(r;\tau) &= w_r^\top S_r^+ w_r\\
&= \left(q_r - B_r^\top K^{-1}\bar{f}\right)^\top \left(C_r + \Lambda_r - B_r^\top K^{-1} B_r\right)^+ \left(q_r - B_r^\top K^{-1}\bar{f}\right).
\end{aligned}
$$

So S_r is the metric of the new block after removing the part already explainable by the old tangent space, w_r is the residualized overlap with the exact projective drift, and $\Delta_{\mathrm{McL}}(r;\tau)$ is residualized force squared divided by residualized norm.

17C.11 The unregularized geometric meaning

Set $\Lambda = \Lambda_c = 0$. For a single real candidate direction $\bar{u}$, define

$$
\bar{u}_\perp := (I - P_{\bar{T}}^{\mathbb{R}})\bar{u}, \qquad \bar{r}_0 := (I - P_{\bar{T}}^{\mathbb{R}})\bar{b}.
$$

Then the exact one-direction gain closes to

$$
\begin{aligned}
\Delta_0 &= \frac{\left(\Re\langle \bar{u}_\perp, \bar{r}_0\rangle\right)^2}{\|\bar{u}_\perp\|^2}\\
&= \frac{\left(\Re\langle (I - P_{\bar{T}}^{\mathbb{R}})\bar{u}, (I - P_{\bar{T}}^{\mathbb{R}})\bar{b}\rangle\right)^2}{\|(I - P_{\bar{T}}^{\mathbb{R}})\bar{u}\|^2}.
\end{aligned}
$$

The numerator uses $\Re\langle \bar{u}_\perp, \bar{r}_0\rangle$ rather than $|\langle \bar{u}_\perp, \bar{r}_0\rangle|$ because one is adding **one real parameter**, not an arbitrary complex tangent line.

17C.12 When should the ansatz adapt?

The intrinsic normalized miss closes to

$$\begin{aligned}\rho_{\text{miss}} &:= \frac{\epsilon_{\text{proj}}^2}{\|\bar{b}\|^2+\varepsilon} = \frac{\|\bar{b}\|^2 - \bar{f}^\top \bar{G}^+ \bar{f}}{\|\bar{b}\|^2+\varepsilon} \\ &= \frac{\mathrm{Var}_\psi(H) - \bar{f}^\top \bar{G}^+ \bar{f}}{\mathrm{Var}_\psi(H)+\varepsilon} = \frac{\langle\psi|(H-E_\psi)^2|\psi\rangle - \bar{f}^\top \bar{G}^+ \bar{f}}{\langle\psi|(H-E_\psi)^2|\psi\rangle+\varepsilon}.\end{aligned}$$

Interpretation:

- $\rho_{\text{miss}} \approx 0$: the current manifold already captures the projective velocity well,
- ρ_{miss} large: the manifold is missing important directions.

A clean rule is therefore

$$\text{adapt if } \rho_{\text{miss}} > \tau_{\text{miss}}.$$

One may also adapt for numerical unreliability or after a maximum interval without any structural update.

17C.13 Which candidates should be admissible?

A candidate may have nonzero gain and still be numerically poor because it is nearly redundant. So before ranking by Δ_λ, one should inspect the residualized geometry:

- compute singular values of S, or of the corresponding whitened residualized block,
- reject candidates whose retained support is too ill-conditioned.

An invariant way to do this is to whiten the old tangent metric. If

$$\bar{G} \approx V\Sigma^2 V^\top$$

on its numerical support, define whitened coordinates

$$z := \Sigma V^\top \dot{\theta}.$$

Then penalizing $\|z\|^2$ is a penalty on **physical tangent speed** rather than coordinate speed. This is the least chart-dependent way to stabilize an overparameterized manifold.

17C.14 Two different residuals should be tracked

The projective residual is

$$\bar{r} := \bar{T}\dot{\theta}_\lambda - \bar{b}.$$

This is the correct gauge-invariant quantity for local candidate scoring and local manifold-growth decisions.

The raw Hilbert-space residual is

$$r_{\text{raw}} := T\dot{\theta}_\lambda - (-iH\psi) = T\dot{\theta}_\lambda + iH\psi.$$

These are related by

$$\bar{r} = Q_\psi r_{\text{raw}}.$$

So:

- use $\bar{r}$ for **local structure growth**,
- use r_{raw} when a rigorous Hilbert-space accumulated-error bound is desired.

17C.15 Why integrated residual is the right branch-pruning quantity

Let a branch be propagated on a fixed scaffold over a segment $[\tau_a, \tau_b]$. Let $|\psi_{\rm ex}\rangle$ be the exact state and $|\psi_{\rm var}\rangle$ the variational state with the same initial condition at τ_a, and define

$$e(\tau) := |\psi_{\rm var}(\tau)\rangle - |\psi_{\rm ex}(\tau)\rangle.$$

Then the error evolution, Duhamel representation, norm bound, and segment cost close to

$$\begin{aligned}
\dot\psi_{\rm ex} &= -iH\psi_{\rm ex}, \qquad\qquad \dot\psi_{\rm var} = T\dot\theta_\lambda, \quad r_{\rm raw} = T\dot\theta_\lambda + iH\psi,\\
\dot e &= -iHe + r_{\rm raw},\\
e(\tau) &= \int_{\tau_a}^{\tau} U(\tau,s)\, r_{\rm raw}(s)\, ds,\\
\|e(\tau)\| &\le \int_{\tau_a}^{\tau} \|r_{\rm raw}(s)\|\, ds \le \sqrt{\tau-\tau_a}\left(\int_{\tau_a}^{\tau} \|r_{\rm raw}(s)\|^2\, ds\right)^{1/2},\\
J_{\rm seg}^{\rm raw} &:= \int_{\tau_a}^{\tau_b} \|r_{\rm raw}(\tau)\|^2\, d\tau = \int_{\tau_a}^{\tau_b} \|T\dot\theta_\lambda(\tau) + iH(\tau)\psi(\tau)\|^2\, d\tau.
\end{aligned}$$

So $J_{\rm seg}^{\rm raw}$ is a natural upper-bound surrogate for accumulated state-vector error. If one wants a fully gauge-invariant pruning metric instead, replace $\|r_{\rm raw}(\tau)\|^2$ by $\|\bar r(\tau)\|^2 = \|Q_\psi r_{\rm raw}(\tau)\|^2$ and interpret the result as a ray/projective error surrogate.

17C.16 The branch objective

A branch $\mathfrak{b}$ is a sequence of scaffolds $\mathcal{O}_0, \mathcal{O}_1, \dots, \mathcal{O}_J$ used over consecutive checkpoint segments, and the cumulative objective closes as

$$\begin{aligned}
J_{\rm dyn}(\mathfrak{b}) &= \sum_{j=0}^{J-1} J_{{\rm seg},j} + \beta\, C_{\rm hw} + \gamma\, N_{\rm adapt} + \mu_E \sum_{j=0}^{J-1}\int_{\tau_j}^{\tau_{j+1}} \left|\frac{d}{d\tau}\langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\rangle_j\right|^2 d\tau\\
&= \sum_{j=0}^{J-1}\int_{\tau_j}^{\tau_{j+1}} \|r_{\rm raw}^{(j)}(\tau)\|^2\, d\tau + \beta\, C_{\rm hw} + \gamma\, N_{\rm adapt} + \mu_E \sum_{j=0}^{J-1}\int_{\tau_j}^{\tau_{j+1}} \left|\frac{d}{d\tau}\langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\rangle_j\right|^2 d\tau\\
&= \sum_{j=0}^{J-1}\int_{\tau_j}^{\tau_{j+1}} \left\|T^{(j)}(\tau)\dot\theta_\lambda^{(j)}(\tau) + iH(\tau)\psi^{(j)}(\tau)\right\|^2 d\tau + \beta\, C_{\rm hw} + \gamma\, N_{\rm adapt} + \mu_E \sum_{j=0}^{J-1}\int_{\tau_j}^{\tau_{j+1}} \left|\frac{d}{d\tau}\langle H(\tau)\rangle_j - \langle \partial_\tau H(\tau)\rangle_j\right|^2 d\tau.
\end{aligned}$$

If one wants a fully gauge-invariant branch surrogate instead, replace $\|r_{\rm raw}^{(j)}(\tau)\|^2$ by $\|\bar r^{(j)}(\tau)\|^2 = \|Q_{\psi^{(j)}} r_{\rm raw}^{(j)}(\tau)\|^2$.

17C.17 The full process, in one coherent sequence

At each physical time τ, with current scaffold $\mathcal{O}$, the full loop can be written compactly as

$$\begin{aligned}
&\text{A: } |\psi\rangle = |\psi(\theta;\mathcal{O})\rangle, \qquad Q_\psi = I - |\psi\rangle\langle\psi|, \qquad \bar T = Q_\psi T, \qquad \bar b = Q_\psi(-iH\psi) = -i(H-E_\psi)\psi,\\
&\text{B: } \bar G = \mathfrak{R}(\bar T^\dagger \bar T), \qquad \bar f = \mathfrak{R}(\bar T^\dagger \bar b), \qquad K = \bar G + \Lambda, \qquad \dot\theta_\lambda = K^+ \bar f,\\
&\quad \epsilon_{\rm proj}^2 = \|\bar b\|^2 - \bar f^\top \bar G^+ \bar f, \qquad \epsilon_{\rm step}^2 = \|\bar T\dot\theta_\lambda - \bar b\|^2, \qquad \bar r = \bar T\dot\theta_\lambda - \bar b, \qquad r_{\rm raw} = T\dot\theta_\lambda + iH\psi,\\
&\text{C: } \rho_{\rm miss} = \frac{\epsilon_{\rm proj}^2}{\|\bar b\|^2 + \varepsilon} = \frac{\epsilon_{\rm proj}^2}{{\rm Var}_\psi(H) + \varepsilon}, \qquad \text{adapt iff } \rho_{\rm miss} > \tau_{\rm miss},\\
&\text{D: } B_r = \mathfrak{R}(\bar T^\dagger \bar U_r), \qquad C_r = \mathfrak{R}(\bar U_r^\dagger \bar U_r), \qquad q_r = \mathfrak{R}(\bar U_r^\dagger \bar b),\\
&\quad S_r = C_r + \Lambda_r - B_r^\top K^+ B_r, \qquad w_r = q_r - B_r^\top K^+ \bar f, \qquad \Delta_{\rm McL}(r;\tau) = w_r^\top S_r^+ w_r.
\end{aligned}$$

For each shortlisted admissible candidate-position record $r = (m, p)$, one then creates a zero-initialized child, propagates it over the next probe segment, and prunes on the accumulated branch objective:

$$\begin{aligned}
&\text{E: } \mathcal{O} \mapsto \mathcal{O}_r^+, \qquad |\psi(\theta, 0; \mathcal{O}_r^+)\rangle = |\psi(\theta; \mathcal{O})\rangle, \\
&\text{F: } J_{\mathrm{seg},r}^{\mathrm{raw}} = \int_{\tau_a}^{\tau_b} \|r_{\mathrm{raw},r}(\tau)\|^2 \, d\tau = \int_{\tau_a}^{\tau_b} \left\| T_r(\tau)\dot{\theta}_{\lambda,r}(\tau) + iH(\tau)\psi_r(\tau) \right\|^2 d\tau, \\
&\text{G: } J_{\mathrm{dyn}}^{\mathrm{new}}(r; \mathfrak{b}) = J_{\mathrm{dyn}}^{\mathrm{old}}(\mathfrak{b}) + J_{\mathrm{seg},r}^{\mathrm{raw}} + \beta\, C_{\mathrm{hw},r} + \gamma\, N_{\mathrm{adapt},r}, \qquad \mathfrak{B}_{\mathrm{next}} = \mathrm{Prune}\Big(\{(\mathcal{O}_r^+, J_{\mathrm{dyn}}^{\mathrm{new}}(r; \mathfrak{b}))\}_r\Big).
\end{aligned}$$

That is the whole piecewise-manifold control loop.

17C.18 One concise interpretation

The method is doing this:

The exact quantum dynamics asks for a projective velocity $\bar{b}$.
The current ansatz only provides a real tangent plane.
McLachlan chooses the closest available tangent velocity.
If the miss is too large, the algorithm enlarges the tangent plane by adding zero-initialized candidate directions.
The local gain formula tells us which candidate most reduces the current miss.
Short probe rollouts then decide which structural choices remain best after actual propagation.

17C.19 Explicit symbolic–mathematical differences between the clean story and the current implementation

The clean story above is the right mathematical idealization. The current implementation differs from that idealization in a few explicit ways:

1. **Sign convention.** The clean story uses

$$\bar{b} = -iQ_\psi H\psi, \qquad \bar{r} = \bar{T}\dot{\theta} - \bar{b}.$$

The implementation instead uses

$$r = \bar{T}\dot{\theta} + iQ_\psi H\psi, \qquad f_i = \Im\langle \bar{\tau}_i | H | \psi\rangle, \qquad \Re\langle \bar{\tau}_i, -iH\psi\rangle = \Im\langle \bar{\tau}_i, H\psi\rangle.$$

1. **Regularization surface.** Rather than a general positive-semidefinite penalty Λ or Λ_c, the implementation uses a scalar $\lambda \geq 0$ on either chart coordinates or whitened metric coordinates:

$$\begin{aligned}
&K_{\mathrm{chart}} = \bar{G} + \lambda I, \\
&\quad \bar{G} \approx V\Sigma^2 V^\top, \qquad z = \Sigma V^\top \dot{\theta}, \qquad (I + \lambda I)z = \Sigma^{-1} V^\top \bar{f}, \qquad \dot{\theta} = V\Sigma^{-1} z.
\end{aligned}$$

1. **Propagation solve versus policy solve.** The clean story presents one stabilized local solve, whereas the implementation may separate the propagated step from the policy/telemetry step:

$$(\dot{\theta}_{\mathrm{prop}}, \epsilon_{\mathrm{step}}^{2,\mathrm{prop}}) \neq (\dot{\theta}_{\mathrm{policy}}, \epsilon_{\mathrm{step}}^{2,\mathrm{policy}}) \quad \text{in general,}$$

even when both are derived from the same local geometry.

1. **Variance normalization and guards.** The clean story writes

$$\rho_{\mathrm{miss}} = \frac{\epsilon_{\mathrm{proj}}^2}{\mathrm{Var}_\psi(H) + \varepsilon}.$$

The implemented ratio surface is closer to

$$\rho_{\mathrm{miss}} = \begin{cases} \epsilon_{\mathrm{proj}}^2 / (\mathrm{Var}_\psi(H) + \varepsilon), & \mathrm{Var}_\psi(H) \geq \nu_{\min}, \\ \text{suppressed}, & \mathrm{Var}_\psi(H) < \nu_{\min}, \end{cases}$$

with an auxiliary guard flag recording the suppressed regime.

1. **Active-parameter windows.** The clean story is written on the full tangent space, but the implementation may restrict to an active suffix or oracle-supplied index set:

$$T \mapsto T_{W_{\text{act}}}, \qquad \dot{\theta} \mapsto (\dot{\theta}_{W_{\text{act}}}, 0), \qquad \bar{G} \mapsto \bar{G}_{W_{\text{act}} W_{\text{act}}},$$

so inactive coordinates are frozen.

1. **Candidate gain evaluation.** The clean story scores candidates by the Schur-complement gain

$$\Delta_\lambda = w^\top S^+ w.$$

The implementation instead zero-initializes one append candidate, rebuilds the augmented local geometry, and scores by direct defect drops

$$\Delta_0 = \epsilon^2_{\text{proj,current}} - \epsilon^2_{\text{proj,aug}}, \qquad \Delta_\lambda = \epsilon^2_{\text{step,current}} - \epsilon^2_{\text{step,aug}}.$$

1. **Position-aware current branch growth.** The clean story allows arbitrary insertion positions and multi-parameter blocks. The helper layer now supports position-aware insertion and position-centered post-insert refit windows,

$$\mathcal{T}(\mathfrak{b}, m, p) = \mathfrak{b}', \qquad \mathcal{O}_{b'} = \mathcal{O}_b \oplus_p m, \qquad \theta_{b'} = \theta_b \oplus_p 0,$$

and the active probe set is built from the append slot, the left boundary, and the active reoptimization window. Hence, when the active window is made maximally large and the probe-position cap is also made large enough, the implementation can evaluate the full insertion set $p \in \{0, \dots, |\theta|\}$ rather than append only. The live surface still remains zero-initialized **single-parameter** growth, but not append-only growth.

1. **Checkpoint-trigger logic.** The clean story says "adapt when ρ_{miss} is too large." The implemented trigger surface is richer:

$$\mathbf{1}_{\text{checkpoint}} = \mathbf{1}_{\text{cadence}} \vee \mathbf{1}(\epsilon^2_{\text{proj}} > \tau_{\text{proj}}) \vee \mathbf{1}(\epsilon^2_{\text{step}} > \tau_{\text{step}}) \vee \mathbf{1}(\rho_{\text{miss}} > \tau_{\text{miss}} \text{ when defined}),$$

together with a model-versus-damping failure classification.

1. **Branch objective surface.** The clean story uses a scalar objective such as

$$J_{\text{scalar}} = \sum J_{\text{seg}} + \beta C_{\text{hw}} + \gamma N_{\text{adapt}}.$$

The implementation instead prunes by the lexicographic key

$$\kappa_{\text{prune}} = (J^{\text{rt}}, J^{\text{res}}, J^{\text{model}}, J^{\text{damp}}, J^{\text{cons}}, |\mathcal{O}|, k, \text{labels}(\mathcal{O}), \text{round}_{10}(\theta), \text{id}),$$

after deduplicating by time index, selected-operator labels, and rounded parameter values.

1. **Projective versus raw residual accumulation.** The clean story distinguishes a projective residual $\bar{r}$ from a raw Hilbert-space residual r_{raw}. The implemented rollout telemetry is closer to

$$(\Delta J^\epsilon, \Delta J^{\text{model}}, \Delta J^{\text{damp}}) = \left(\int \epsilon^2_{\text{step}}, \int \epsilon^2_{\text{proj}}, \int \max(\epsilon^2_{\text{step}} - \epsilon^2_{\text{proj}}, 0) \right),$$

rather than a separately stored raw Hilbert-space residual bound.

1. **Geometry source.** The clean story is written as if the tangent vectors and $H\psi$ are assembled directly every time. The implementation may instead start from externally supplied local data:

$$(\bar{G}, \bar{f}, \text{Var}_\psi(H)) = \mathcal{O}_{\text{geom}}(\psi, H, \mathcal{O}),$$

after which the host solves the McLachlan system on that returned geometry.

So the mathematical backbone is the same, but the implemented restricted surface closes schematically to

projective policy geometry + scalar chart/whitened damping + append-only zero-init growth
+ checkpointed short-horizon rollout + lexicographic prune key + optional oracle-supplied local ge

17C.20 Discrete spectra and density-sector observables for HH checkpoint dynamics A controller trajectory row r_n (index $n = 0, \dots, N-1$) carries either `time` or `physical_time`, and stores raw lattice observables under keys such as `site_occupations`, `site_occupations_exact`, `staggered`, `doublon`, and `doublon_exact`. The spectral stack in the repository works directly on these rows and does not re-simulate dynamics.

$$t_n := \mathrm{row}_n[\mathrm{time}_\star], \qquad \mathrm{time}_\star \in \{\texttt{time}, \texttt{physical_time}\}.$$

For each site $j \in \{0, \dots, L-1\}$, define spin-resolved occupations

$$n_j^\uparrow(t_n) = \langle \psi(t_n) | \hat{n}_{j,\uparrow} | \psi(t_n) \rangle, \quad n_j^\downarrow(t_n) = \langle \psi(t_n) | \hat{n}_{j,\downarrow} | \psi(t_n) \rangle, \quad n_j(t_n) = n_j^\uparrow(t_n) + n_j^\downarrow(t_n),$$

with $\hat{n}_{j,\sigma} = (I - \hat{Z}_{j,\sigma})/2$ under this codebase's convention. Equivalently, in the statevector path used by `hh_realtime_checkpoint_controller.py`, these are evaluated as

$$n_j^\sigma(t_n) = \sum_s p_s \, b_j^\sigma(s), \qquad D(t_n) = \sum_s p_s \sum_{j=0}^{L-1} b_j^\uparrow(s) b_j^\downarrow(s),$$

where $p_s = |\psi_s|^2$ in the computational basis and $b_j^\sigma(s) \in \{0, 1\}$ are basis bits. The repository therefore records

$$\texttt{site_occupations}[n, j] = n_j(t_n), \qquad \texttt{staggered}[n] = \frac{1}{L} \sum_{j=0}^{L-1} (-1)^j n_j(t_n), \qquad \texttt{doublon}[n] = D(t_n),$$

and the corresponding `_exact` fields for the driven exact trajectory.

$$D(t_n) = \sum_{j=0}^{L-1} n_j^\uparrow(t_n) n_j^\downarrow(t_n).$$

Here j denotes the lattice-site index, while n denotes the discrete time-sample index.

From $\{n_j(t_n)\}$ one obtains the secondary diagnostics used in the spectrum module:

$$\delta n_j(t_n) := n_j(t_n) - \frac{1}{L} \sum_{m=0}^{L-1} n_m(t_n), \qquad d_{ij}(t_n) := n_i(t_n) - n_j(t_n), \qquad m(t_n) := \frac{1}{L} \sum_{j=0}^{L-1} (-1)^j n_j(t_n).$$

(For $L = 2$, $\delta n_0 = \frac{1}{2} d_{01}$ and $\delta n_1 = -\frac{1}{2} d_{01}$.)

For spectral post-processing, let $x_n := x(t_n)$ denote any sampled scalar observable. The implemented V1 pipeline first infers or assumes a uniform step $t_{n+1} - t_n = \Delta t$, then detrends and re-centers the trace, applies either no window or a Hann window, and finally evaluates the one-sided real FFT. In this notation, $\tilde{x}_n$ denotes the detrended and re-centered signal: for constant detrending it is x_n with its sample mean removed, whereas for linear detrending it is the residual after subtracting the least-squares line in t_n, re-centered exactly as in the code path. The windowed sequence is $y_n := w_n \tilde{x}_n$, with $w_n = 1$ for no window and w_n equal to the standard Hann taper for Hann windowing.

$$Y_k = \sum_{n=0}^{N-1} y_n e^{-i2\pi kn/N}, \qquad k = 0, \dots, \lfloor N/2 \rfloor,$$

which is the one-sided coefficient returned by the real-FFT stage (implemented with `np.fft.rfft`) for a real sampled trace. The associated frequency grid is

$$\omega_k = \frac{2\pi k}{N\Delta t}, \qquad f_k = \frac{k}{N\Delta t}, \qquad \omega_{\mathrm{Ny}} = \frac{\pi}{\Delta t}, \qquad \Delta\omega = \frac{2\pi}{N\Delta t}.$$

The reported one-sided amplitude A_k is obtained from $|Y_k|$ by normalizing with the window-weight sum and doubling positive-frequency bins away from DC and Nyquist:

$$A_k = \begin{cases} \dfrac{|Y_k|}{\sum_{n=0}^{N-1} w_n}, & k = 0, \\ \dfrac{2|Y_k|}{\sum_{n=0}^{N-1} w_n}, & 1 \le k < N/2, \\ \dfrac{|Y_k|}{\sum_{n=0}^{N-1} w_n}, & k = N/2 \text{ when } N \text{ is even.} \end{cases}$$

Error-time signals are also treated as observables:

$$x_n^{\mathrm{err}} = x_n^{\mathrm{ctrl}} - x_n^{\mathrm{exact}} \quad \Rightarrow \quad \text{apply the same FFT-processing machinery to } x_n^{\mathrm{err}}.$$

When the drive frequency ω_d is known, the same centered trace is also fit to drive-locked harmonics:

$$x(t) \approx c_0 + \sum_{\ell=1}^{K_{\max}} (a_\ell \cos(\ell\omega_d t) + b_\ell \sin(\ell\omega_d t)), \quad R_\ell = \sqrt{a_\ell^2 + b_\ell^2},\ \phi_\ell = \mathrm{atan2}(b_\ell, a_\ell).$$

Here R_ℓ is the amplitude of the ℓ-th drive-locked harmonic and ϕ_ℓ is its phase.

The spectral stack is therefore a full transform layer on top of trajectory observables: raw site-resolved occupations, staggered density, doublon, and pair imbalance are each converted to sampled series, then detrend-window-FFT processed in the same one-sided-amplitude form, with controller-versus-exact residuals using the same operators.

18. Final Primitive-Closed Summary

The symbolic substitution chain is linear and closed:

1. choose the fermion ordering $p(i, \sigma)$,
2. write the Jordan-Wigner ladder primitives $\hat{c}_p^\dagger$ and $\hat{c}_p$,
3. reduce number operators to $(I - Z_p)/2$,
4. substitute them into $\hat{H}_t$, $\hat{H}_U$, and $\hat{H}_v$,
5. write $\hat{b}_i$, $\hat{b}_i^\dagger$, $\hat{n}_{b,i}$, $\hat{x}_i$, and $\hat{P}_i$,
6. substitute them into $\hat{H}_{\mathrm{ph}}$ and $\hat{H}_g$,
7. reduce the drive term to explicit identity-plus-Z coefficients,
8. propagate those operators into statevector primitives,
9. build fixed and adaptive variational manifolds from the same operators,
10. define projective McLachlan tangent geometry on those manifolds,
11. define branch growth, probe rollout, and beam pruning on top of the local defect geometry,
12. define continuation and handoff on top of those manifolds, with historical replay paths kept in the appendix,
13. define benchmark and noise-validation contracts relative to the same exact targets.

This is the point of the symbolic duplicate: the same mathematical skeleton remains, but implementation labels are stripped away so the document can serve as a clean theory-facing reference.

19. Data

This final chapter records empirical evidence for the selector and pruning claims developed in the main body. The goal is not to replace the symbolic formulas above, but to show that the live Phase-3 HH selector surfaces described in §§11 and 17 materially change the observed outcomes on matched runs.

19.1 Matched novelty ablation

The freshest clean ablation available is a matched four-run HH study with

$$L = 2, \qquad t = 1, \qquad U = 4, \qquad \omega_0 = 1, \qquad g = 0.5, \qquad n_{\mathrm{ph,max}} = 1,$$

using binary bosons, blocked ordering, open boundaries, the same seed, the same compiled backend path, the same windowed reoptimization policy, the same live pruning machinery, and the same rescue/symmetry/lifetime settings. The only changes are:

1. operator pool: `pareto_lean_l2` versus `full_meta`,
2. novelty exponent: $\gamma_N = 1$ versus $\gamma_N = 0$.

Writing the final filtered-energy error as

$$|\Delta E| = \left|E_{\text{final}} - E_{\text{exact,filtered}}\right|,$$

the matched runs give:

Pool	γ_N	E_{final}	$\lvert\Delta E\rvert$	Depth	Parameters	Stop reason	Prune kept
`pareto_lean_l2`	1.0	0.158724082365	5.37107234644 \times 10^{-5}	14	25	`drop_plateau`	4 / 5
`pareto_lean_l2`	0.0	0.486960214364	3.28624231024 \times 10^{-1}	14	23	`drop_plateau`	4 / 5
`full_meta`	1.0	0.158724082365	5.37107234644 \times 10^{-5}	13	28	`drop_plateau`	5 / 5
`full_meta`	0.0	0.486990909513	3.28652300539 \times 10^{-1}	13	22	`drop_plateau`	5 / 5

If

$$R_{\text{nov}}(\mathcal{G}) = \frac{|\Delta E|_{\gamma_N=0,\mathcal{G}}}{|\Delta E|_{\gamma_N=1,\mathcal{G}}},$$

then the two matched pool ratios are

$$R_{\text{nov}}(\texttt{pareto_lean_l2}) \approx 5.84376 \times 10^3, \qquad R_{\text{nov}}(\texttt{full_meta}) \approx 5.84431 \times 10^3.$$

So, on this matched HH benchmark, turning novelty weighting off worsens the final filtered-energy error by roughly a factor of 5.8×10^3 in both a lean pool and a heavy pool.

19.2 Interpretation of the novelty ablation

Several points survive this comparison cleanly.

1. **Novelty-on recovers the low-error line.** In both pools, $\gamma_N = 1$ reaches the same final error scale,
$$|\Delta E| \approx 5.62 \times 10^{-5}.$$
2. **Novelty-off collapses to a much worse terminal scaffold.** In both pools, $\gamma_N = 0$ ends near
$$E_{\text{final}} \approx 0.487, \qquad |\Delta E| \approx 3.28 \times 10^{-1},$$
far above the filtered exact target near 0.158668.
3. **This is not merely a depth effect.** The on/off pairs stop at the same logical depth within each pool, so the dominant difference is the operator-ranking trajectory rather than a simple "more steps" explanation.
4. **This is not merely a pruning-on versus pruning-off effect.** Pruning stays active in all four runs. What changes is the upstream ranking rule. In particular, novelty telemetry is still assembled when $\gamma_N = 0$; what is ablated is the multiplicative use of novelty inside the selector score, not the underlying geometric information itself.

These runs therefore support the claim that the reduced-tangent-space novelty factor is doing real work inside the ADAPT ranking rule rather than serving as passive telemetry.

19.3 Pool-family redundancy

The same data story also sharpens the pool-reduction claim. For the studied HH regime

$$L = 2, \qquad n_{\mathrm{ph,max}} = 1,$$

the heavy `full_meta` pool has size 46, while the lean `pareto_lean_l2` pool has size 11. Hence the pool-reduction fraction is

$$\eta_{\mathrm{pool}} = 1 - \frac{11}{46} = \frac{35}{46} \approx 0.7609.$$

Equivalently, the lean pool removes about 76% of the heavy-pool candidates while still retaining the same practical final energy scale,

$$E_{\mathrm{final}} \approx 0.15872408, \qquad |\Delta E| \approx 5.62 \times 10^{-5},$$

on the current-code lean rerun.

The generator classes discussed in this L=2 comparison may be represented symbolically by

$$\begin{aligned}
A^{\mathrm{sing}}_{ia,\sigma} &= i\left(\hat{c}^\dagger_{a\sigma}\hat{c}_{i\sigma} - \hat{c}^\dagger_{i\sigma}\hat{c}_{a\sigma}\right), \\
A^{\mathrm{dbl}}_{ijab} &= i\left(\hat{c}^\dagger_a\hat{c}^\dagger_b\hat{c}_j\hat{c}_i - \mathrm{h.c.}\right), \\
G^{(\mathrm{cloud})}_i &= \tilde{n}_i \sum_{r\in\mathcal{N}(i)} \alpha_{ir} P_r, \\
G^{(\mathrm{hd})}_{ij} &= T^{(+)}_{ij}(P_i - P_j), \\
G^{(\mathrm{dd})}_i &= D_i P_i, \\
G^{(\mathrm{od})}_{ij} &= J^{(-)}_{ij}(x_i - x_j), \\
G^{(2)}_{ij} &= T^{(+)}_{ij}(x_i - x_j)^2, \\
x_i &= \hat{b}_i + \hat{b}^\dagger_i, \qquad P_i = i(\hat{b}^\dagger_i - \hat{b}_i).
\end{aligned}$$

On this $L = 2$ benchmark, the retained/useful classes were A^{sing}, the P-type polaronic classes $G^{(\mathrm{cloud})}_i$, $G^{(\mathrm{hd})}_{ij}$, and $G^{(\mathrm{dd})}_i$, together with the quadrature seeds. By contrast, A^{dbl}, HVA, and the x-type / higher-order channels $G^{(\mathrm{od})}_{ij}$ and $G^{(2)}_{ij}$ were not needed to preserve the achieved error scale.

Empirically, the following heavy-pool families are absent from the lean pool yet are not needed to retain that achieved accuracy scale on this benchmark:

- lifted UCCSD doubles,
- HH unit terms,
- HVA layers,
- `paop_cloud_x`,
- `paop_disp`,
- `paop_dbl`,
- `paop_dbl_x`,
- `paop_curdrag`,
- `paop_hop2`.

This should be read as a scoped data claim, not as a universal proof: for the present HH $L = 2$, $n_{\mathrm{ph,max}} = 1$ target problem, the heavy pool carries substantial operator-family redundancy relative to the observed energy objective.

19.3.1 Weak-coupling $L = 2$ class-prune redo

A second $L = 2$ class-prune line at weaker coupling sharpens the same conclusion. For

$$L = 2, \qquad t = 1, \qquad U = 0.5, \qquad \omega_0 = 1, \qquad g = 0.2, \qquad n_{\mathrm{ph,max}} = 1,$$

using binary bosons, blocked ordering, open boundaries, the compiled backend path, full-window reoptimization, beam width $B = 3$, and the direct `phase3_v1` HH selector, the heavy parent `full_meta` run reached

$$E_{\mathrm{parent}} = -0.7763497696348323, \qquad |\Delta E|_{\mathrm{parent}} = 2.7761269927 \times 10^{-5}, \qquad F_{\mathrm{parent}} = 0.9999861889450502,$$

at logical depth $d_{\text{parent}} = 7$.

From that parent, a class-first pruning loop removed one operator family at a time and reran fresh ADAPT after every proposed removal. A removal was accepted only if the rerun satisfied

$$|\Delta E| \leq 5 \times 10^{-4}, \qquad F \geq F_{\text{parent}} - 10^{-4}.$$

The accepted removals were

- `hh_termwise_unit`,
- `hva_layer`,
- `paop_cloud_x`,
- `paop_curdrag`,
- `paop_dbl`,
- `paop_dbl_p`,
- `paop_dbl_x`,
- `paop_disp`,
- `paop_hop2`,
- `paop_hopdrag`,
- `hh_termwise_quadrature`.

Hence the final validated keep-set was simply

$$\mathcal{G}_{\text{lean,weak}} = \{\texttt{uccsd_sing},\ \texttt{uccsd_dbl},\ \texttt{paop_cloud_p}\}.$$

A fresh rerun using only this class-filtered pool then produced

$$E_{\text{lean}} = -0.7763497696348325, \qquad |\Delta E|_{\text{lean}} = 2.7761269927 \times 10^{-5}, \qquad F_{\text{lean}} = 0.9999861889437017,$$

at logical depth $d_{\text{lean}} = 4$. Thus, to numerical precision,

$$E_{\text{lean}} \approx E_{\text{parent}}, \qquad |\Delta E|_{\text{lean}} \approx |\Delta E|_{\text{parent}}, \qquad F_{\text{lean}} \approx F_{\text{parent}},$$

while the logical depth drops from 7 to 4.

A separate optimizer check on this same weak-coupling direct Phase-3 beam surface asked a narrower question: whether SPSA, still run locally/noiselessly with the pre-transpile proxy-cost surface (`phase3_backend_cost_mode=proxy`) and still using the same reduced class-filtered pool with full-window refits, could at least reach the practical noise-threshold regime even if it did not exactly recover the Powell basin. The best completed local SPSA artifact on that surface was

$$E_{\text{SPSA,local}} = -0.7763064877481185, \qquad |\Delta E|_{\text{SPSA,local}} = 7.1043156641 \times 10^{-5}, \qquad F_{\text{SPSA,local}} = 0.9999653825260001,$$

from the direct beam-enabled class-filtered run `20260328_hh_l2_u05_g02_local_beam_b3_spsa_case4_direct_m3200_a0p1_c0`
This did not match the Powell/class-pruned reference value $2.7761269927 \times 10^{-5}$, but it did enter the sub-10^{-4} regime and therefore satisfied the more practical gate of crossing below the anticipated noise floor. In that sense the local SPSA question at weak coupling was answered positively: exact-basin recovery remained unresolved, but threshold-level local recovery was achieved strongly enough to justify a direct noisy/backend follow-on on the same algorithmic surface.

A direct noisy transfer follow-on then kept the same reduced pool $\{\texttt{uccsd_sing}, \texttt{uccsd_dbl}, \texttt{paop_cloud_p}\}$, the same beam/full-window `phase3_v1` surface, and the same SPSA recipe $(a, c, A, \texttt{maxiter}) = (0.1, 0.02, 5.0, 3200)$, but turned on `backend_scheduled` FakeMarrakesh scoring with `expectation_v1`, `512` shots, `1` repeat, and no mitigation; the exact command is recorded at `artifacts/agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrakesh_`
During that supervised run, the best observed heartbeat reached

$$E_{\text{SPSA,noisy,best}} = -0.7762934955401648, \qquad |\Delta E|_{\text{SPSA,noisy,best}} = 8.4035364595 \times 10^{-5},$$

at depth 95, as preserved in `artifacts/agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrakesh_spsa_case4`
So, before final max-depth termination, the same reduced-pool direct Phase-3 beam SPSA lane already crossed the practical noisy sub-10^{-4} regime under transpiled FakeMarrakesh selection/scoring.

19.3.2 Live four-run supervised status block

```
Objective<As of March 28, 2026 at 10:59:12 CDT, all 4 noisy
  reduced-pool backend_scheduled SPSA runs are still genuinely
  running with live child PIDs and no stderr in artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrake
  sh_spsa_case4_recipe_attempt4_dropstop/progress.json,
  artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrake
  sh_spsa_case4_recipe_attempt5_dropstop_lifetimeoff/
  progress.json, artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b4_c3_k4_fake_m
  arrakesh_spsa_case4_recipe_attempt6_dropstop_lifetimeoff/
  sh_spsa_case4_recipe_attempt7_dropstop_transpile_single/
  progress.json. Best observed noisy gaps so far are: attempt4
  3.3871153069520155e-05 at depth 76 in artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrake
  sh_spsa_case4_recipe_attempt4_dropstop/logs/stdout.log,
  attempt6 3.733262477279009e-05 at depth 49 in artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b4_c3_k4_fake_m
  arrakesh_spsa_case4_recipe_attempt6_dropstop_lifetimeoff/
  logs/stdout.log, attempt7 3.899366517123859e-05 at depth 67
  in artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrake
  sh_spsa_case4_recipe_attempt7_dropstop_transpile_single/logs/
  stdout.log, and attempt5 1.0855648531227224e-04 at depth 67
  in artifacts/
  agent_runs/20260328_hh_l2_u05_g02_phase3_beam_b3_fake_marrake
  sh_spsa_case4_recipe_attempt5_dropstop_lifetimeoff/logs/
  stdout.log.>
  Why/Intent<So the current result ordering is: baseline clean-
  stop attempt4 best, then wider-beam+lifetime-off attempt6,
  then transpile_single_v1 attempt7, with lifetime-off only
  attempt5 trailing. None has written a final JSON yet because
  the clean stop is intentionally gated by --adapt-drop-min-
  depth 96, so even though three runs are already well below
  the practical 1e-4 threshold, the normal drop_plateau exit
  cannot fire before depth 96.>
  Suggested Next step/how this fits into broader picture<I
  recommend we keep these running until the first one crosses
  depth 96 and exits normally, because that will give us the
  first clean final JSON on this surface without interrupting a
  good basin. If the ranking stays similar, attempt4 should
  finish first, while attempt6 and attempt7 already look
  scientifically competitive enough to justify keeping them
  alive as the main comparison axes.>
```

The rejected removals show which families remained indispensable under this contract:

- removing `paop_cloud_p` gave $|\Delta E| \approx 1.0813 \times 10^{-2}$ and $F \approx 0.996632$,
- removing `uccsd_sing` gave $|\Delta E| \approx 2.19934$ and $F \approx 0.214233$,
- removing `uccsd_dbl` gave $|\Delta E| \approx 1.31029 \times 10^{-2}$ and $F \approx 0.996691$.

So at this weaker-coupling $L = 2$ point the validated lean pool is even smaller than the earlier heavy/lean $U = 4$, $g = 0.5$ line: the heavy search may still visit quadrature seeds, but the final class-level support needed to preserve the achieved energy/fidelity scale is just

$$\texttt{uccsd_sing} \cup \texttt{uccsd_dbl} \cup \texttt{paop_cloud_p}.$$

For the backend-facing follow-up at this same weak-coupling point, the Heron/Marrakesh optimization should be done as a **transpile-only** sweep rather than through the noisy staged wrapper: run `python -m pipelines.hardcoded.adapt_circuit_cost` on the promoted locked scaffold `hh_l2_u05_g02_full_meta_class_pruned_lean4` over `\texttt{optimization\_level}\in\{1,2\}` and `\texttt{seed\_transpiler}\in\{0,\dots,9\}`, then rank by compiled two-qubit count, depth, and size. On `\texttt{FakeMarrakesh}` the winning setting was `\texttt{optimization\_level}=2`, `\texttt{seed\_transpiler}=5`, reducing the same 4-operator, 6-runtime-term scaffold from about `25` two-qubit gates, depth `63`, size `109` to `18` two-qubit gates, depth `43`, size `83` without changing the physics or operator order.

19.4 Relation to live Phase-3 selector/pruning implementation

The canonical Phase-3 target contract is the one stated in §§11.4.1–11.4.3. Current live repo code still lags that contract in two visible ways.

First, the live selector path still applies a hard uncertainty-adjusted gradient pre-cap before the cheap screen and still uses a raw remaining-depth / remaining-evaluations proxy inside the lifetime-burden term. The corrected canonical contract in §11.4.1 instead ranks the full admissible candidate-position pool with the horizon-smoothed cheap score $S_{3,\mathrm{cheap}}(r;t)$, forms a horizon-sized coarse shortlist $\mathcal{C}_{\mathrm{cheap}}$, and uses the useful horizon

$$H_t = \min\bigl(D_{\mathrm{left}}(t), \widehat{N}_{\mathrm{rem}}(t)\bigr)$$

inside the lifetime multiplier. Here $\widehat{N}_{\mathrm{rem}}(t)$ is controller telemetry and therefore a ranking/burden aid, not a hard admissibility theorem.

Second, the live pruning path still ranks removals primarily by small amplitude together with weaker prior proxy-benefit tie-breaks and accepts removals through a standalone post-refit regression threshold. The corrected canonical contract in §11.4.3 instead separates prune permissibility from local expendability, ranks permissible removals by frozen-ablation loss, and uses retained admitted-step gain as the principal energetic safety condition. Thus frozen ablation is to be read as a local, solver-relative expendability heuristic, while the gain-retention inequality is the actual prune safety condition.

So the honest status note is: current code still trails the canonical Phase-3 mathematics, but the intended direction of alignment is

$$\text{live implementation} \longrightarrow \text{corrected canonical contract of §§11.4.1–11.4.3},$$

not the reverse.

19.5 Data provenance note

The novelty-ablation numbers in this chapter come from the fresh direct `adapt_pipeline` artifacts `pi_ablate_pareto_lean_l2_n` `pi_ablate_pareto_lean_l2_novelty_off.json`, `pi_ablate_full_meta_novelty_on.json`, and `pi_ablate_full_meta_novel` all dated `2026-03-24`. The heavy-versus-lean pool-family comparison is anchored by the `2026-03-21` comparison bundle summarized in `pareto_lean_l2_vs_heavy_L2_ecut1_comparison.md` together with the corresponding lean rerun artifact. The data are therefore matched and current enough for PI-facing evidence.

The honesty caveat is the same one stated in the reporting notes: these matched ablations share the same physics, seed, and selector family, but they are not claimed to be byte-for-byte reproductions of every earlier March 21/22 runtime-resolved default. That caveat does not weaken the on/off comparisons above, because the resolved settings were shared inside each matched pair.

19.6 Machine-agent redo note for the successful L=2 pruning session

For the validated HH prune line, keep the problem fixed at

$$L = 2, \qquad t = 1, \qquad U = 4, \qquad \omega_0 = 1, \qquad g = 0.5, \qquad n_{\mathrm{ph,max}} = 1,$$

with the saved fullhorse parent artifact as the starting point. A machine agent can reproduce the session in three steps from repo root:

```
python -m pipelines.hardcoded.hh_prune_nighthawk \
  --input-json artifacts/json/hh_backend_adapt_fullhorse_powell_20260322T194155Z_fakenighthawk.json \
  --mode both \
  --prune-threshold 1e-4 \
  --output-json artifacts/json/hh_prune_nighthawk_<timestamp>.json

python -m pipelines.hardcoded.hh_prune_nighthawk \
  --mode export_fixed_scaffolds \
  --source-artifact-json artifacts/json/hh_prune_nighthawk_aggressive_5op.json

python -m pipelines.hardcoded.adapt_circuit_cost \
  --artifact-json artifacts/json/hh_prune_nighthawk_gate_pruned_7term.json \
  --backend-name ibm_marrakesh \
  --seed-transpiler 7 --optimization-level 1 \
  --sweep-backends
```

The redo is not complete unless it records **fidelity as well as energy**. For each surviving scaffold and exported fixed scaffold, save at least: `abs_delta_e`, exact-state fidelity (`exact_state_fidelity`, `local_exact_state_fidelity`, or `expected_local_exact_state_fidelity`), retained exyz labels, backend, transpiler seed, optimization level, compiled `count_2q`, compiled depth, and compiled size. The validated executable anchor is the `gate_pruned_7term` artifact, not `circuit_optimized_7term`: the trustworthy target is

$$|\Delta E| \approx 4.23 \times 10^{-4}, \qquad F \approx 0.9998,$$

with Marrakesh compile cost near `25` two-qubit gates and depth `63-75`. A leaner 6-term executable may reach about `14` two-qubit gates and depth `48`, but its exact regression is materially worse, near

$$|\Delta E| \approx 4.61 \times 10^{-3}.$$

19.7 Direct noisy `backend_scheduled` L=2 fastprobe note

For the same HH point

$$L = 2, \qquad t = 1, \qquad U = 4, \qquad \omega_0 = 1, \qquad g = 0.5, \qquad n_{\mathrm{ph,max}} = 1,$$

a useful data line is the direct local noisy `phase3_v1` fastprobe on `FakeNighthawk` with `backend_scheduled` oracle evaluation, `512` shots, `2` repeats, and optional `mthree` readout mitigation. The March 26 command family is recorded in `artifacts/runstate/hh_phase3_v1_local_noisy_backend_scheduled_fastprobe_retry2_20260326_cmd.sh` and the stabilized readout follow-up `artifacts/runstate/hh_phase3_v1_local_noisy_backend_scheduled_fastprobe_retry` with the no-readout comparator in `artifacts/runstate/hh_phase3_v1_local_noisy_backend_scheduled_noreadout_retry1`

These probes were intentionally weak (`\texttt{adapt\_max\_depth}=1`) and were stopped after the first scout evaluation timing split was captured, so they are debug-quality runtime evidence rather than convergence-quality VQE runs. The first measured `plus` evaluation on the readout+mthree path gave

$$T_{\mathrm{total,readout}} \approx 37.04\,\mathrm{s}, \qquad T_{\mathrm{term}} \approx 26.39\,\mathrm{s}, \qquad T_{\mathrm{calib}} \approx 10.64\,\mathrm{s},$$

with readout-apply time negligible at about `0.01s`. The no-readout comparator gave

$$T_{\mathrm{total,no\,readout}} \approx 26.84\,\mathrm{s}, \qquad T_{\mathrm{term}} \approx 26.83\,\mathrm{s}.$$

So on this $L = 2$ HH debug line the dominant cost is backend-scheduled term execution itself, while readout mitigation contributes a secondary first-hit warmup rather than the main runtime burden. The practical lesson is that `backend_scheduled` is honest enough for a very small late shortlist or confirmation pass, but too expensive to act as the broad pre-shortlist scout surface.

19.8 Fixed-scaffold compile-control scout note

For the locked Marrakesh/Heron 6-term HH scaffold `artifacts/json/hh_marrakesh_fixed_scaffold_6term_drop_eyezee_20` the useful local compile-control test is a transpile-only scout on `FakeMarrakesh` with fixed circuit/parameters,

shots=4096, oracle_repeats=8, readout/mthree mitigation, and grid (optimization_level, seed_transpiler) $\in \{1,2\} \times \{0,\dots,4\}$, ranking by $\Delta E = E_{\text{noisy}} - E_{\text{ideal}}$ then by compiled cost; among the saved 8/10 completed candidates in artifacts/agent_runs/20260328_direct_fixed_scaffold_compile_scout_fullaccess_attempt2_timeout7200/j the best-so-far setting was opt1_seed4 with $\Delta E \approx 9.9899 \times 10^{-2}$, two-qubit count 14, and depth 48, whereas the completed lower-depth opt2_* candidates kept the same two-qubit count 14 but had depth 38 and worse ΔE: opt2_seed0\approx 1.1726\times 10^{-1}, opt2_seed1\approx 1.2245\times 10^{-1}, opt2_seed2\approx 1.2029\times 10^{-1}, so lower compiled depth alone did not improve noisy accuracy and the current hardware-facing compile preset is (optimization_level=1, seed_transpiler=4).

As an $L = 2$ run-planning prior rather than a theorem, the earlier local FakeMarrakesh ablations were mixed between readout-only and twirl while DD usually underperformed, but the completed 12-cell saved-$\theta^\star$ mitigation matrix in artifacts/json/20260328_fixed_scaffold_saved_theta_mitigation_matrix_attempt5_timeout86400_delta_logs_fixe now supersedes that partial prior: the top three cells were opt2_seed0__zne_on__twirl_dd with $\Delta E \approx -6.6431 \times 10^{-4}$ and stderr $\approx 1.2322 \times 10^{-2}$ at 14 two-qubit gates and depth 38, opt1_seed4__zne_on__twirl_dd with $\Delta E \approx 8.2990 \times 10^{-3}$ and stderr $\approx 2.6104 \times 10^{-2}$ at 14 two-qubit gates and depth 48, and opt2_seed0__zne_on__twirl with $\Delta E \approx 1.4807 \times 10^{-2}$ and stderr $\approx 1.1157 \times 10^{-2}$ at 14 two-qubit gates and depth 38. For a real-QPU first pass on this locked 6-term scaffold, the current best local starting lane is therefore (optimization_level=2, seed_transpiler=0) with ZNE on + twirl + DD, with (optimization_level=1, seed_transpiler=4) plus the same mitigation stack as the nearest fallback if one wants to hedge against ZNE-fit variance while staying on the same two-qubit count.

For the weak-coupling locked scaffold artifacts/json/useful/L2/hh_l2_u05_g02_full_meta_class_pruned_lean4_locked_s the completed 4-cell readout+mthree shortlist in artifacts/json/20260329_weak_fixed_scaffold_top4_mitigation_readou had top two cells opt2_seed5__zne_on__twirl_dd with $\Delta E \approx 1.0800 \times 10^{-2}$ and stderr $\approx 4.7014 \times 10^{-3}$, and opt2_seed5__zne_on__dd with $\Delta E \approx 1.6984 \times 10^{-2}$ and stderr $\approx 1.9163 \times 10^{-3}$, both at 18 two-qubit gates and depth 43.

The later readout-on versus readout-off ablations showed that base readout mitigation was helping rather than hurting on both fixed-scaffold winner lines. For the strong-coupling local winner, the same opt2_seed0__zne_on__twirl_dd lane moved from $\Delta E \approx -6.6431 \times 10^{-4}$ with readout on in artifacts/json/20260328_fixed_scaffold_saved_theta_mitigati to $\Delta E \approx 5.4456 \times 10^{-2}$ with readout off in artifacts/json/20260329_strong_fixed_scaffold_winner_readout_off_ablatio at the same compiled cost (14 two-qubit gates, depth 38). For the weak-coupling 18-two-qubit compile opt2_seed5, the analogous readout-off follow-up in artifacts/json/20260330_weak_fixed_scaffold_top2_readout_off_ablation_opt2s gave opt2_seed5__zne_on__twirl_dd at $\Delta E \approx 5.6830 \times 10^{-2}$ and opt2_seed5__zne_on__dd at $\Delta E \approx 5.8131 \times 10^{-2}$, versus the readout-on values $\approx 1.0800 \times 10^{-2}$ and $\approx 1.6984 \times 10^{-2}$ respectively, so turning readout off worsened the best weak lanes by about 4.60×10^{-2} and 4.11×10^{-2}.

The surprising regime comparison is therefore not in the noiseless scaffold bias but in the noisy response. The strong locked scaffold had $E_{\text{ideal}} - E_{\text{exact}} \approx 4.6131 \times 10^{-3}$ from artifacts/json/hh_marrakesh_fixed_scaffold_6term_drop_eyezee_2 while the weak locked scaffold had $E_{\text{ideal}} - E_{\text{exact}} \approx 2.7761 \times 10^{-5}$ from artifacts/json/useful/L2/hh_l2_u05_g02_full_meta_ Thus the weak scaffold was much closer to exact in the noiseless sense, yet still produced a worse noisy ΔE after the current suppression stack; the deficit is therefore a noise-response / mitigation-effect issue rather than a larger ideal-state approximation error.

19.9 Heron transpile Pareto front and noisy replay for fixed L=2 scaffolds

All scaffolds share $L = 2$, $n_{\text{ph,max}} = 1$, binary bosons, blocked ordering, open boundaries. Each was transpiled to FakeMarrakesh over (optimization_level, seed_transpiler) $\in \{1,2\} \times \{0,\dots,9\}$; the table reports the best compiled two-qubit count per scaffold. Term-pruned variants start from the 7-term locked scaffold and drop Pauli rotation terms, then reoptimize θ via Powell.

Scaffold	$\lvert\Delta E\rvert$	CZ gates	Depth	F	Coupling	Source
weak_lean4_locked (4 ops)	2.78×10^{-5}	18	43	0.9999	U=0.5, g=0.2	class-prune
aggressive_5op (12 terms)	5.62×10^{-5}	26	72	—	U=4, g=0.5	op-prune
gate_pruned_7term (7 terms)	4.23×10^{-4}	21	51	0.9998	U=4, g=0.5	term-prune

Scaffold	$\lvert\Delta E\rvert$	CZ gates	Depth	F	Coupling	Source
`6term_drop_eyezee` (6 terms)	4.61×10^{-3}	14	38	0.9985	U=4, g=0.5	term-prune
`6term_drop_eyeeez` (6 terms)	4.61×10^{-3}	14	56	0.9985	U=4, g=0.5	term-prune
`5term_drop_both_drop` (5 terms)	6.92×10^{-3}	12	48	0.9986	U=4, g=0.5	term-prune

The Pareto front over ($|\Delta E|$, CZ count) contains two points when both coupling regimes are pooled:

$$(2.78 \times 10^{-5},\ 18\ \mathrm{CZ})_{\mathrm{weak}}, \qquad (4.61 \times 10^{-3},\ 14\ \mathrm{CZ})_{\mathrm{strong}}.$$

Within strong coupling alone, the three-point front is $\{(5.62 \times 10^{-5}, 26),\ (4.61 \times 10^{-3}, 14),\ (6.92 \times 10^{-3}, 12)\}$. The dominated scaffolds (`pruned_scaffold_11op` at 51 CZ, `ultra_lean_6op`/`readapt_6op` at 38 CZ, `gate_pruned_7term` at 21 CZ) offer no accuracy benefit relative to cheaper alternatives.

Noisy replay on `FakeMarrakesh` (`backend_scheduled`) for the strong-coupling Pareto-optimal 6-term scaffold (14 CZ, depth 38, `optimization_level=2`, `seed_transpiler=0`):

Configuration	E_{noisy}	$E_{\mathrm{noisy}} - E_{\mathrm{exact}}$
Saved θ^* + readout/mthree	0.2773	+0.119
Saved θ^* + readout + gate twirling	0.2208	+0.062
SPSA-128 best + readout + gate twirling	0.1368	−0.022
Saved θ^* ideal (noiseless)	0.1633	+0.005
Exact GS	0.1587	0

Mitigation stack: readout/mthree plus local 2Q Pauli twirling, symmetry `verify_only`, no ZNE, no DD. SPSA: 128 iterations, 1024 shots, 1 repeat, seed 19. The SPSA-optimized noisy energy undershoots exact by 0.022, indicating noise-induced bias below the variational floor; the ideal energy at the SPSA-best θ is 0.189, confirming parameter drift. At the noiseless-optimal θ^*, gate twirling halves the noise bias from 0.119 to 0.062.

Reproduction:

```
python -m pipelines.exact_bench.heron_pareto_sweep --backend FakeMarrakesh --coupling both
python -m pipelines.hardcoded.hh_staged_noise \
  --L 2 --t 1.0 --u 4.0 --g-ep 0.5 --omega0 1.0 --n-ph-max 1 \
  --ordering blocked --boundary open --boson-encoding binary \
  --include-fixed-scaffold-noisy-replay \
  --fixed-final-state-json artifacts/json/hh_marrakesh_fixed_scaffold_6term_drop_eyezee_20260323T171528Z.j
  --use-fake-backend --backend-name FakeMarrakesh \
  --fixed-scaffold-runtime-transpile-optimization-level 2 \
  --fixed-scaffold-runtime-seed-transpiler 0 \
  --final-method SPSA --final-maxiter 128 --final-seed 19 \
  --mitigation readout --local-readout-strategy mthree --local-gate-twirling \
  --symmetry-mitigation-mode verify_only \
  --shots 1024 --oracle-repeats 1 --noise-seed 7 \
  --smoke-test-intentionally-weak --skip-pdf
```

Artifacts: `artifacts/json/heron_pareto_front_20260327T223120Z.json`, `artifacts/json/hh_6term_pareto_readout_twi

Appendix A. Drive, Exact Propagation, and Propagator Semantics

A.1 Static drive labels and time-dependent coefficients

Let $\mathcal{L}_{\text{drive}}$ denote the fixed set of density operators appearing in the drive. Then the drive always has the form

$$\hat{H}_{\text{drive}}(t) = \sum_{\lambda \in \mathcal{L}_{\text{drive}}} c_\lambda(t) P_\lambda,$$

with fixed labels P_λ and time-dependent coefficients $c_\lambda(t)$.

A.2 Reference coefficient map

For a site-density drive,

$$\Delta c[Z_{p(i,\uparrow)}](t) = \Delta c[Z_{p(i,\downarrow)}](t) = -\frac{1}{2} v_i(t),$$

and the identity shift is

$$\Delta c[I](t) = \sum_i v_i(t)$$

if one chooses to retain the scalar offset explicitly.

A.3 Propagator surfaces

A common propagator menu is:

- second-order Suzuki,
- piecewise exact propagation.

A.4 Reference propagator versus reported trajectory propagator

Write the reported propagator as $U_{\text{traj}}(t)$ and the high-accuracy reference propagator as $U_{\text{ref}}(t)$. These need not be the same approximation.

A.4.1 Static reference branch

In the static case, one may take

$$U_{\text{ref}}(t) = e^{-it\hat{H}}.$$

A.4.2 Drive-enabled reference branch

For time-dependent $\hat{H}(t)$, a refined piecewise-constant reference is

$$U_{\text{ref}}(t_{n+1}, t_n) \approx e^{-i\Delta t \hat{H}(t_n^*)},$$

with t_n^* sampled on a refined grid and the full reference obtained as an ordered product over the refined slices.

A.5 Filtered exact manifold and subspace fidelity

Let Π_{sector} denote the projector onto the target symmetry sector. Then the filtered exact energy is

$$E_{\text{exact,filtered}} = \min_{\psi \in \Pi_{\text{sector}} \mathcal{H}} \langle \psi | \hat{H} | \psi \rangle,$$

and a sector-aware fidelity may be defined by

$$F_{\text{sector}}(t) = \frac{|\langle \psi_{\text{exact}}(t) | \Pi_{\text{sector}} | \psi(t) \rangle|^2}{\langle \psi(t) | \Pi_{\text{sector}} | \psi(t) \rangle}.$$

A.7 Branch-resolved observable contracts

If several trajectory branches are carried simultaneously, each observable should be indexed by branch label b:

$$\mathcal{O}_b(t), \qquad b \in \{\mathrm{var}, \mathrm{traj}, \mathrm{ref}, \mathrm{noisy}, \mathrm{ideal}\}.$$

Typical observables are energy, density, doublon density, and fidelity.

A.8 State import and handoff contract

A portable state bundle can be written abstractly as

$$\mathcal{B} = (\Lambda, \theta, a, E, \mathcal{C}),$$

where Λ is the physical manifest, θ are variational parameters when available, a is an amplitude map for the working state, E stores energy summaries, and $\mathcal{C}$ is continuation metadata.

A.9 Highest-tier continuation surfaces

At the highest continuation tier, $\mathcal{C}$ may include split-event summaries, motif data, symmetry diagnostics, replay policy data, and limited branch histories. These are naturally viewed as auxiliary mathematical metadata rather than as part of the operator algebra itself.

Appendix B. Continuation and Handoff Contract

B.1 Handoff state bundle

A handoff bundle may contain:

- the parameter manifest Λ,
- the current energy summary,
- the normalized statevector amplitudes,
- an exact comparison target,
- optional continuation metadata.

One convenient symbolic amplitude map is

$$\{b_{N_q-1} \cdots b_0 \mapsto (\Re a_b, \Im a_b)\}.$$

B.2 Continuation block

The continuation payload can be treated abstractly as

$$\mathcal{C} = (m, \mathcal{S}, \mathcal{M}_{\mathrm{opt}}, \Gamma, \Sigma, \mathcal{L}, \mathcal{U}, \Xi, \mathcal{R}, \mathcal{P}),$$

where:

- m is the continuation mode,
- $\mathcal{S}$ is inherited manifold-structure data,
- $\mathcal{M}_{\mathrm{opt}}$ is auxiliary optimizer state,
- Γ is selected-generator metadata,
- Σ is split-event data,
- $\mathcal{L}$ is a motif library,
- $\mathcal{U}$ is motif usage,
- Ξ is symmetry information,
- $\mathcal{R}$ is rescue or recovery history,
- $\mathcal{P}$ is optional historical replay-policy metadata when such provenance is still being carried.

When controller forecast telemetry is persisted, quantities such as $D_{\mathrm{left}}(t)$, $\widehat{N}_{\mathrm{rem}}(t)$, and H_t belong naturally to this policy/provenance layer. They remain heuristic controller state rather than operator-level observables.

B.3 General handoff map

Let $\mathcal{H}_{A\to B}$ be a handoff map from source manifold $\mathcal{M}_A$ to target manifold $\mathcal{M}_B$:

$$\mathcal{H}_{A\to B} : (|\psi_A\rangle, \theta_A, \mathcal{C}_A) \mapsto (|\psi_B^{(0)}\rangle, \theta_B^{(0)}, \mathcal{C}_B).$$

The map is intentionally abstract here: it records state, coordinates, and continuation payload transfer between manifolds without committing the main body to any obsolete replay-specific burn-in policy.

B.4 Symmetry and tier-three payload note

Symmetry information belongs in the handoff if later stages will use it for validation, postselection, or projector-renormalized estimation. Mathematically, the carried object is a projector or a list of commuting symmetry generators.

B.5 Historical warm/refine chain note

The obsolete warm-start → refine → ADAPT → replay path is kept only in Appendix A.2 and is not part of the active main-body symbolic contract.

Appendix C. Cross-Checks and Exact-Benchmark Contracts

C.1 Trial matrix by problem family

For the Hubbard model, a natural benchmark trial set is

$$\mathcal{T}_{\text{Hub}} = \{\text{layerwise, excitation-based, adaptive(exc), adaptive(full-H)}\}.$$

For the HH model, the canonical sector-preserving core set is

$$\mathcal{T}_{\text{HH}} = \{\text{HH-layerwise, HH-physical-termwise, adaptive(full-H)}\}.$$

A seed-surface extension can enlarge $\mathcal{T}_{\text{HH}}$ by adding polaronic refinements built on the same sector-preserving fixed-family surface.

C.2 Auto-scaled parameter resolution

Let

$$S_{\text{trot}}(L, n_{\text{ph,max}}), \quad R_{\text{restart}}(L, n_{\text{ph,max}}), \quad I_{\max}(L, n_{\text{ph,max}})$$

be monotone nondecreasing control maps for Trotter steps, restart count, and optimizer effort. The key contract is monotonicity with problem size, not a single universal constant setting.

C.3 Exact target and shared reference-state construction

Within a benchmark matrix, all trial families should be compared against the same exact target and the same conserved-sector reference construction. Thus every trial is judged against the same $E_{\text{exact,sector}}$ and the same physical Hamiltonian instance.

C.4 Per-trial energy and trajectory semantics

Each trial should at minimum record:

- the variational ground-state energy,
- the exact comparison target,
- trajectory observables on a shared time grid when dynamics are studied,
- fidelity or overlap relative to a common reference branch.

C.5 Artifact contracts

Every report artifact should begin with a concise parameter manifest listing:

- model family,
- ansatz family,
- drive enabled or disabled,
- t, U, ω_0, g, and $n_{\mathrm{ph,max}}$,
- any propagation parameters needed to reproduce the result.

C.6 HH seed-surface preset and resource proxies

A seed-surface comparison is naturally evaluated not only by energy but also by simple resource proxies such as

$$C_{\mathrm{2q}}, \qquad D_{\mathrm{proxy}}, \qquad C_{\mathrm{1q}}.$$

One can then compare improvement per marginal resource,

$$\frac{\Delta E}{C_{\mathrm{2q}}}, \qquad \frac{\Delta E}{D_{\mathrm{proxy}}}.$$

Appendix D. Noise Validation, Symmetry-Mitigation, and Parity Contracts

D.1 Filtered versus full exact targets

The filtered target is

$$E_{\mathrm{exact,filtered}} = \min_{\psi \in \mathcal{H}_{\mathrm{sector}}} \langle \psi | \hat{H} | \psi \rangle,$$

while the unrestricted target is

$$E_{\mathrm{exact,full}} = \min \mathrm{spec}(\hat{H}).$$

The two coincide only if the physical and computational sectors match exactly.

D.2 Noisy-minus-ideal observable deltas

For any observable $\mathcal{O}(t)$, define

$$\Delta\mathcal{O}(t) = \mathcal{O}_{\mathrm{noisy}}(t) - \mathcal{O}_{\mathrm{ideal}}(t).$$

In particular,

$$\Delta E(t) = E_{\mathrm{noisy}}(t) - E_{\mathrm{ideal}}(t), \qquad \Delta D(t) = D_{\mathrm{noisy}}(t) - D_{\mathrm{ideal}}(t).$$

D.3 Anchor parity gate

If a new idealized path is compared against a fixed anchor trajectory, a strict parity gate takes the form

$$\max_{t \in \mathcal{T}} |\mathcal{O}_{\mathrm{new}}(t) - \mathcal{O}_{\mathrm{anchor}}(t)| \leq \varepsilon_{\mathrm{parity}}.$$

This is an anchor-based equality test, not merely a qualitative agreement statement.

D.4 Same-anchor paired comparison semantics

Two noisy methods should be compared against the same ideal anchor, on the same time grid, with the same initial state, so that differences can be attributed to the noise or mitigation model rather than to different starting conditions.

D.5 Symmetry-mitigation mode surface

Three increasingly strong symmetry surfaces are:

1. verify-only sector diagnostics,
2. bitstring postselection,
3. projector-renormalized estimators.

If Π is the projector onto the target symmetry sector, the renormalized estimator is

$$\langle O \rangle_{\Pi} = \frac{\langle \psi | \Pi O \Pi | \psi \rangle}{\langle \psi | \Pi | \psi \rangle}.$$

D.6 Parameter manifest and provenance digest

Even in a symbolic setting, it is useful to view each artifact as carrying a manifest $\mathfrak{M}$ and a provenance digest $h(\mathfrak{M})$. The purpose is not algebraic; it is to guarantee that the reported mathematics can be traced back to a unique physical parameter set.

Appendix E. Spec-only or not-yet-formalized surfaces kept out of the main body

E.1 Broader continuation-theory items

Several topics belong naturally in a later symbolic pass rather than in the present core manuscript:

- multi-source motif transfer and motif tiling,
- richer split-event formalism,
- more detailed rescue dynamics,
- exact handoff-policy closure beyond the present abstract maps,
- measured-geometry and backend-specific closure beyond the exact symbolic statevector surface.

E.2 Obsolete historical warm-start → refine → ADAPT → replay path

The historical staged path can be written symbolically as

$$\begin{aligned}
\mathcal{M}_{\text{warm}} &= \{\, U_{\text{warm}}(\theta_{\text{w}}) |\phi_0\rangle : \theta_{\text{w}} \in \mathbb{R}^{m_{\text{w}}} \,\}, \\
\mathcal{M}_{\text{refine}} &= \{\, U_{\text{refine}}(\theta_{\text{r}}) U_{\text{warm}}(\theta_{\text{w}}^*) |\phi_0\rangle : \theta_{\text{r}} \in \mathbb{R}^{m_{\text{r}}} \,\}, \\
\mathcal{M}_{\text{replay}}(\mathcal{O}_*) &= \{\, U_{\text{replay}}(\varphi; \mathcal{O}_*) |\phi_0\rangle : \varphi \in \mathbb{R}^{m_{\text{rep}}} \,\}.
\end{aligned}$$

The corresponding historical handoff chain is

$$\begin{aligned}
(\theta_{\text{w}}^*, |\psi_{\text{warm}}\rangle) \in \mathcal{M}_{\text{warm}} &\mapsto (\theta_{\text{r}}^*, |\psi_{\text{refine}}\rangle) \in \mathcal{M}_{\text{refine}} \\
&\mapsto (\mathcal{O}_*, \theta_*^{\text{adapt}}, |\psi_*\rangle) \\
&\mapsto (\varphi^{(0)}, |\psi_{\text{replay}}^{(0)}\rangle) = \mathcal{H}_{\text{replay}}(|\psi_*\rangle, \theta_*^{\text{adapt}}, \mathcal{O}_*, \mathcal{C}_*) \in \mathcal{M}_{\text{replay}}(\mathcal{O}_*).
\end{aligned}$$

A typical replay-style historical policy is:

1. preserve inherited scaffold parameters,
2. initialize residual parameters at zero,
3. optionally freeze inherited coordinates for a burn-in stage,
4. then unfreeze and refit in a controlled window.

This path is kept for archival provenance only and is not part of the active main-body symbolic contract.

E.3 Drive-amplitude comparison path

A drive-amplitude comparison path can be expressed symbolically by comparing several amplitudes $A_0, A_1, \ldots$ at fixed Hamiltonian parameters and examining

$$\Delta E_A(t), \qquad \Delta D_A(t), \qquad F_A(t)$$

across the amplitude family. The formal structure is clear even if the exact report layout is deferred.

E.4 Appendix rule

The rule of this symbolic manuscript is therefore:

1. main body = closed mathematical definitions and comparison contracts,
2. appendix = promising but not yet fully formalized extensions.

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Addendum (2026-04-04): manuscript-aligned direct HH ADAPT / `phase3_v1` sweep notes

This addendum records the new manuscript-aligned direct HH ADAPT behavior on the public `phase3_v1` surface, with `runtime_split=off` treated as the canonical manuscript/public setting.

Physics point for the main recovery study. - `L=2` - `t=1.0` - `U=4.0` - `g_ep=0.5` - `omega0=1.0` - `n_ph_max=1` - `problem=hh` - binary bosons - blocked ordering - open boundary

Authority targets for interpretation. - Manuscript matched-ADAPT target from §19.1: `E = 0.1587240823603703`, `|ΔE| = 5.6178234644e-05`, `stop_reason = drop_plateau`. - Fixed-scaffold anchor from §19.6: `|ΔE|  4.23e-04`, `F  0.9998`.

Current status on the new manuscript-aligned surface The best live strong-coupling beam line on the direct public surface is `beam_runtime_split-off__batching-off__lifetime-off__profile-tight__gamma-1.0` with - `E = 0.15872511715315915` - `|ΔE| = 5.721302743347256e-05` - `ansatz_depth = 20` - `logical_num_parameters = 20` - `num_parameters = 48` - `stop_reason = drop_plateau` - `exact_state_fidelity = 0.9999715655261739`

This line clearly beats the fixed-scaffold anchor, but it does not yet numerically match the manuscript matched-ADAPT target. The residual gap to the manuscript matched-ADAPT line is `1.0347927889e-06` in `|ΔE|`.

A historical diagnostic reoptimization run with windowed `COBYLA` reached - `E = 0.15872468271899687` - `|ΔE| = 5.6778593271189504e-05` - residual gap to the manuscript matched-ADAPT line: `6.0035862640e-07`

This is useful as a numerical diagnostic, but it is not the forward recipe for new runs. Going forward, the intended optimizer family is `SPSA`, not `COBYLA`.

Executed run families and outcomes

1. Strong-coupling beam sweep.

- Artifact root: `artifacts/agent_runs/20260403_hh_l2_u4_g05_beam_sweep_stage1/`
- Best case: `beam_runtime_split-off__batching-off__lifetime-off__profile-tight__gamma-1.0`
- Result: `|ΔE| = 5.721302743347256e-05`
- Interpretation: beam search nearly recovers the manuscript matched-ADAPT quality.

2. Strong-coupling runtime sweep.

- Artifact root: `artifacts/agent_runs/20260403_hh_l2_u4_g05_runtime_sweep_stage1/`
- Best case still bad: `|ΔE| = 0.3282923042519191`
- Interpretation: the single-trajectory runtime line collapses into the same bad basin seen in the manuscript novelty-off failure scale.

3. Phase-3 beam-vs-batching diagnostic.

- Artifact root: `artifacts/agent_runs/20260403_hh_l2_u4_g05_phase3_diag_beam_vs_batch_v1/`
- Single-trajectory cases stayed at `|ΔE|  0.3282923`.
- Beam `b3,c2,k3` immediately recovered `|ΔE| = 5.721302743347256e-05`.
- Interpretation: the decisive lever is frontier diversity from beam search, not batching.

4. Beam-capacity recovery slice.

- Artifact root: `artifacts/agent_runs/20260403_hh_l2_u4_g05_phase3_beam_capacity_recovery_v1/`
- `b3,c2,k3` and `b4,c2,k4` tied at `|ΔE| = 5.721302743347256e-05`.
- Interpretation: larger beam width by itself does not improve the recovered solution once the beam is already large enough.

5. Prune/shortlist micro-sweep.

- Artifact root: `artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_prune_shortlist_micro_v1/`
- All cases were numerically identical to saved precision.
- Interpretation: `phase1_prune_fraction`, `phase1_prune_max_candidates`, and shortlist width are not the active driver of the remaining manuscript gap on this beam-fixed line.

6. Reoptimization slice.

- Artifact root: `artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_reopt_slice_v1/`
- Windowed `COBYLA` improved the gap to `|ΔE| = 5.6778593271189504e-05`.
- `POWELL` windowed stayed at the original beam winner.
- `POWELL` periodic/full refits got worse.
- `append_only` failed badly at `|ΔE| = 0.6244148638396572`.
- Interpretation: optimizer/refit policy can move the line, but the `COBYLA` improvement should be treated as diagnostic only, not as the new manuscript-forward recipe.

7. Score-weight slice.

- Artifact root: `artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_score_weight_slice_v1/`
- `near=0.90` gave a small improvement to `|ΔE| = 5.721150236287498e-05` and also reduced the scaffold to `logical_num_parameters = 19`, `num_parameters = 46`, `ansatz_depth = 19`.
- Other `rho` and `lambda_H` variants were essentially flat at the saved precision.

8. Seed robustness slice.

- Artifact root: `artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_seed_robustness_slice_v1/`
- Seeds `5, 7, 11, 13, 17` all reproduced the same `POWELL` baseline result exactly to saved precision.
- Interpretation: the current beam-fixed `POWELL` plateau is deterministic across this seed set.

9. Weak-coupling diagnostic lanes.

- Canonical weak sweep root: `artifacts/agent_runs/20260403_hh_l2_u05_g02_weak_logical_scaffold_canonical_sw`
- Weak beam sweep root: `artifacts/agent_runs/20260403_hh_l2_u05_g02_beam_sweep_stage1_parent_full_meta/`
- Best weak result in both lanes stayed at `|ΔE|  0.0131029358626`.
- Parent-quality target from §19.3.1 is much smaller, so weak coupling remains a diagnostic regime rather than a recovered manuscript-quality regime.

Good parameters versus bad parameters Parameters/choices that consistently helped or were at least required for recovery. - Direct HH ADAPT on the public `phase3_v1` surface. - `runtime_split=off`. - Beam search on. - Beam at or above `live_branches=3`, `children_per_parent=2`, `terminated_keep=3`. - `gamma_N=1.0` on the public strong-coupling line. - `batching=off` was slightly better than `batching=on` in the best recovered beam lane. - `lifetime_cost_mode=off` was at least as good as `phase3_v1` within the explored resolution. - `near=0.90` was the only score-weight variation that gave a measurable improvement in the explored `POWELL` lane.

Parameters/choices that were neutral or nearly inert in the explored region. - Beam width beyond the `b3,c2,k3` threshold. - `profile=tight` versus `profile=wide`. - `phase1_prune_fraction` in the explored `0.15` to `0.35` range. - `phase1_prune_max_candidates` in the explored `4` to `8` range. - shortlist width in the explored micro-sweep. - Most `rho` and `lambda_H` score-weight perturbations.

Parameters/choices that consistently hurt. - Single-trajectory runtime mode with no beam. - Runtime line behavior that reproduces `|ΔE|  0.3282923`. - `runtime_split=shortlist_pauli_children_v1` relative to `off` on this manuscript/public lane. - `POWELL` periodic/full refits relative to the windowed baseline. - `append_only` refit policy, which failed catastrophically.

Practical interpretation For the manuscript-aligned direct HH ADAPT path at strong coupling, the main qualitative lesson is now clear: beam-induced frontier diversity is the key ingredient that recovers the near-manuscript solution, while pruning pressure, shortlist width, profile choice, and modest score-weight changes are second-order. The live public-surface recipe already beats the fixed-scaffold anchor by a large margin, but it still does not numerically match the manuscript matched-ADAPT target.

2026-04-04 update: public-CLI runtime weights and reduced-pool sweeps After exposing the manuscript-facing selector/runtime knobs on the public direct `phase3_v1` CLI, two additional conclusions emerged for the same strong-coupling $L = 2$ point.

1. Public-CLI `full_meta + SPSA` improved once frontier ratios were decoupled from the near-degenerate batch shell.

- Artifact root: `artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_public_weights_frontier_spsa_v1/`
- Best public-CLI `full_meta + SPSA` line:

$$E = 0.15878350593338042, \qquad |\Delta E| = 1.1560180765 \times 10^{-4},$$

  with `logical_num_parameters=21`, `num_parameters=53`, `ansatz_depth=21`, and `fidelity=0.9999480673`.
- The corresponding `weights_zero` line was only slightly worse,

$$|\Delta E| = 1.1722877711 \times 10^{-4},$$

  while the tighter coupled-frontier setting

$$(\text{phase2 frontier}, \text{phase3 frontier}) = (0.98, 0.98)$$

  reproduced the older worse basin

$$|\Delta E| = 1.4383993770 \times 10^{-4}.$$

- So the main gain from that patch was not a delicate burden-weight choice; it was fixing the implementation coupling between shortlist frontiering and the batch-shell threshold so that `phase2_frontier_ratio`, `phase3_frontier_ratio`, and `eta_{nd}` became genuinely independent controls.

2. The reduced-pool `pareto_lean + SPSA` lane remained rigid.

- Artifact roots:
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_pool_family_spsa_fixed_v1/
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_pareto_lean_gamma_spsa_fixed_v1/
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_phase3_pareto_lean_drop_policy_spsa_fixed_v1/
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_pareto_lean_new_cli_weights_spsa_v1/
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_pareto_lean_compat_proxy_spsa_v1/
  - artifacts/agent_runs/20260404_hh_l2_u4_g05_pareto_lean_frontier_ratio_spsa_v1/
- Baseline reduced-pool result:

$$E = 0.1587895750333393, \qquad |\Delta E| = 1.2167090761 \times 10^{-4},$$

  with `logical_num_parameters=21`, `num_parameters=52`, `ansatz_depth=21`, and `fidelity=0.9999443566`.
- This value persisted across all tested variations of:
  - `gamma_N` in the explored `0.75` to `1.5` range,
  - drop-policy parameters,
  - motif bonus,
  - duplicate penalty,
  - leakage penalty `eta_L`,
  - compatibility-prescreen weights,
  - remaining-evaluations proxy mode,
  - frontier ratios from `(0.85,0.85)` up through `(0.95,0.95)` and the asymmetric `(0.90,0.75)`, `(0.90,0.95)`, `(0.95,0.90)` cases.
- Loosening the phase-2 frontier too far did hurt: `(0.75,0.75)` gave

$$|\Delta E| = 1.5234589586 \times 10^{-4},$$

  and `(0.75,0.90)` gave

$$|\Delta E| = 1.5457582580 \times 10^{-4}.$$

- Hence the reduced `pareto_lean` line appears structurally plateaued under the currently exposed runtime-law knobs: it is not rescued by more local tuning of `gamma_N`, batching pressure, motif/repetition heuristics, or compatibility/proxy terms.

3. Cost-aware ranking versus raw-energy ranking now separate.

- On raw strong-coupling energy error alone, the best historical rerank observed in these sweeps was the diagnostic `COBYLA` line

$$E = 0.15872468271899687, \qquad |\Delta E| = 5.6778593271 \times 10^{-5},$$

  with `logical_num_parameters=20`, `num_parameters=50`, `ansatz_depth=20`.
- However, the best currently observed $|\Delta E|/K$-style point among runs that reported logical scaffold size was the `near=0.90` strong-coupling line from the score-weight slice:

$$E = 0.15872511562808855, \qquad |\Delta E| = 5.7211502363 \times 10^{-5},$$

  with

$$K_{\text{logical}} = 19, \qquad K_{\text{runtime}} = 46, \qquad d = 19.$$

- So the present picture is:
  - best raw-$|\Delta E|$ line: slightly better than the old beam winner, but achieved on a non-preferred optimizer branch,
  - best cost-aware line: slightly worse in $|\Delta E|$ but leaner in both logical scaffold size and runtime parameter count,
  - best reduced-pool SPSA line: still stuck near 1.22×10^{-4} and therefore materially behind the broad-beam public lane.

2026-04-04 update: completed reduced-pool `L=2/L=3` matrix The overnight reduced-pool matrix `artifacts/agent_runs/20260404_hh_phase3_reduced_pool_matrix_v1/` completed through the strong confirmation and weak-transfer waves. Its campaign summary and final strong templates are recorded in

- artifacts/agent_runs/20260404_hh_phase3_reduced_pool_matrix_v1/campaign_summary.json,
- artifacts/agent_runs/20260404_hh_phase3_reduced_pool_matrix_v1/final_strong_templates_by_L.json,
- artifacts/agent_runs/20260404_hh_phase3_reduced_pool_matrix_v1/matrix_leaderboard.json.

Data field: the completed matrix contained 212 successful cases, 36 failed cases, and 184 unique successful configurations.

For the strong-coupling $L = 2$ target point, the best matrix result was

$$E = 0.15877612622031317, \qquad |\Delta E| = 1.0822209459 \times 10^{-4},$$

with

$$K_{\text{logical}} = 20, \qquad K_{\text{runtime}} = 52, \qquad d = 20, \qquad F = 0.9999519211,$$

from the confirmed pareto_lean template with

$$(\text{beam}, \text{frontier}) = (2, 2, 2;\ 0.85, 0.85).$$

This improved over the earlier reduced-pool SPSA plateau near 1.2167×10^{-4}, but it still did not recover the manuscript-quality strong-coupling target at $5.6178234644 \times 10^{-5}$ and did not beat the earlier broad-lane live recovery line near $5.7213027433 \times 10^{-5}$.

For the strong-coupling $L = 3$ point, the reduced-pool matrix failed badly. Its best confirmed case was

$$E = -0.2564741176143438, \qquad |\Delta E| = 5.0141481774 \times 10^{-1},$$

with fidelity only

$$F \approx 5.81 \times 10^{-8}.$$

The best matrix lane there was uccsd_paop_lf_full, not the more aggressively reduced families. Hence the completed reduced-pool matrix should be read as negative evidence for the current public reduced execution surface at $L = 3$, not as evidence that the historical $L = 3$ HH target itself is unattainable.

For weak coupling, the transferred $L = 2$ reduced-pool line remained poor:

$$E = -0.7632521982953032, \qquad |\Delta E| = 1.3125332609 \times 10^{-2},$$

so the matrix did not close the weak-coupling parent-quality gap there. By contrast, the transferred $L = 3$ weak baseline was numerically much better,

$$E = -1.0439438365685594, \qquad |\Delta E| = 4.1313925567 \times 10^{-4},$$

with fidelity

$$F = 0.9998116001.$$

That $L = 3$ weak result is still not a manuscript anchor, but it does show that the reduced families are not uniformly broken across all $L = 3$ HH regimes; the main failure is the strong-coupling $L = 3$ execution surface.

To make that contrast explicit, the surviving historical broad-pool $L = 3$ reference from artifacts/reports/hh_heavy_scaffold was

$$E = 0.24502564220194084, \qquad |\Delta E| = 8.4942074026 \times 10^{-5},$$

while the matched lean comparison report artifacts/reports/pareto_lean_vs_heavy_L3_comparison.md recorded the lean/class-pruned line at about

$$|\Delta E| \approx 1.15 \times 10^{-4}.$$

Both are therefore qualitatively incompatible with the present reconstructed direct-surface reruns, which all collapsed to the same bad basin.

In particular, the explicit three-way reproduction run artifacts/agent_runs/20260404_hh_l3_u4_g05_historical_surface_ reconstructed the documented historical settings

phase3_v1, POWELL, phase1 shortlist $= 256$, phase2 shortlist $= 128$, split = shortlist_pauli_children_v1,

and compared

- `full_meta`,
- class-filtered `full_meta`,
- `pareto_lean_l3`.

All three produced the same failed result,

$$E = -0.25838912148429044, \qquad |\Delta E| = 5.0332982161 \times 10^{-1},$$

with negligible fidelity. Hence the current issue is not simply which reduced family is chosen. Rather, some additional execution-surface ingredient from the older successful $L = 3$ lane is still missing from the present direct CLI reconstruction.

Appendix: raw HF-start direct ADAPT burdens versus historical post-processed / fixed-scaffold submissions To avoid conflating the heavy raw recent ADAPT replay artifacts with the older locked fixed-manifold scaffold family, the following tables record the exact transpile-style cost snapshots discussed during the April 7 accounting pass.

These counts are all on the same local hardware proxy surface, namely `FakeMarrakesh`, and they should be interpreted as scaffold burden rather than full campaign burden. In particular, the recent raw ADAPT ladders below are *not* the old locked 6/7-term artifacts.

Throughout this appendix, **raw HF-start direct ADAPT** means `adapt_ref_json = null` with no later JSON prune/recovery substitution. The post-processed `31/37/57`-2Q readapt/prune artifacts listed later are comparison-only and are not the object a fresh real-QPU ADAPT run from HF would execute.

A. Raw HF-start direct ADAPT / replay scaffold burden snapshots

Case	Logical blocks	Runtime rotations	Raw `cx`	Transpiled 2Q	Depth	Size
Noisy weak raw run (`20260407` ladder)	19	47	142	221	631	922
Noisy mid raw run (`20260407` ladder)	21	53	154	256	697	1084
Noisy high raw run (`20260407` ladder)	23	56	160	262	764	1113
Noiseless Math front-page anchor (`w5_l2_strong_pareto_lean_b2c2k2_f0p85_0p85_f1_seed5`)	20	52	152	218	633	939

Interpretation: the validated front-page `pareto_lean + phase3_v1` deployment anchor is still a heavy raw HF-start ADAPT replay scaffold on Marrakesh (`218` two-qubit gates), i.e. it sits near the recent weak raw-noise lane and far above the post-processed prune/readapt and locked fixed-scaffold families. The lighter `81`-2Q direct HF-start winner is the separate scientific winner listed on the front page.

B. Historical post-processed or fixed-scaffold runtime submissions The next table isolates the $L = 2$ runtime submissions that were actually sent to the real `ibm_marrakesh` backend. The listed two-qubit counts are obtained by transpiling the exact submitted scaffold artifacts on `FakeMarrakesh` with the same transpiler settings recorded in the runtime-result JSON.

Real-QPU submission	Runtime source artifact	Runtime transpile settings	Transpiled 2Q	Depth	Notes
hh_gatepruned7_fixedtheta_marrakesh_20260328T142797Z_direct	fixed/hh_gatepruned7_fixedtheta.artifact.json	backend=ibm_marrakesh seed=42	25	63	Locked 7-term fixed-theta Marrakesh submission
hh_readapt5_energy_marrakesh_20260323T153243Z	artifacts/ADAPT/hh/hh_readapt_5.artifact.json	backend=ibm_marrakesh	37	126	Post-prune/recovery runtime candidate; not fresh HF-start ADAPT
hh_gatepruned6_fixedtheta_marrakesh_20260323T171528Z	fixed/hh_gatepruned6_fixedtheta.artifact.json	backend=ibm_marrakesh seed=42	14	48	Locked 6-term fixed-theta Marrakesh submission

So the historical real-QPU gate counts for the three recorded Marrakesh submissions were

$$25, \qquad 37, \qquad 14$$

two-qubit gates respectively for the locked 7-term, readapt-5, and locked 6-term submissions.